Part I

Linear Algebra

Chapter 1

Matrices

1.1 Definition of a Matrix

Definition 1.1.1 (Matrix) A rectangular array of numbers is called a matrix.

We shall mostly be concerned with matrices having real numbers as entries.

The horizontal arrays of a matrix are called its ROWS and the vertical arrays are called its COLUMNS. A matrix having m rows and n columns is said to have the order $m \times n$.

A matrix A of ORDER $m \times n$ can be represented in the following form:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix},$$

where a_{ij} is the entry at the intersection of the ith row and j^{th} column.

In a more concise manner, we also denote the matrix A by $[a_{ij}]$ by suppressing its order.

Remark 1.1.2 *Some books also use* $\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$ *to represent a matrix.*

Let $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 5 & 6 \end{bmatrix}$. Then $a_{11} = 1,\ a_{12} = 3,\ a_{13} = 7,\ a_{21} = 4,\ a_{22} = 5,$ and $a_{23} = 6$.

A matrix having only one column is called a COLUMN VECTOR; and a matrix with only one row is called a ROW VECTOR.

WHENEVER A VECTOR IS USED, IT SHOULD BE UNDERSTOOD FROM THE CONTEXT WHETHER IT IS A ROW VECTOR OR A COLUMN VECTOR.

Definition 1.1.3 (Equality of two Matrices) Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ having the same order $m \times n$ are equal if $a_{ij} = b_{ij}$ for each $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$.

In other words, two matrices are said to be equal if they have the same order and their corresponding entries are equal.

Example 1.1.4 The linear system of equations $2x + 3y = 5$ and $3x + 2y = 5$ can be identified with the matrix $\begin{bmatrix} 2 & 3 & : & 5 \\ 3 & 2 & : & 5 \end{bmatrix}$.

1.1.1 Special Matrices

Definition 1.1.5

1. A matrix in which each entry is zero is called a zero-matrix, denoted by $\mathbf{0}$. For example,
$$\mathbf{0}_{2\times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } \mathbf{0}_{2\times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

2. A matrix having the number of rows equal to the number of columns is called a square matrix. Thus, its order is $m \times m$ (for some m) and is represented by m only.

3. In a square matrix, $A = [a_{ij}]$, of order n, the entries $a_{11}, a_{22}, \ldots, a_{nn}$ are called the diagonal entries and form the principal diagonal of A.

4. A square matrix $A = [a_{ij}]$ is said to be a diagonal matrix if $a_{ij} = 0$ for $i \neq j$. In other words, the non-zero entries appear only on the principal diagonal. For example, the zero matrix $\mathbf{0}_n$ and $\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$ are a few diagonal matrices.

   A diagonal matrix D of order n with the diagonal entries $d_1, d_2, \ldots, d_n$ is denoted by $D = \text{diag}(d_1, \ldots, d_n)$.

   If $d_i = d$ for all $i = 1, 2, \ldots, n$ then the diagonal matrix D is called a **scalar matrix**.

5. A square matrix $A = [a_{ij}]$ with $a_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$ is called the identity matrix, denoted by I_n.

   For example, $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, and $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

   The subscript n is suppressed in case the order is clear from the context or if no confusion arises.

6. A square matrix $A = [a_{ij}]$ is said to be an upper triangular matrix if $a_{ij} = 0$ for $i > j$.

   A square matrix $A = [a_{ij}]$ is said to be an lower triangular matrix if $a_{ij} = 0$ for $i < j$.

   A square matrix A is said to be triangular if it is an upper or a lower triangular matrix.

   For example $\begin{bmatrix} 2 & 1 & 4 \\ 0 & 3 & -1 \\ 0 & 0 & -2 \end{bmatrix}$ is an upper triangular matrix. An upper triangular matrix will be represented by $\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$.

1.2 Operations on Matrices

Definition 1.2.1 (Transpose of a Matrix) The transpose of an $m \times n$ matrix $A = [a_{ij}]$ is defined as the $n \times m$ matrix $B = [b_{ij}]$, with $b_{ij} = a_{ji}$ for $1 \leq i \leq m$ and $1 \leq j \leq n$. The transpose of A is denoted by A^t.

That is, by the transpose of an $m \times n$ matrix A, we mean a matrix of order $n \times m$ having the rows of A as its columns and the columns of A as its rows.

For example, if $A = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 2 \end{bmatrix}$ then $A^t = \begin{bmatrix} 1 & 0 \\ 4 & 1 \\ 5 & 2 \end{bmatrix}$.

Thus, the transpose of a row vector is a column vector and vice-versa.

Theorem 1.2.2 For any matrix A, we have $(A^t)^t = A$.

PROOF. Let $A = [a_{ij}]$, $A^t = [b_{ij}]$ and $(A^t)^t = [c_{ij}]$. Then, the definition of transpose gives

$$c_{ij} = b_{ji} = a_{ij} \quad \text{for all} \quad i, j$$

and the result follows. □

Definition 1.2.3 (Addition of Matrices) let $A = [a_{ij}]$ and $B = [b_{ij}]$ be are two $m \times n$ matrices. Then the sum $A + B$ is defined to be the matrix $C = [c_{ij}]$ with $c_{ij} = a_{ij} + b_{ij}$.

Note that, we define the sum of two matrices only when the order of the two matrices are same.

Definition 1.2.4 (Multiplying a Scalar to a Matrix) Let $A = [a_{ij}]$ be an $m \times n$ matrix. Then for any element $k \in \mathbb{R}$, we define $kA = [ka_{ij}]$.

For example, if $A = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 2 \end{bmatrix}$ and $k = 5$, then $5A = \begin{bmatrix} 5 & 20 & 25 \\ 0 & 5 & 10 \end{bmatrix}$.

Theorem 1.2.5 Let A, B and C be matrices of order $m \times n$, and let $k, \ell \in \mathbb{R}$. Then

1. $A + B = B + A$ (commutativity).
2. $(A + B) + C = A + (B + C)$ (associativity).
3. $k(\ell A) = (k\ell)A$.
4. $(k + \ell)A = kA + \ell A$.

PROOF. Part 1.
Let $A = [a_{ij}]$ and $B = [b_{ij}]$. Then

$$A + B = [a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}] = [b_{ij} + a_{ij}] = [b_{ij}] + [a_{ij}] = B + A$$

as real numbers commute.

The reader is required to prove the other parts as all the results follow from the properties of real numbers. □

Exercise 1.2.6 1. Suppose $A + B = A$. Then show that $B = \mathbf{0}$.

2. Suppose $A + B = \mathbf{0}$. Then show that $B = (-1)A = [-a_{ij}]$.

Definition 1.2.7 (Additive Inverse) Let A be an $m \times n$ matrix.

1. Then there exists a matrix B with $A + B = \mathbf{0}$. This matrix B is called the additive inverse of A, and is denoted by $-A = (-1)A$.
2. Also, for the matrix $\mathbf{0}_{m\times n}$, $A + \mathbf{0} = \mathbf{0} + A = A$. Hence, the matrix $\mathbf{0}_{m\times n}$ is called the additive identity.

1.2.1 Multiplication of Matrices

Definition 1.2.8 (Matrix Multiplication / Product) Let $A = [a_{ij}]$ be an $m \times n$ matrix and $B = [b_{ij}]$ be an $n \times r$ matrix. The product AB is a matrix $C = [c_{ij}]$ of order $m \times r$, with

$$c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

Observe that the product AB is defined if and only if
THE NUMBER OF COLUMNS OF A = THE NUMBER OF ROWS OF B.

For example, if $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 3 \\ 1 & 0 & 4 \end{bmatrix}$ then

$$AB = \begin{bmatrix} 1+0+3 & 2+0+0 & 1+6+12 \\ 2+0+1 & 4+0+0 & 2+12+4 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 19 \\ 3 & 4 & 18 \end{bmatrix}.$$

Note that in this example, while AB is defined, the product BA is not defined. However, for square matrices A and B of the same order, both the product AB and BA are defined.

Definition 1.2.9 Two square matrices A and B are said to commute if $AB = BA$.

Remark 1.2.10 *1. Note that if A is a square matrix of order n then $AI_n = I_nA$. Also for any $d \in \mathbb{R}$, the matrix dI_n commutes with every square matrix of order n. The matrices dI_n for any $d \in \mathbb{R}$ are called* SCALAR *matrices.*

2. In general, the matrix product is not commutative. For example, consider the following two matrices $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$. Then check that the matrix product

$$AB = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = BA.$$

Theorem 1.2.11 Suppose that the matrices A, B and C are so chosen that the matrix multiplications are defined.

1. Then $(AB)C = A(BC)$. That is, the matrix multiplication is associative.
2. For any $k \in \mathbb{R}$, $(kA)B = k(AB) = A(kB)$.
3. Then $A(B + C) = AB + AC$. That is, multiplication distributes over addition.
4. If A is an $n \times n$ matrix then $AI_n = I_nA = A$.
5. For any square matrix A of order n and $D = \text{diag}(d_1, d_2, \ldots, d_n)$, we have
   - the first row of DA is d_1 times the first row of A;
   - for $1 \leq i \leq n$, the ith row of DA is d_i times the ith row of A.

   A similar statement holds for the columns of A when A is multiplied on the right by D.

PROOF. Part 1. Let $A = [a_{ij}]_{m\times n}$, $B = [b_{ij}]_{n\times p}$ and $C = [c_{ij}]_{p\times q}$. Then

$$(BC)_{kj} = \sum_{\ell=1}^{p} b_{k\ell}c_{\ell j} \quad \text{and} \quad (AB)_{i\ell} = \sum_{k=1}^{n} a_{ik}b_{k\ell}.$$

Therefore,

$$\begin{aligned} \big(A(BC)\big)_{ij} &= \sum_{k=1}^{n} a_{ik}\big(BC\big)_{kj} = \sum_{k=1}^{n} a_{ik}\big(\sum_{\ell=1}^{p} b_{k\ell}c_{\ell j}\big) \\ &= \sum_{k=1}^{n}\sum_{\ell=1}^{p} a_{ik}\big(b_{k\ell}c_{\ell j}\big) = \sum_{k=1}^{n}\sum_{\ell=1}^{p}\big(a_{ik}b_{k\ell}\big)c_{\ell j} \\ &= \sum_{\ell=1}^{p}\big(\sum_{k=1}^{n} a_{ik}b_{k\ell}\big)c_{\ell j} = \sum_{\ell=1}^{t}\big(AB\big)_{i\ell}c_{\ell j} \\ &= \big((AB)C\big)_{ij}. \end{aligned}$$

Part 5. For all $j = 1, 2, \ldots, n$, we have

$$(DA)_{ij} = \sum_{k=1}^{n} d_{ik}a_{kj} = d_ia_{ij}$$

as $d_{ik} = 0$ whenever $i \neq k$. Hence, the required result follows.

The reader is required to prove the other parts. □

Exercise 1.2.12 1. Let A and B be two matrices. If the matrix addition $A + B$ is defined, then prove that $(A + B)^t = A^t + B^t$. Also, if the matrix product AB is defined then prove that $(AB)^t = B^tA^t$.

2. Let $A = [a_1, a_2, \ldots, a_n]$ and $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$. Compute the matrix products AB and BA.

3. Let n be a positive integer. Compute A^n for the following matrices:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

   Can you guess a formula for A^n and prove it by induction?

4. Find examples for the following statements.
   (a) Suppose that the matrix product AB is defined. Then the product BA need not be defined.
   (b) Suppose that the matrix products AB and BA are defined. Then the matrices AB and BA can have different orders.
   (c) Suppose that the matrices A and B are square matrices of order n. Then AB and BA may or may not be equal.

1.3 Some More Special Matrices

Definition 1.3.1 1. A matrix A over $\mathbb{R}$ is called symmetric if $A^t = A$ and skew-symmetric if $A^t = -A$.

2. A matrix A is said to be orthogonal if $AA^t = A^tA = I$.

Example 1.3.2 1. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & -1 \\ 3 & -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix}$. Then A is a symmetric matrix and B is a skew-symmetric matrix.

2. Let $A = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \end{bmatrix}$. Then A is an orthogonal matrix.

3. Let $A = [a_{ij}]$ be an $n \times n$ matrix with $a_{ij} = \begin{cases} 1 & \text{if } i = j + 1 \\ 0 & \text{otherwise} \end{cases}$. Then $A^n = \mathbf{0}$ and $A^\ell \neq \mathbf{0}$ for $1 \leq \ell \leq n - 1$. The matrices A for which a positive integer k exists such that $A^k = \mathbf{0}$ are called NILPOTENT matrices. The least positive integer k for which $A^k = \mathbf{0}$ is called the ORDER OF NILPOTENCY.

4. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. Then $A^2 = A$. The matrices that satisfy the condition that $A^2 = A$ are called IDEMPOTENT matrices.

Exercise 1.3.3 1. Show that for any square matrix A, $S = \frac{1}{2}(A + A^t)$ is symmetric, $T = \frac{1}{2}(A - A^t)$ is skew-symmetric, and $A = S + T$.

2. Show that the product of two lower triangular matrices is a lower triangular matrix. A similar statement holds for upper triangular matrices.

3. Let A and B be symmetric matrices. Show that AB is symmetric if and only if $AB = BA$.

4. Show that the diagonal entries of a skew-symmetric matrix are zero.

5. Let A, B be skew-symmetric matrices with $AB = BA$. Is the matrix AB symmetric or skew-symmetric?

6. Let A be a symmetric matrix of order n with $A^2 = \mathbf{0}$. Is it necessarily true that $A = \mathbf{0}$?

7. Let A be a nilpotent matrix. Show that there exists a matrix B such that $B(I + A) = I = (I + A)B$.

1.3.1 Submatrix of a Matrix

Definition 1.3.4 A matrix obtained by deleting some of the rows and/or columns of a matrix is said to be a submatrix of the given matrix.

For example, if $A = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 2 \end{bmatrix}$, a few submatrices of A are

$$[1], [2], \begin{bmatrix} 1 \\ 0 \end{bmatrix}, [1\ 5], \begin{bmatrix} 1 & 5 \\ 0 & 2 \end{bmatrix}, A.$$

But the matrices $\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}$ are not submatrices of A. (The reader is advised to give reasons.)

Miscellaneous Exercises

Exercise 1.3.5 1. Complete the proofs of Theorems 1.2.5 and 1.2.11.

2. Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. Geometrically interpret $\mathbf{y} = A\mathbf{x}$ and $\mathbf{y} = B\mathbf{x}$.

3. Consider the two coordinate transformations
$$\begin{aligned} x_1 &= a_{11}y_1 + a_{12}y_2 \\ x_2 &= a_{21}y_1 + a_{22}y_2 \end{aligned} \quad \text{and} \quad \begin{aligned} y_1 &= b_{11}z_1 + b_{12}z_2 \\ y_2 &= b_{21}z_1 + b_{22}z_2 \end{aligned}.$$

(a) Compose the two transformations to express x_1, x_2 in terms of z_1, z_2.

(b) If $\mathbf{x}^t = [x_1,\ x_2]$, $\mathbf{y}^t = [y_1,\ y_2]$ and $\mathbf{z}^t = [z_1,\ z_2]$ then find matrices A, B and C such that $\mathbf{x} = A\mathbf{y}$, $\mathbf{y} = B\mathbf{z}$ and $\mathbf{x} = C\mathbf{z}$.

(c) Is $C = AB$?

4. For a square matrix A of order n, we define **trace** of A, denoted by tr (A) as

$$\text{tr } (A) = a_{11} + a_{22} + \cdots a_{nn}.$$

Then for two square matrices, A and B of the same order, show the following:

(a) tr $(A + B) =$ tr $(A) +$ tr (B).

(b) tr $(AB) =$ tr (BA).

5. Show that, there do not exist matrices A and B such that $AB - BA = cI_n$ for any $c \neq 0$.

6. Let A and B be two $m \times n$ matrices and let $\mathbf{x}$ be an $n \times 1$ column vector.

(a) Prove that if $A\mathbf{x} = \mathbf{0}$ for all $\mathbf{x}$, then A is the zero matrix.

(b) Prove that if $A\mathbf{x} = B\mathbf{x}$ for all $\mathbf{x}$, then $A = B$.

7. Let A be an $n \times n$ matrix such that $AB = BA$ for all $n \times n$ matrices B. Show that $A = \alpha I$ for some $\alpha \in \mathbb{R}$.

8. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}$. Show that there exist infinitely many matrices B such that $BA = I_2$. Also, show that there does not exist any matrix C such that $AC = I_3$.

1.3.1 Block Matrices

Let A be an $n \times m$ matrix and B be an $m \times p$ matrix. Suppose $r < m$. Then, we can decompose the matrices A and B as $A = [P\ Q]$ and $B = \begin{bmatrix} H \\ K \end{bmatrix}$; where P has order $n \times r$ and H has order $r \times p$. That is, the matrices P and Q are submatrices of A and P consists of the first r columns of A and Q consists of the last $m - r$ columns of A. Similarly, H and K are submatrices of B and H consists of the first r rows of B and K consists of the last $m - r$ rows of B. We now prove the following important theorem.

Theorem 1.3.6 Let $A = [a_{ij}] = [P\ Q]$ and $B = [b_{ij}] = \begin{bmatrix} H \\ K \end{bmatrix}$ be defined as above. Then

$$AB = PH + QK.$$

PROOF. First note that the matrices PH and QK are each of order $n \times p$. The matrix products PH and QK are valid as the order of the matrices P, H, Q and K are respectively, $n \times r$, $r \times p$, $n \times (m - r)$ and $(m-r) \times p$. Let $P = [P_{ij}]$, $Q = [Q_{ij}]$, $H = [H_{ij}]$, and $K = [k_{ij}]$. Then, for $1 \leq i \leq n$ and $1 \leq j \leq p$, we have

$$\begin{aligned} (AB)_{ij} &= \sum_{k=1}^{m} a_{ik}b_{kj} = \sum_{k=1}^{r} a_{ik}b_{kj} + \sum_{k=r+1}^{m} a_{ik}b_{kj} \\ &= \sum_{k=1}^{r} P_{ik}H_{kj} + \sum_{k=r+1}^{m} Q_{ik}K_{kj} \\ &= (PH)_{ij} + (QK)_{ij} = (PH + QK)_{ij}. \end{aligned}$$

□

Theorem 1.3.6 is very useful due to the following reasons:

1. The order of the matrices P, Q, H and K are smaller than that of A or B.

2. It may be possible to block the matrix in such a way that a few blocks are either identity matrices or zero matrices. In this case, it may be easy to handle the matrix product using the block form.

3. Or when we want to prove results using induction, then we may assume the result for $r \times r$ submatrices and then look for $(r + 1) \times (r + 1)$ submatrices, etc.

For example, if $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$, Then

$$AB = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}[e\ f] = \begin{bmatrix} a+2c & b+2d \\ 2a+5c & 2b+5d \end{bmatrix}.$$

If $A = \left[\begin{array}{ccc} 0 & -1 & 2 \\ 3 & 1 & 4 \\ -2 & 5 & -3 \end{array}\right]$, then A can be decomposed as follows:

$A = \left[\begin{array}{c|cc} 0 & -1 & 2 \\ \hline 3 & 1 & 4 \\ -2 & 5 & -3 \end{array}\right]$, or $A = \left[\begin{array}{cc|c} 0 & -1 & 2 \\ 3 & 1 & 4 \\ \hline -2 & 5 & -3 \end{array}\right]$, or

$A = \left[\begin{array}{cc|c} 0 & -1 & 2 \\ \hline 3 & 1 & 4 \\ -2 & 5 & -3 \end{array}\right]$ and so on.

Suppose $A = \begin{array}{c} \\ n_1 \\ n_2 \end{array}\begin{array}{c} \begin{array}{cc} m_1 & m_2 \end{array} \\ \begin{bmatrix} P & Q \\ R & S \end{bmatrix} \end{array}$ and $B = \begin{array}{c} \\ r_1 \\ r_2 \end{array}\begin{array}{c} \begin{array}{cc} s_1 & s_2 \end{array} \\ \begin{bmatrix} E & F \\ G & H \end{bmatrix} \end{array}$. Then the matrices P, Q, R, S and E, F, G, H, are called the blocks of the matrices A and B, respectively.

Even if $A + B$ is defined, the orders of P and E **may not be same** and hence, we may not be able to add A and B in the block form. But, if $A + B$ and $P + E$ is defined then $A + B = \begin{bmatrix} P+E & Q+F \\ R+G & S+H \end{bmatrix}$.

Similarly, if the product AB is defined, the product PE need not be defined. Therefore, we can talk of matrix product AB as block product of matrices, if both the products AB and PE are defined. And in this case, we have $AB = \begin{bmatrix} PE+QG & PF+QH \\ RE+SG & RF+SH \end{bmatrix}$.

That is, **once a partition of A is fixed, the partition of B has to be properly chosen for purposes of block addition or multiplication.**

Exercise 1.3.7 1. Compute the matrix product AB using the block matrix multiplication for the matrices $A = \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ \hline 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array}\right]$ and $B = \left[\begin{array}{cc|cc} 1 & 2 & 2 & 1 \\ 1 & 1 & 2 & 1 \\ \hline 1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 1 \end{array}\right]$.

2. Let $A = \begin{bmatrix} P & Q \\ R & S \end{bmatrix}$. If P, Q, R and S are symmetric, what can you say about A? Are P, Q, R and S symmetric, when A is symmetric?

3. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices. Suppose $\mathbf{a}_1,\ \mathbf{a}_2,\ \ldots,\ \mathbf{a}_n$ are the rows of A and $\mathbf{b}_1,\ \mathbf{b}_2,\ \ldots,\ \mathbf{b}_p$ are the columns of B. If the product AB is defined, then show that

$$AB = [A\mathbf{b}_1,\ A\mathbf{b}_2,\ \ldots,\ A\mathbf{b}_p] = \begin{bmatrix} \mathbf{a}_1B \\ \mathbf{a}_2B \\ \vdots \\ \mathbf{a}_nB \end{bmatrix}.$$

[That is, left multiplication by A, is same as multiplying each column of B by A. Similarly, right multiplication by B, is same as multiplying each row of A by B.]

1.4 Matrices over Complex Numbers

Here the entries of the matrix are complex numbers. All the definitions still hold. One just needs to look at the following additional definitions.

Definition 1.4.1 (Conjugate Transpose of a Matrix) 1. Let A be an $m \times n$ matrix over $\mathbb{C}$. If $A = [a_{ij}]$ then the Conjugate of A, denoted by $\overline{A}$, is the matrix $B = [b_{ij}]$ with $b_{ij} = \overline{a_{ij}}$.

For example, Let $A = \begin{bmatrix} 1 & 4+3i & i \\ 0 & 1 & i-2 \end{bmatrix}$. Then

$$\overline{A} = \begin{bmatrix} 1 & 4-3i & -i \\ 0 & 1 & -i-2 \end{bmatrix}.$$

2. Let A be an $m \times n$ matrix over $\mathbb{C}$. If $A = [a_{ij}]$ then the Conjugate Transpose of A, denoted by A^*, is the matrix $B = [b_{ij}]$ with $b_{ij} = \overline{a_{ji}}$.

For example, Let $A = \begin{bmatrix} 1 & 4+3i & i \\ 0 & 1 & i-2 \end{bmatrix}$. Then

$$A^* = \begin{bmatrix} 1 & 0 \\ 4-3i & 1 \\ -i & -i-2 \end{bmatrix}.$$

3. A square matrix A over $\mathbb{C}$ is called Hermitian if $A^* = A$.

4. A square matrix A over $\mathbb{C}$ is called skew-Hermitian if $A^* = -A$.

5. A square matrix A over $\mathbb{C}$ is called unitary if $A^*A = AA^* = I$.

6. A square matrix A over $\mathbb{C}$ is called Normal if $AA^* = A^*A$.

Remark 1.4.2 *If $A = [a_{ij}]$ with $a_{ij} \in \mathbb{R}$, then $A^* = A^t$.*

Exercise 1.4.3 1. Give examples of Hermitian, skew-Hermitian and unitary matrices that have entries with non-zero imaginary parts.

2. Restate the results on **transpose** in terms of **conjugate transpose**.

3. Show that for any square matrix A, $S = \frac{A+A^*}{2}$ is Hermitian, $T = \frac{A-A^*}{2}$ is skew-Hermitian, and $A = S + T$.

4. Show that if A is a complex triangular matrix and $AA^* = A^*A$ then A is a diagonal matrix.

Chapter 2

Linear System of Equations

2.1 Introduction

Let us look at some examples of linear systems.

1. Suppose $a, b \in \mathbb{R}$. Consider the system $ax = b$.
   (a) If $a \neq 0$ then the system has a UNIQUE SOLUTION $x = \frac{b}{a}$.
   (b) If $a = 0$ and
      i. $b \neq 0$ then the system has NO SOLUTION.
      ii. $b = 0$ then the system has INFINITE NUMBER OF SOLUTIONS, namely all $x \in \mathbb{R}$.
2. We now consider a system with 2 equations in 2 unknowns.
   Consider the equation $ax + by = c$. If one of the coefficients, a or b is non-zero, then this linear equation represents a line in $\mathbb{R}^2$. Thus for the system
   $$a_1x + b_1y = c_1 \text{ and } a_2x + b_2y = c_2,$$
   the set of solutions is given by the points of intersection of the two lines. There are three cases to be considered. Each case is illustrated by an example.
   (a) UNIQUE SOLUTION
   $x + 2y = 1$ and $x + 3y = 1$. The unique solution is $(x, y)^t = (1, 0)^t$.
   Observe that in this case, $a_1b_2 - a_2b_1 \neq 0$.
   (b) INFINITE NUMBER OF SOLUTIONS
   $x + 2y = 1$ and $2x + 4y = 2$. The set of solutions is $(x, y)^t = (1 - 2y, y)^t = (1, 0)^t + y(-2, 1)^t$ with y arbitrary. In other words, both the equations represent the same line.
   Observe that in this case, $a_1b_2 - a_2b_1 = 0$, $a_1c_2 - a_2c_1 = 0$ and $b_1c_2 - b_2c_1 = 0$.
   (c) NO SOLUTION
   $x + 2y = 1$ and $2x + 4y = 3$. The equations represent a pair of parallel lines and hence there is no point of intersection.
   Observe that in this case, $a_1b_2 - a_2b_1 = 0$ but $a_1c_2 - a_2c_1 \neq 0$.
3. As a last example, consider 3 equations in 3 unknowns.
   A linear equation $ax + by + cz = d$ represent a plane in $\mathbb{R}^3$ provided $(a, b, c) \neq (0, 0, 0)$. As in the case of 2 equations in 2 unknowns, we have to look at the points of intersection of the given three planes. Here again, we have three cases. The three cases are illustrated by examples.

   (a) UNIQUE SOLUTION
   Consider the system $x+y+z = 3$, $x+4y+2z = 7$ and $4x+10y-z = 13$. The unique solution to this system is $(x, y, z)^t = (1, 1, 1)^t$; *i.e.* THE THREE PLANES INTERSECT AT A POINT.
   (b) INFINITE NUMBER OF SOLUTIONS
   Consider the system $x + y + z = 3$, $x + 2y + 2z = 5$ and $3x + 4y + 4z = 11$. The set of solutions to this system is $(x, y, z)^t = (1, 2 - z, z)^t = (1, 2, 0)^t + z(0, -1, 1)^t$, with z arbitrary: THE THREE PLANES INTERSECT ON A LINE.
   (c) NO SOLUTION
   The system $x + y + z = 3$, $x + 2y + 2z = 5$ and $3x + 4y + 4z = 13$ has no solution. In this case, we get three parallel lines as intersections of the above planes taken two at a time.
   The readers are advised to supply the proof.

2.2 Definition and a Solution Method

Definition 2.2.1 (Linear System) A linear system of m equations in n unknowns $x_1, x_2, \ldots, x_n$ is a set of equations of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \qquad\qquad & \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned} \tag{2.2.1}$$

where for $1 \leq i \leq n$, and $1 \leq j \leq m$; $a_{ij}, b_i \in \mathbb{R}$. Linear System (2.2.1) is called HOMOGENEOUS if $b_1 = 0 = b_2 = \cdots = b_m$ and NON-HOMOGENEOUS otherwise.

We rewrite the above equations in the form $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \text{ and } \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

The matrix A is called the COEFFICIENT matrix and the block matrix $[A \;\; \mathbf{b}]$, is the AUGMENTED matrix of the linear system (2.2.1).

Remark 2.2.2 *Observe that the ith row of the augmented matrix $[A \;\; \mathbf{b}]$ represents the ith equation and the jth column of the coefficient matrix A corresponds to coefficients of the jth variable x_j. That is, for $1 \leq i \leq m$ and $1 \leq j \leq n$, the entry a_{ij} of the coefficient matrix A corresponds to the ith equation and j^{th} variable x_j..*

For a system of linear equations $A\mathbf{x} = \mathbf{b}$, the system $A\mathbf{x} = \mathbf{0}$ is called the ASSOCIATED HOMOGENEOUS SYSTEM.

Definition 2.2.3 (Solution of a Linear System) A solution of the linear system $A\mathbf{x} = \mathbf{b}$ is a column vector $\mathbf{y}$ with entries $y_1, y_2, \ldots, y_n$ such that the linear system (2.2.1) is satisfied by substituting y_i in place of x_i.

That is, if $\mathbf{y}^t = [y_1, y_2, \ldots, y_n]$ then $A\mathbf{y} = \mathbf{b}$ holds.

Note: The zero n-tuple $\mathbf{x} = \mathbf{0}$ is always a solution of the system $A\mathbf{x} = \mathbf{0}$, and is called the TRIVIAL solution. A non-zero n-tuple $\mathbf{x}$, if it satisfies $A\mathbf{x} = \mathbf{0}$, is called a NON-TRIVIAL solution.

2.2.1 A Solution Method

Example 2.2.4 Let us solve the linear system $x + 7y + 3z = 11$, $x + y + z = 3$, and $4x + 10y - z = 13$.
Solution:

1. The above linear system and the linear system
   $$\begin{aligned} x + y + z &= 3 \quad \text{Interchange the first two equations.} \\ x + 7y + 3z &= 11 \\ 4x + 10y - z &= 13 \end{aligned} \tag{2.2.2}$$
   have the same set of solutions. (why?)
2. Eliminating x from 2nd and 3rd equation, we get the linear system
   $$\begin{aligned} x + y + z &= 3 \\ 6y + 2z &= 8 \quad \text{(obtained by subtracting the first equation from the second equation.)} \\ 6y - 5z &= 1 \quad \text{(obtained by subtracting 4 times the first equation from the third equation.)} \end{aligned} \tag{2.2.3}$$
   This system and the system (2.2.2) has the same set of solution. (why?)
3. Eliminating y from the last two equations of system (2.2.3), we get the system
   $$\begin{aligned} x + y + z &= 3 \\ 6y + 2z &= 8 \\ 7z &= 7 \quad \text{obtained by subtracting the third equation from the second equation.} \end{aligned} \tag{2.2.4}$$
   which has the same set of solution as the system (2.2.3). (why?)
4. The system (2.2.4) and system
   $$\begin{aligned} x + y + z &= 3 \\ 3y + z &= 4 \quad \text{divide the second equation by 2} \\ z &= 1 \quad \text{divide the second equation by 2} \end{aligned} \tag{2.2.5}$$
   has the same set of solution. (why?)
5. Now, $z = 1$ implies $y = \frac{4-1}{3} = 1$ and $x = 3 - (1 + 1) = 1$. Or in terms of a vector, the set of solution is $\{ (x, y, z)^t : (x, y, z) = (1, 1, 1)\}$.

2.3 Row Operations and Equivalent Systems

Definition 2.3.1 (Elementary Operations) The following operations 1, 2 and 3 are called elementary operations.

1. interchange of two equations, say "interchange the ith and j^{th} equations";
   (compare the system (2.2.2) with the original system.)

2. multiply a non-zero constant throughout an equation, say "multiply the kth equation by $c \neq 0$";
   (compare the system (2.2.5) and the system (2.2.4).)
3. replace an equation by itself plus a constant multiple of another equation, say "replace the kth equation by k^{th} equation plus c times the jth equation".
   (compare the system (2.2.3) with (2.2.2) or the system (2.2.4) with (2.2.3).)

Observations:

1. In the above example, observe that the elementary operations helped us in getting a linear system (2.2.5), which was easily solvable.
2. Note that at Step 1, if we interchange the first and the second equation, we get back to the linear system from which we had started. This means the operation at Step 1, has an inverse operation. In other words, INVERSE OPERATION sends us back to the step where we had precisely started. It will be a useful exercise for the reader to IDENTIFY THE INVERSE OPERATIONS at each step in Example 2.2.4.

So, in Example 2.2.4, the application of a finite number of elementary operations helped us to obtain a simpler system whose solution can be obtained directly. That is, after applying a finite number of elementary operations, a simpler linear system is obtained which can be easily solved. Note that the three elementary operations defined above, have corresponding INVERSE operations, namely,

1. "interchange the ith and j^{th} equations",
2. "divide the kth equation by $c \neq 0$";
3. "replace the kth equation by k^{th} equation minus c times the jth equation".

It will be a useful exercise for the reader to IDENTIFY THE INVERSE OPERATIONS at each step in Example 2.2.4.

Definition 2.3.2 (Equivalent Linear Systems) Two linear systems are said to be equivalent if one can be obtained from the other by a finite number of elementary operations.

The linear systems at each step in Example 2.2.4 are equivalent to each other and also to the original linear system.

Lemma 2.3.3 Let $C\mathbf{x} = \mathbf{d}$ be the linear system obtained from the linear system $A\mathbf{x} = \mathbf{b}$ by a single elementary operation. Then the linear systems $A\mathbf{x} = \mathbf{b}$ and $C\mathbf{x} = \mathbf{d}$ have the same set of solutions.

PROOF. We prove the result for the elementary operation "the kth equation is replaced by k^{th} equation plus c times the jth equation." The reader is advised to prove the result for other elementary operations.

In this case, the systems $A\mathbf{x} = \mathbf{b}$ and $C\mathbf{x} = \mathbf{d}$ vary only in the kth equation. Let $(\alpha_1, \alpha_2, \ldots, \alpha_n)$ be a solution of the linear system $A\mathbf{x} = b$. Then substituting for α_i's in place of x_i's in the kth and j^{th} equations, we get

$$a_{k1}\alpha_1 + a_{k2}\alpha_2 + \cdots a_{kn}\alpha_n = b_k, \text{ and } a_{j1}\alpha_1 + a_{j2}\alpha_2 + \cdots a_{jn}\alpha_n = b_j.$$

Therefore,

$$(a_{k1} + ca_{j1})\alpha_1 + (a_{k2} + ca_{j2})\alpha_2 + \cdots + (a_{kn} + ca_{jn})\alpha_n = b_k + cb_j. \tag{2.3.1}$$

But then the kth equation of the linear system $C\mathbf{x} = \mathbf{d}$ is

$$(a_{k1} + ca_{j1})x_1 + (a_{k2} + ca_{j2})x_2 + \cdots + (a_{kn} + ca_{jn})x_n = b_k + cb_j. \tag{2.3.2}$$

Therefore, using Equation (2.3.1), $(\alpha_1, \alpha_2, \ldots, \alpha_n)$ is also a solution for the kth Equation (2.3.2).

Use a similar argument to show that if $(\beta_1, \beta_2, \ldots, \beta_n)$ is a solution of the linear system $C\mathbf{x} = \mathbf{d}$ then it is also a solution of the linear system $A\mathbf{x} = \mathbf{b}$.

Hence, we have the proof in this case. □

Lemma 2.3.3 is now used as an induction step to prove the main result of this section (Theorem 2.3.4).

Theorem 2.3.4 Two equivalent systems have the same set of solutions.

PROOF. Let n be the number of elementary operations performed on $A\mathbf{x} = \mathbf{b}$ to get $C\mathbf{x} = \mathbf{d}$. We prove the theorem by induction on n.

If $n = 1$, Lemma 2.3.3 answers the question. If $n > 1$, assume that the theorem is true for $n = m$. Now, suppose $n = m + 1$. Apply the Lemma 2.3.3 again at the "last step" (that is, at the $(m+1)^{\text{th}}$ step from the mth step) to get the required result using induction. □

Let us formalise the above section which led to Theorem 2.3.4. For solving a linear system of equations, we applied elementary operations to equations. It is observed that in performing the elementary operations, the calculations were made on the COEFFICIENTS (numbers). The variables $x_1, x_2, \ldots, x_n$ and the sign of equality (that is, " $=$ ") are not disturbed. Therefore, in place of looking at the system of equations as a whole, we just need to work with the coefficients. These coefficients when arranged in a rectangular array gives us the augmented matrix $[A \;\; \mathbf{b}]$.

Definition 2.3.5 (Elementary Row Operations) The elementary row operations are defined as:

1. interchange of two rows, say "interchange the ith and j^{th} rows", denoted R_{ij};
2. multiply a non-zero constant throughout a row, say "multiply the kth row by $c \neq 0$", denoted $R_k(c)$;
3. replace a row by itself plus a constant multiple of another row, say "replace the kth row by k^{th} row plus c times the jth row", denoted $R_{kj}(c)$.

Exercise 2.3.6 Find the INVERSE row operations corresponding to the elementary row operations that have been defined just above.

Definition 2.3.7 (Row Equivalent Matrices) Two matrices are said to be row-equivalent if one can be obtained from the other by a finite number of elementary row operations.

Example 2.3.8 The three matrices given below are row equivalent.

$$\begin{bmatrix} 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 1 & 1 & 1 & 3 \end{bmatrix} \xrightarrow{R_{12}} \begin{bmatrix} 2 & 0 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \end{bmatrix} \xrightarrow{R_1(1/2)} \begin{bmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \end{bmatrix}.$$

Whereas the matrix $\begin{bmatrix} 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 1 & 1 & 1 & 3 \end{bmatrix}$ is not row equivalent to the matrix $\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 2 & 3 & 5 \\ 1 & 1 & 1 & 3 \end{bmatrix}$.

2.3.1 Gauss Elimination Method

Definition 2.3.9 (Forward/Gauss Elimination Method) Gaussian elimination is a method of solving a linear system $A\mathbf{x} = \mathbf{b}$ (consisting of m equations in n unknowns) by bringing the augmented matrix

$$[A \;\; \mathbf{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

to an upper triangular form

$$\left[\begin{array}{cccc|c} c_{11} & c_{12} & \cdots & c_{1n} & d_1 \\ 0 & c_{22} & \cdots & c_{2n} & d_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & c_{mn} & d_m \end{array}\right].$$

This elimination process is also called the forward elimination method.

The following examples illustrate the Gauss elimination procedure.

Example 2.3.10 Solve the linear system by Gauss elimination method.

$$\begin{aligned} y + z &= 2 \\ 2x + 3z &= 5 \\ x + y + z &= 3 \end{aligned}$$

Solution: In this case, the augmented matrix is $\begin{bmatrix} 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 1 & 1 & 1 & 3 \end{bmatrix}$. The method proceeds along the following steps.

1. Interchange 1st and 2nd equation (or R_{12}).

$$\begin{aligned} 2x + 3z &= 5 \\ y + z &= 2 \\ x + y + z &= 3 \end{aligned} \qquad \begin{bmatrix} 2 & 0 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \end{bmatrix}.$$

2. Divide the 1st equation by 2 (or $R_1(1/2)$).

$$\begin{aligned} x + \tfrac{3}{2}z &= \tfrac{5}{2} \\ y + z &= 2 \\ x + y + z &= 3 \end{aligned} \qquad \begin{bmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \end{bmatrix}.$$

3. Add -1 times the 1st equation to the 3rd equation (or $R_{31}(-1)$).

$$\begin{aligned} x + \tfrac{3}{2}z &= \tfrac{5}{2} \\ y + z &= 2 \\ y - \tfrac{1}{2}z &= \tfrac{1}{2} \end{aligned} \qquad \begin{bmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

4. Add -1 times the 2nd equation to the 3rd equation (or $R_{32}(-1)$).

$$\begin{aligned} x + \tfrac{3}{2}z &= \tfrac{5}{2} \\ y + z &= 2 \\ -\tfrac{3}{2}z &= -\tfrac{3}{2} \end{aligned} \qquad \begin{bmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -\frac{3}{2} & -\frac{3}{2} \end{bmatrix}.$$

5. Multiply the 3rd equation by $\frac{-2}{3}$ (or $R_3(-\frac{2}{3})$).

$$\begin{aligned} x + \tfrac{3}{2}z &= \tfrac{5}{2} \\ y + z &= 2 \\ z &= 1 \end{aligned} \qquad \begin{bmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

The last equation gives $z = 1$, the second equation now gives $y = 1$. Finally the first equation gives $x = 1$. Hence the set of solutions is $(x, y, z)^t = (1, 1, 1)^t$, A UNIQUE SOLUTION.

Example 2.3.11 Solve the linear system by Gauss elimination method.

$$\begin{aligned} x + y + z &= 3 \\ x + 2y + 2z &= 5 \\ 3x + 4y + 4z &= 11 \end{aligned}$$

Solution: In this case, the augmented matrix is $\begin{bmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & 2 & 5 \\ 3 & 4 & 4 & 11 \end{bmatrix}$ and the method proceeds as follows:

1. Add -1 times the first equation to the second equation.

$$\begin{aligned} x + y + z &= 3 \\ y + z &= 2 \\ 3x + 4y + 4z &= 11 \end{aligned} \qquad \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 3 & 4 & 4 & 11 \end{bmatrix}.$$

2. Add -3 times the first equation to the third equation.

$$\begin{aligned} x + y + z &= 3 \\ y + z &= 2 \\ y + z &= 2 \end{aligned} \qquad \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{bmatrix}.$$

3. Add -1 times the second equation to the third equation

$$\begin{aligned} x + y + z &= 3 \\ y + z &= 2 \end{aligned} \qquad \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Thus, the set of solutions is $(x, y, z)^t = (1, 2 - z, z)^t = (1, 2, 0)^t + z(0, -1, 1)^t$, with z arbitrary. In other words, the system has INFINITE NUMBER OF SOLUTIONS.

Example 2.3.12 Solve the linear system by Gauss elimination method.

$$\begin{aligned} x + y + z &= 3 \\ x + 2y + 2z &= 5 \\ 3x + 4y + 4z &= 12 \end{aligned}$$

Solution: In this case, the augmented matrix is $\begin{bmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & 2 & 5 \\ 3 & 4 & 4 & 12 \end{bmatrix}$ and the method proceeds as follows:

1. Add -1 times the first equation to the second equation.

$$\begin{aligned} x + y + z &= 3 \\ y + z &= 2 \\ 3x + 4y + 4z &= 12 \end{aligned} \qquad \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 3 & 4 & 4 & 12 \end{bmatrix}.$$

2. Add -3 times the first equation to the third equation.

$$\begin{aligned} x + y + z &= 3 \\ y + z &= 2 \\ y + z &= 3 \end{aligned} \qquad \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 3 \end{bmatrix}.$$

3. Add -1 times the second equation to the third equation

$$\begin{aligned} x + y + z &= 3 \\ y + z &= 2 \\ 0 &= 1 \end{aligned} \qquad \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The third equation in the last step is

$$0x + 0y + 0z = 1.$$

This can never hold for any value of x, y, z. Hence, the system has NO SOLUTION.

Remark 2.3.13 *Note that to solve a linear system, $A\mathbf{x} = \mathbf{b}$, one needs to apply only the elementary row operations to the augmented matrix $[A \;\; \mathbf{b}]$.*

2.4 Row Reduced Echelon Form of a Matrix

Definition 2.4.1 (Row Reduced Form of a Matrix) A matrix C is said to be in the row reduced form if

1. THE FIRST NON-ZERO ENTRY IN EACH ROW OF C IS 1;
2. THE COLUMN CONTAINING THIS 1 HAS ALL ITS OTHER ENTRIES ZERO.

A matrix in the row reduced form is also called a ROW REDUCED MATRIX.

Example 2.4.2

1. One of the most important examples of a row reduced matrix is the $n \times n$ identity matrix, I_n. Recall that the $(i, j)^{\text{th}}$ entry of the identity matrix is

$$I_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}.$$

δ_{ij} is usually referred to as the Kronecker delta function.

2. The matrices $\begin{bmatrix} 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ are also in row reduced form.

3. The matrix $\begin{bmatrix} 1 & 0 & 0 & 0 & 5 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ is not in the row reduced form. (why?)

Definition 2.4.3 (Leading Term, Leading Column) For a row-reduced matrix, the first non-zero entry of any row is called a LEADING TERM. The columns containing the leading terms are called the LEADING COLUMNS.

Definition 2.4.4 (Basic, Free Variables) Consider the linear system $A\mathbf{x} = \mathbf{b}$ in n variables and m equations. Let $[C \;\; \mathbf{d}]$ be the row-reduced matrix obtained by applying the Gauss elimination method to the augmented matrix $[A \;\; \mathbf{b}]$. Then the variables corresponding to the leading columns in the first n columns of $[C \;\; \mathbf{d}]$ are called the BASIC variables. The variables which are not basic are called FREE variables.

The free variables are called so as they can be assigned arbitrary values and the value of the basic variables can then be written in terms of the free variables.

Observation: In Example 2.3.11, the solution set was given by

$$(x, y, z)^t = (1, 2 - z, z)^t = (1, 2, 0)^t + z(0, -1, 1)^t, \;\text{ with } z \text{ arbitrary.}$$

That is, we had two basic variables, x and y, and z as a free variable.

Remark 2.4.5 *It is very important to observe that if there are r non-zero rows in the row-reduced form of the matrix then there will be r leading terms. That is, there will be r leading columns. Therefore,* IF THERE ARE r LEADING TERMS AND n VARIABLES, THEN THERE WILL BE r BASIC VARIABLES AND $n - r$ FREE VARIABLES.

2.4.1 Gauss-Jordan Elimination

We now start with Step 5 of Example 2.3.10 and apply the elementary operations once again. But this time, we start with the 3rd row.

I. Add -1 times the third equation to the second equation (or $R_{23}(-1)$).

$$\begin{array}{rl} x + \frac{3}{2}z & = \frac{5}{2} \\ y & = 2 \\ z & = 1 \end{array} \qquad \begin{bmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

II. Add $\frac{-3}{2}$ times the third equation to the first equation (or $R_{13}(-\frac{3}{2})$).

$$\begin{array}{rl} x & = 1 \\ y & = 1 \\ z & = 1 \end{array} \qquad \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

III. From the above matrix, we directly have the set of solution as $(x, y, z)^t = (1, 1, 1)^t$.

Definition 2.4.6 (Row Reduced Echelon Form of a Matrix) A matrix C is said to be in the row reduced echelon form if

1. C is already in the row reduced form;
2. The rows consisting of all zeros comes below all non-zero rows; and
3. the leading terms appear from left to right in successive rows. That is, for $1 \le \ell \le k$, let i_ℓ be the leading column of the ℓth row. Then $i_1 < i_2 < \cdots < i_k$.

Example 2.4.7 Suppose $A = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ are in row reduced form. Then the corresponding matrices in the row reduced echelon form are respectively, $\begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$.

Definition 2.4.8 (Row Reduced Echelon Matrix) A matrix which is in the row reduced echelon form is also called a row reduced echelon matrix.

Definition 2.4.9 (Back Substitution/Gauss-Jordan Method) The procedure to get to Step II of Example 2.3.10 from Step 5 of Example 2.3.10 is called the back substitution.

The elimination process applied to obtain the row reduced echelon form of the augmented matrix is called the Gauss-Jordan elimination.

That is, the Gauss-Jordan elimination method consists of both the forward elimination and the backward substitution.

Method to get the row-reduced echelon form of a given matrix A

Let A be an $m \times n$ matrix. Then the following method is used to obtain the row-reduced echelon form the matrix A.

Step 1: Consider the first column of the matrix A.

If all the entries in the first column are zero, move to the second column.

Else, find a row, say i^{th} row, which contains a non-zero entry in the first column. Now, interchange the first row with the ith row. Suppose the non-zero entry in the $(1,1)$-position is $\alpha \neq 0$. Divide the whole row by α so that the $(1,1)$-entry of the new matrix is 1. Now, use the 1 to make all the entries below this 1 equal to 0.

Step 2: If all entries in the first column after the first step are zero, consider the right $m \times (n-1)$ submatrix of the matrix obtained in step 1 and proceed as in step 1.

Else, forget the first row and first column. Start with the lower $(m-1) \times (n-1)$ submatrix of the matrix obtained in the first step and proceed as in step 1.

Step 3: Keep repeating this process till we reach a stage where all the entries below a particular row, say r, are zero. Suppose at this stage we have obtained a matrix C. Then C has the following form:

1. THE FIRST NON-ZERO ENTRY IN EACH ROW of C is 1. These 1's are the leading terms of C and the columns containing these leading terms are the leading columns.
2. THE ENTRIES OF C BELOW THE LEADING TERM ARE ALL ZERO.

Step 4: Now use the leading term in the rth row to make all entries in the rth leading column equal to zero.

Step 5: Next, use the leading term in the $(r-1)^{\text{th}}$ row to make all entries in the $(r-1)^{\text{th}}$ leading column equal to zero and continue till we come to the first leading term or column.

The final matrix is the row-reduced echelon form of the matrix A.

Remark 2.4.10 *Note that the row reduction involves only row operations and proceeds from* LEFT TO RIGHT. *Hence, if A is a matrix consisting of first s columns of a matrix C, then the row reduced form of A will be the first s columns of the row reduced form of C.*

The proof of the following theorem is beyond the scope of this book and is omitted.

Theorem 2.4.11 The row reduced echelon form of a matrix is unique.

Exercise 2.4.12 1. Solve the following linear system.

(a) $x + y + z + w = 0$, $x - y + z + w = 0$ and $-x + y + 3z + 3w = 0$.
(b) $x + 2y + 3z = 1$ and $x + 3y + 2z = 1$.
(c) $x + y + z = 3$, $x + y - z = 1$ and $x + y + 7z = 6$.
(d) $x + y + z = 3$, $x + y - z = 1$ and $x + y + 4z = 6$.
(e) $x + y + z = 3$, $x + y - z = 1$, $x + y + 4z = 6$ and $x + y - 4z = -1$.

2. Find the row-reduced echelon form of the following matrices.

$$1.\ \begin{bmatrix} -1 & 1 & 3 & 5 \\ 1 & 3 & 5 & 7 \\ 9 & 11 & 13 & 15 \\ -3 & -1 & 13 & \end{bmatrix}, \qquad 2.\ \begin{bmatrix} 10 & 8 & 6 & 4 \\ 2 & 0 & -2 & -4 \\ -6 & -8 & -10 & -12 \\ -2 & -4 & -6 & -8 \end{bmatrix}$$

2.4.2 Elementary Matrices

Definition 2.4.13 A square matrix E of order n is called an **elementary matrix** if it is obtained by applying exactly one elementary row operation to the identity matrix, I_n.

Remark 2.4.14 *There are three types of elementary matrices.*

1. *E_{ij}, which is obtained by the application of the elementary row operation R_{ij} to the identity matrix, I_n. Thus, the $(k,\ell)^{th}$ entry of E_{ij} is $(E_{ij})_{(k,\ell)} = \begin{cases} 1 & \text{if } k = \ell \text{ and } \ell \neq i, j \\ 1 & \text{if } (k,\ell) = (i,j) \text{ or } (k,\ell) = (j,i) \\ 0 & \text{otherwise} \end{cases}$.*
2. *$E_k(c)$, which is obtained by the application of the elementary row operation $R_k(c)$ to the identity matrix, I_n. The $(i,j)^{th}$ entry of $E_k(c)$ is $(E_k(c))_{(i,j)} = \begin{cases} 1 & \text{if } i = j \text{ and } i \neq k \\ c & \text{if } i = j = k \\ 0 & \text{otherwise} \end{cases}$.*
3. *$E_{ij}(c)$, which is obtained by the application of the elementary row operation $R_{ij}(c)$ to the identity matrix, I_n. The $(k,\ell)^{th}$ entry of $E_{ij}(c)$ is $(E_{ij})_{(k,\ell)} \begin{cases} 1 & \text{if } k = \ell \\ c & \text{if } (k,\ell) = (i,j) \\ 0 & \text{otherwise} \end{cases}$.*

In particular,

$$E_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix},\ E_1(c) = \begin{bmatrix} c & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},\ \text{and } E_{23}(c) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}.$$

Example 2.4.15 1. Let $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 0 & 3 & 4 \\ 3 & 4 & 5 & 6 \end{bmatrix}$. Then

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 0 & 3 & 4 \\ 3 & 4 & 5 & 6 \end{bmatrix} \xrightarrow{R_{23}} \begin{bmatrix} 1 & 2 & 3 & 0 \\ 3 & 4 & 5 & 6 \\ 2 & 0 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} A = E_{23}A.$$

That is, interchanging the two rows of the matrix A is same as multiplying on the left by the corresponding elementary matrix. In other words, we see that the left multiplication of elementary matrices to a matrix results in elementary row operations.

2. Consider the augmented matrix $[A \;\; \mathbf{b}] = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 1 & 1 & 1 & 3 \end{bmatrix}$. Then the result of the steps given below is same as the matrix product

$$E_{23}(-1)E_{12}(-1)E_3(1/3)E_{32}(2)E_{23}E_{21}(-2)E_{13}[A \;\; \mathbf{b}].$$

$$\begin{bmatrix} 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 1 & 1 & 1 & 3 \end{bmatrix} \xrightarrow{R_{13}} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \\ 0 & 1 & 1 & 2 \end{bmatrix} \xrightarrow{R_{21}(-2)} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -2 & 1 & -1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \xrightarrow{R_{23}} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & -2 & 1 & -1 \end{bmatrix}$$

$$\xrightarrow{R_{32}(2)} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 3 & 3 \end{bmatrix} \xrightarrow{R_3(1/3)} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_{12}(-1)} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow{R_{23}(-1)} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Now, consider an $m \times n$ matrix A and an elementary matrix E of order n. Then multiplying by E on the right to A corresponds to applying column transformation on the matrix A. Therefore, for each elementary matrix, there is a corresponding column transformation. We summarize:

Definition 2.4.16 The column transformations obtained by right multiplication of elementary matrices are called **elementary column operations**.

Example 2.4.17 Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \\ 3 & 4 & 5 \end{bmatrix}$ and consider the elementary column operation f which interchanges the second and the third column of A. Then $f(A) = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 0 \\ 3 & 5 & 4 \end{bmatrix} = A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = AE_{23}$.

Exercise 2.4.18 1. Let e be an elementary row operation and let $E = e(I)$ be the corresponding elementary matrix. That is, E is the matrix obtained from I by applying the elementary row operation e. Show that $e(A) = EA$.

2. Show that the Gauss elimination method is same as multiplying by a series of elementary matrices on the left to the augmented matrix.

   Does the Gauss-Jordan method also corresponds to multiplying by elementary matrices on the left? Give reasons.

3. Let A and B be two $m \times n$ matrices. Then prove that the two matrices A, B are row-equivalent if and only if $B = PA$, where P is product of elementary matrices. When is this P unique?

2.5 Rank of a Matrix

In previous sections, we solved linear systems using Gauss elimination method or the Gauss-Jordan method. In the examples considered, we have encountered three possibilities, namely

1. existence of a unique solution,
2. existence of an infinite number of solutions, and

3. no solution.

Based on the above possibilities, we have the following definition.

Definition 2.5.1 (Consistent, Inconsistent) A linear system is called CONSISTENT if it admits a solution and is called INCONSISTENT if it admits no solution.

The question arises, as to whether there are conditions under which the linear system $A\mathbf{x} = \mathbf{b}$ is consistent. The answer to this question is in the affirmative. To proceed further, we need a few definitions and remarks.

Recall that the row reduced echelon form of a matrix is unique and therefore, the number of non-zero rows is a unique number. Also, note that the number of non-zero rows in either the row reduced form or the row reduced echelon form of a matrix are same.

Definition 2.5.2 (Row rank of a Matrix) The number of non-zero rows in the row reduced form of a matrix is called the row-rank of the matrix.

By the very definition, it is clear that row-equivalent matrices have the same row-rank. For a matrix A, we write 'row-rank (A)' to denote the row-rank of A.

Example 2.5.3 1. Determine the row-rank of $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$.

Solution: To determine the row-rank of A, we proceed as follows.

(a) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix} \xrightarrow{R_{21}(-2), R_{31}(-1)} \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$.

(b) $\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{R_2(-1), R_{32}(1)} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$.

(c) $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \xrightarrow{R_3(1/2), R_{12}(-2)} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

(d) $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_{23}(-1), R_{13}(1)} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

The last matrix in Step 1d is the row reduced form of A which has 3 non-zero rows. Thus, row-rank$(A) = 3$. This result can also be easily deduced from the last matrix in Step 1b.

2. Determine the row-rank of $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.

Solution: Here we have

(a) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 0 \end{bmatrix} \xrightarrow{R_{21}(-2), R_{31}(-1)} \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{bmatrix}$.

(b) $\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{bmatrix} \xrightarrow{R_2(-1), R_{32}(1)} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$.

From the last matrix in Step 2b, we deduce row-rank$(A) = 2$.

Remark 2.5.4 *Let $A\mathbf{x} = \mathbf{b}$ be a linear system with m equations and n unknowns. Then the row-reduced echelon form of A agrees with the first n columns of $[A \ \ \mathbf{b}]$, and hence*

$$\textit{row-rank}(A) \leq \textit{row-rank}([A \ \ \mathbf{b}]).$$

The reader is advised to supply a proof.

Remark 2.5.5 *Consider a matrix A. After application of a finite number of elementary column operations (see Definition 2.4.16) to the matrix A, we can have a matrix, say B, which has the following properties:*

1. *The first nonzero entry in each column is* 1.
2. *A column containing only* 0*'s comes after all columns with at least one non-zero entry.*
3. *The first non-zero entry (the leading term) in each non-zero column moves down in successive columns.*

Therefore, we can define **column-rank** of A as the number of non-zero columns in B. It will be proved later that

$$\text{row-rank}(A) = \text{column-rank}(A).$$

Thus we are led to the following definition.

Definition 2.5.6 The number of non-zero rows in the row reduced form of a matrix A is called the **rank** of A, denoted rank (A).

Theorem 2.5.7 Let A be a matrix of rank r. Then there exist elementary matrices $E_1, E_2, \ldots, E_s$ and $F_1, F_2, \ldots, F_\ell$ such that

$$E_1 E_2 \ldots E_s \, A \, F_1 F_2 \ldots F_\ell = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}.$$

PROOF. Let C be the row reduced echelon matrix obtained by applying elementary row operations to the given matrix A. As rank$(A) = r$, the matrix C will have the first r rows as the non-zero rows. So by Remark 2.4.5, C will have r leading columns, say $i_1, i_2, \ldots, i_r$. Note that, for $1 \leq s \leq r$, the i_s^{th} column will have 1 in the sth row and zero elsewhere.

We now apply column operations to the matrix C. Let D be the matrix obtained from C by successively interchanging the sth and i_s^{th} column of C for $1 \leq s \leq r$. Then the matrix D can be written in the form $\begin{bmatrix} I_r & B \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$, where B is a matrix of appropriate size. As the $(1,1)$ block of D is an identity matrix, the block $(1,2)$ can be made the zero matrix by application of column operations to D. This gives the required result. □

Exercise 2.5.8 1. Determine the ranks of the coefficient and the augmented matrices that appear in Part 1 and Part 2 of Exercise 2.4.12.

2. For any matrix A, prove that rank(A) = rank(A^t).

3. Let A be an $n \times n$ matrix with rank$(A) = n$. Then prove that A is row-equivalent to I_n.

2.6 Existence of Solution of $A\mathbf{x} = \mathbf{b}$

We try to understand the properties of the set of solutions of a linear system through an example, using the Gauss-Jordan method. Based on this observation, we arrive at the existence and uniqueness results for the linear system $A\mathbf{x} = \mathbf{b}$. This example is more or less a motivation.

2.6.1 Example

Consider a linear system $A\mathbf{x} = \mathbf{b}$ which after the application of the Gauss-Jordan method reduces to a matrix $[C \ \ \mathbf{d}]$ with

$$[C \ \ \mathbf{d}] = \begin{bmatrix} 1 & 0 & 2 & -1 & 0 & 0 & 2 & 8 \\ 0 & 1 & 1 & 3 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

For this particular matrix $[C \ \ \mathbf{d}]$, we want to see the set of solutions. We start with some observations.
Observations:

1. The number of non-zero rows in C is 4. This number is also equal to the number of non-zero rows in $[C \ \ \mathbf{d}]$.
2. The first non-zero entry in the non-zero rows appear in columns $1, 2, 5$ and 6.
3. Thus, the respective variables x_1, x_2, x_5 and x_6 are the basic variables.
4. The remaining variables, x_3, x_4 and x_7 are free variables.
5. We assign arbitrary constants k_1, k_2 and k_3 to the free variables x_3, x_4 and x_7, respectively.

Hence, we have the set of solutions as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{bmatrix} = \begin{bmatrix} 8 - 2k_1 + k_2 - 2k_3 \\ 1 - k_1 - 3k_2 - 5k_3 \\ k_1 \\ k_2 \\ 2 + k_3 \\ 4 - k_3 \\ k_3 \end{bmatrix}$$

$$= \begin{bmatrix} 8 \\ 1 \\ 0 \\ 0 \\ 2 \\ 4 \\ 0 \end{bmatrix} + k_1 \begin{bmatrix} -2 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 1 \\ -3 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + k_3 \begin{bmatrix} -2 \\ -5 \\ 0 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix},$$

where k_1, k_2 and k_3 are arbitrary.

Let $\mathbf{u}_0 = \begin{bmatrix} 8 \\ 1 \\ 0 \\ 0 \\ 2 \\ 4 \\ 0 \end{bmatrix}$, $\mathbf{u}_1 = \begin{bmatrix} -2 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} 1 \\ -3 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ and $\mathbf{u}_3 = \begin{bmatrix} -2 \\ -5 \\ 0 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$.

Then it can easily be verified that $C\mathbf{u}_0 = \mathbf{d}$, and for $1 \leq i \leq 3$, $C\mathbf{u}_i = \mathbf{0}$.

A similar idea is used in the proof of the next theorem and is omitted. The interested readers can read the proof in Appendix 14.1.

2.6.2 Main Theorem

Theorem 2.6.1 [Existence and Non-existence] Consider a linear system $A\mathbf{x} = \mathbf{b}$, where A is a $m \times n$ matrix, and $\mathbf{x}$, $\mathbf{b}$ are vectors with orders $n \times 1$, and $m \times 1$, respectively. Suppose rank $(A) = r$ and rank$([A \ \ \mathbf{b}]) = r_a$. Then exactly one of the following statement holds:

1. if $r_a = r < n$, the set of solutions of the linear system is an infinite set and has the form

$$\{\mathbf{u}_0 + k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + \cdots + k_{n-r}\mathbf{u}_{n-r} \ : \ k_i \in \mathbb{R}, \ 1 \leq i \leq n-r\},$$

where $\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{n-r}$ are $n \times 1$ vectors satisfying $A\mathbf{u}_0 = \mathbf{b}$ and $A\mathbf{u}_i = \mathbf{0}$ for $1 \leq i \leq n-r$.

2. if $r_a = r = n$, the solution set of the linear system has a unique $n \times 1$ vector $\mathbf{x}_0$ satisfying $A\mathbf{x}_0 = \mathbf{b}$.
3. If $r < r_a$, the linear system has no solution.

Remark 2.6.2 *Let A be an $m \times n$ matrix and consider the linear system $A\mathbf{x} = \mathbf{b}$. Then by Theorem 2.6.1, we see that the linear system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if*

$$\textit{rank } (A) = \textit{rank}([A \ \ \mathbf{b}]).$$

The following corollary of Theorem 2.6.1 is a very important result about the homogeneous linear system $A\mathbf{x} = \mathbf{0}$.

Corollary 2.6.3 Let A be an $m \times n$ matrix. Then the homogeneous system $A\mathbf{x} = \mathbf{0}$ has a non-trivial solution if and only if rank$(A) < n$.

PROOF. Suppose the system $A\mathbf{x} = \mathbf{0}$ has a non-trivial solution, $\mathbf{x}_0$. That is, $A\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{x}_0 \neq \mathbf{0}$. Under this assumption, we need to show that rank$(A) < n$. On the contrary, assume that rank$(A) = n$. So,

$$n = \text{rank}(A) = \text{rank}([A \ \ \mathbf{0}]) = r_a.$$

Also $A\mathbf{0} = \mathbf{0}$ implies that $\mathbf{0}$ is a solution of the linear system $A\mathbf{x} = \mathbf{0}$. Hence, by the uniqueness of the solution under the condition $r = r_a = n$ (see Theorem 2.6.1), we get $\mathbf{x}_0 = \mathbf{0}$. A contradiction to the fact that $\mathbf{x}_0$ was a given non-trivial solution.

Now, let us assume that rank$(A) < n$. Then

$$r_a = \text{rank}([A \ \ \mathbf{0}]) = \text{rank}(A) < n.$$

So, by Theorem 2.6.1, the solution set of the linear system $A\mathbf{x} = \mathbf{0}$ has infinite number of vectors $\mathbf{x}$ satisfying $A\mathbf{x} = \mathbf{0}$. From this infinite set, we can choose any vector $\mathbf{x}_0$ that is different from $\mathbf{0}$. Thus, we have a solution $\mathbf{x}_0 \neq \mathbf{0}$. That is, we have obtained a non-trivial solution $\mathbf{x}_0$. □

We now state another important result whose proof is immediate from Theorem 2.6.1 and Corollary 2.6.3.

Proposition 2.6.4 Consider the linear system $A\mathbf{x} = \mathbf{b}$. Then the two statements given below cannot hold together.

1. The system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$.
2. The system $A\mathbf{x} = \mathbf{0}$ has a non-trivial solution.

Remark 2.6.5 *1. Suppose $\mathbf{x}_1, \mathbf{x}_2$ are two solutions of $A\mathbf{x} = \mathbf{0}$. Then $k_1\mathbf{x}_1 + k_2\mathbf{x}_2$ is also a solution of $A\mathbf{x} = \mathbf{0}$ for any $k_1, k_2 \in \mathbb{R}$.*

2. If $\mathbf{u}, \mathbf{v}$ are two solutions of $A\mathbf{x} = \mathbf{b}$ then $\mathbf{u} - \mathbf{v}$ is a solution of the system $A\mathbf{x} = \mathbf{0}$. That is, $\mathbf{u} - \mathbf{v} = \mathbf{x}_h$ for some solution $\mathbf{x}_h$ of $A\mathbf{x} = \mathbf{0}$. That is, any two solutions of $A\mathbf{x} = \mathbf{b}$ differ by a solution of the associated homogeneous system $A\mathbf{x} = \mathbf{0}$.

In conclusion, for $\mathbf{b} \neq \mathbf{0}$, the set of solutions of the system $A\mathbf{x} = \mathbf{b}$ is of the form, $\{\mathbf{x}_0 + \mathbf{x}_h\}$; where $\mathbf{x}_0$ is a particular solution of $A\mathbf{x} = \mathbf{b}$ and $\mathbf{x}_h$ is a solution $A\mathbf{x} = \mathbf{0}$.

2.6.3 Exercises

Exercise 2.6.6 1. For what values of c and k-the following systems have i) no solution, ii) a unique solution and iii) infinite number of solutions.

(a) $x + y + z = 3,\ x + 2y + cz = 4,\ 2x + 3y + 2cz = k.$

(b) $x + y + z = 3,\ x + y + 2cz = 7,\ x + 2y + 3cz = k.$

(c) $x + y + 2z = 3,\ x + 2y + cz = 5,\ x + 2y + 4z = k.$

(d) $kx + y + z = 1,\ x + ky + z = 1,\ x + y + kz = 1.$

(e) $x + 2y - z = 1,\ 2x + 3y + kz = 3,\ x + ky + 3z = 2.$

(f) $x - 2y = 1,\ x - y + kz = 1,\ ky + 4z = 6.$

2. Find the condition on a, b, c so that the linear system

$$x + 2y - 3z = a,\ 2x + 6y - 11z = b,\ x - 2y + 7z = c$$

is consistent.

3. Let A be an $n \times n$ matrix. If the system $A^2\mathbf{x} = \mathbf{0}$ has a non trivial solution then show that $A\mathbf{x} = \mathbf{0}$ also has a non trivial solution.

2.7 Invertible Matrices

2.7.1 Inverse of a Matrix

Definition 2.7.1 (Inverse of a Matrix) Let A be a square matrix of order n.

1. A square matrix B is said to be a LEFT INVERSE of A if $BA = I_n$.
2. A square matrix C is called a RIGHT INVERSE of A, if $AC = I_n$.
3. A matrix A is said to be INVERTIBLE (or is said to have an INVERSE) if there exists a matrix B such that $AB = BA = I_n$.

Lemma 2.7.2 Let A be an $n \times n$ matrix. Suppose that there exist $n \times n$ matrices B and C such that $AB = I_n$ and $CA = I_n$, then $B = C$.

PROOF. Note that

$$C = CI_n = C(AB) = (CA)B = I_nB = B.$$

□

Remark 2.7.3 *1. From the above lemma, we observe that if a matrix A is invertible, then the inverse is unique.*

2. As the inverse of a matrix A is unique, we denote it by A^{-1}. That is, $AA^{-1} = A^{-1}A = I$.

Theorem 2.7.4 Let A and B be two matrices with inverses A^{-1} and B^{-1}, respectively. Then

1. $(A^{-1})^{-1} = A$.
2. $(AB)^{-1} = B^{-1}A^{-1}$.
3. prove that $(A^t)^{-1} = (A^{-1})^t$.

PROOF. Proof of Part 1.
By definition $AA^{-1} = A^{-1}A = I$. Hence, if we denote A^{-1} by B, then we get $AB = BA = I$. This again by definition, implies $B^{-1} = A$, or equivalently $(A^{-1})^{-1} = A$.

Proof of Part 2.
Verify that $(AB)(B^{-1}A^{-1}) = I = (B^{-1}A^{-1})(AB)$. Hence, the result follows by definition.

Proof of Part 3.
We know $AA^{-1} = A^{-1}A = I$. Taking transpose, we get

$$(AA^{-1})^t = (A^{-1}A)^t = I^t \iff (A^{-1})^tA^t = A^t(A^{-1})^t = I.$$

Hence, by definition $(A^t)^{-1} = (A^{-1})^t$. □

Exercise 2.7.5 1. If A is a symmetric matrix, is the matrix A^{-1} symmetric?

2. Show that every elementary matrix is invertible. Is the inverse of an elementary matrix, also an elementary matrix?
3. Let $A_1, A_2, \ldots, A_r$ be invertible matrices. Prove that the product $A_1A_2\cdots A_r$ is also an invertible matrix.
4. If P and Q are invertible matrices and PAQ is defined then show that rank $(PAQ) =$ rank (A).
5. Find matrices P and Q which are product of elementary matrices such that $B = PAQ$ where $A = \begin{bmatrix} 2 & 4 & 8 \\ 1 & 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.
6. Let A and B be two matrices. Show that
   (a) if $A + B$ is defined, then rank$(A + B) \leq$ rank$(A) +$ rank(B),
   (b) if AB is defined, then rank$(AB) \leq$ rank(A) and rank$(AB) \leq$ rank(B).
7. Let A be any matrix of rank r. Then show that there exists invertible matrices B_i, C_i such that $B_1A = \begin{bmatrix} R_1 & R_2 \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$, $AC_1 = \begin{bmatrix} S_1 & \mathbf{0} \\ S_3 & \mathbf{0} \end{bmatrix}$, $B_2AC_2 = \begin{bmatrix} A_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$, and $B_3AC_3 = \begin{bmatrix} I_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$. Also, prove that the matrix A_1 is an $r \times r$ invertible matrix.

8. Let A be an $m \times n$ matrix of rank r. Then A can be written as $A = BC$, where both B and C have rank r and B is a matrix of size $m \times r$ and C is a matrix of size $r \times n$.
9. Let A and B be two matrices such that AB is defined and rank $(A) =$ rank (AB). Then show that $A = ABX$ for some matrix X. Similarly, if BA is defined and rank $(A) =$ rank (BA), then $A = YBA$ for some matrix Y. *[Hint: Choose non-singular matrices P, Q and R such that $PAQ = \begin{bmatrix} A_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$ and $P(AB)R = \begin{bmatrix} C & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$. Define $X = R\begin{bmatrix} C^{-1}A_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}Q^{-1}$.]*
10. Let $A = [a_{ij}]$ be an invertible matrix and let $B = [p^{i-j}a_{ij}]$ for some nonzero real number p. Find the inverse of B.
11. If matrices B and C are invertible and the involved partitioned products are defined, then show that

$$\begin{bmatrix} A & B \\ C & \mathbf{0} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{0} & C^{-1} \\ B^{-1} & -B^{-1}AC^{-1} \end{bmatrix}.$$

12. Suppose A is the inverse of a matrix B. Partition A and B as follows:

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix},\ B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}.$$

If A_{11} is invertible and $P = A_{22} - A_{21}(A_{11}^{-1}A_{12})$, then show that

$$B_{11} = A_{11}^{-1} + (A_{11}^{-1}A_{12})P^{-1}(A_{21}A_{11}^{-1}),\ B_{21} = -P^{-1}(A_{21}A_{11}^{-1}),\ B_{12} = -(A_{11}^{-1}A_{12})P^{-1},$$

and $B_{22} = P^{-1}$.

2.7.2 Equivalent conditions for Invertibility

Definition 2.7.6 A square matrix A or order n is said to be of **full rank** if rank $(A) = n$.

Theorem 2.7.7 For a square matrix A of order n, the following statements are equivalent.

1. A is invertible.
2. A is of full rank.
3. A is row-equivalent to the identity matrix.
4. A is a product of elementary matrices.

PROOF. $1 \implies 2$

Let if possible rank$(A) = r < n$. Then there exists an invertible matrix P (a product of elementary matrices) such that $PA = \begin{bmatrix} B_1 & B_2 \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$, where B_1 is an $r \times r$ matrix. Since A is invertible, let $A^{-1} = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}$, where C_1 is an $r \times n$ matrix. Then

$$P = PI_n = P(AA^{-1}) = (PA)A^{-1} = \begin{bmatrix} B_1 & B_2 \\ \mathbf{0} & \mathbf{0} \end{bmatrix}\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} B_1C_1 + B_2C_2 \\ \mathbf{0} \end{bmatrix}. \tag{2.7.1}$$

Thus the matrix P has $n - r$ rows as zero rows. Hence, P cannot be invertible. A contradiction to P being a product of invertible matrices. Thus, A is of full rank.

$2 \implies 3$

Suppose A is of full rank. This implies, the row reduced echelon form of A has all non-zero rows. But A has as many columns as rows and therefore, the last row of the row reduced echelon form of A will be $(0, 0, \ldots, 0, 1)$. Hence, the row reduced echelon form of A is the identity matrix.

$3 \implies 4$

Since A is row-equivalent to the identity matrix there exist elementary matrices $E_1, E_2, \ldots, E_k$ such that $A = E_1E_2\cdots E_kI_n$. That is, A is product of elementary matrices.

$4 \implies 1$

Suppose $A = E_1E_2\cdots E_k$; where the E_i's are elementary matrices. We know that elementary matrices are invertible and product of invertible matrices is also invertible, we get the required result. □

The ideas of Theorem 2.7.7 will be used in the next subsection to find the inverse of an invertible matrix. The idea used in the proof of the first part also gives the following important Theorem. We repeat the proof for the sake of clarity.

Theorem 2.7.8 Let A be a square matrix of order n.

1. Suppose there exists a matrix B such that $AB = I_n$. Then A^{-1} exists.
2. Suppose there exists a matrix C such that $CA = I_n$. Then A^{-1} exists.

PROOF. Suppose that $AB = I_n$. We will prove that the matrix A is of full rank. That is, rank $(A) = n$.

Let if possible, rank$(A) = r < n$. Then there exists an invertible matrix P (a product of elementary matrices) such that $PA = \begin{bmatrix} C_1 & C_2 \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$. Let $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$, where B_1 is an $r \times n$ matrix. Then

$$P = PI_n = P(AB) = (PA)B = \begin{bmatrix} C_1 & C_2 \\ \mathbf{0} & \mathbf{0} \end{bmatrix}\begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} C_1B_1 + C_2B_2 \\ \mathbf{0} \end{bmatrix}. \tag{2.7.2}$$

Thus the matrix P has $n - r$ rows as zero rows. So, P cannot be invertible. A contradiction to P being a product of invertible matrices. Thus, rank $(A) = n$. That is, A is of full rank. Hence, using Theorem 2.7.7, A is an invertible matrix. That is, $BA = I_n$ as well.

Using the first part, it is clear that the matrix C in the second part, is invertible. Hence

$$AC = I_n = CA.$$

Thus, A is invertible as well. □

Remark 2.7.9 *This theorem implies the following: "if we want to show that a square matrix A of order n is invertible, it is enough to show the existence of*

1. either a matrix B such that $AB = I_n$

2. or a matrix C such that $CA = I_n$.

Theorem 2.7.10 The following statements are equivalent for a square matrix A of order n.

1. A is invertible.
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.
3. $A\mathbf{x} = \mathbf{b}$ has a solution $\mathbf{x}$ for every $\mathbf{b}$.

PROOF. $1 \implies 2$

Since A is invertible, by Theorem 2.7.7 A is of full rank. That is, for the linear system $A\mathbf{x} = \mathbf{0}$, the number of unknowns is equal to the rank of the matrix A. Hence, by Theorem 2.6.1 the system $A\mathbf{x} = \mathbf{0}$ has a unique solution $\mathbf{x} = \mathbf{0}$.

$2 \implies 1$

Let if possible A be non-invertible. Then by Theorem 2.7.7, the matrix A is not of full rank. Thus by Corollary 2.6.3, the linear system $A\mathbf{x} = \mathbf{0}$ has infinite number of solutions. This contradicts the assumption that $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.

$1 \implies 3$

Since A is invertible, for every $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.

$3 \implies 1$

For $1 \le i \le n$, define $\mathbf{e}_i = (0, \ldots, 0, \underbrace{1}_{i^{\text{th}} \text{ position}}, 0, \ldots, 0)^t$, and consider the linear system $A\mathbf{x} = \mathbf{e}_i$. By assumption, this system has a solution $\mathbf{x}_i$ for each i, $1 \le i \le n$. Define a matrix $B = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n]$. That is, the ith column of B is the solution of the system $A\mathbf{x} = \mathbf{e}_i$. Then

$$AB = A[\mathbf{x}_1, \mathbf{x}_2 \ldots, \mathbf{x}_n] = [A\mathbf{x}_1, A\mathbf{x}_2 \ldots, A\mathbf{x}_n] = [\mathbf{e}_1, \mathbf{e}_2 \ldots, \mathbf{e}_n] = I_n.$$

Therefore, by Theorem 2.7.8, the matrix A is invertible. □

Exercise 2.7.11 1. Show that a triangular matrix A is invertible if and only if each diagonal entry of A is non-zero.

2. Let A be a 1×2 matrix and B be a 2×1 matrix having positive entries. Which of BA or AB is invertible? Give reasons.

3. Let A be an $n \times m$ matrix and B be an $m \times n$ matrix. Prove that the matrix $I - BA$ is invertible if and only if the matrix $I - AB$ is invertible.

2.7.3 Inverse and Gauss-Jordan Method

We first give a consequence of Theorem 2.7.7 and then use it to find the inverse of an invertible matrix.

Corollary 2.7.12 Let A be an invertible $n \times n$ matrix. Suppose that a sequence of elementary row-operations reduces A to the identity matrix. Then the same sequence of elementary row-operations when applied to the identity matrix yields A^{-1}.

PROOF. Let A be a square matrix of order n. Also, let $E_1, E_2, \ldots, E_k$ be a sequence of elementary row operations such that $E_1E_2 \cdots E_kA = I_n$. Then $E_1E_2 \cdots E_kI_n = A^{-1}$. This implies $A^{-1} = E_1E_2 \cdots E_k$. □

Summary: Let A be an $n \times n$ matrix. Apply the Gauss-Jordan method to the matrix $[A \;\; I_n]$. Suppose the row reduced echelon form of the matrix $[A \;\; I_n]$ is $[B \;\; C]$. If $B = I_n$, then $A^{-1} = C$ or else A is not invertible.

Example 2.7.13 Find the inverse of the matrix $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ using Gauss-Jordan method.

Solution: Consider the matrix $\begin{bmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$. A sequence of steps in the Gauss-Jordan method are:

1. $\begin{bmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1(1/2)} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$

2. $\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} \xrightarrow{R_{21}(-1)} \\ \xrightarrow{R_{31}(-1)} \end{matrix} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \end{bmatrix}$

3. $\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \end{bmatrix} \xrightarrow{R_2(2/3)} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \end{bmatrix} \xrightarrow{R_{32}(-1/2)} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & \frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 \end{bmatrix}$

5. $\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & \frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 \end{bmatrix} \xrightarrow{R_3(3/4)} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{bmatrix}$

6. $\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{bmatrix} \begin{matrix} \xrightarrow{R_{23}(-1/3)} \\ \xrightarrow{R_{13}(-1/2)} \end{matrix} \begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{5}{8} & \frac{1}{8} & \frac{-3}{8} \\ 0 & 1 & 0 & \frac{-1}{4} & \frac{3}{4} & \frac{-1}{4} \\ 0 & 0 & 1 & \frac{-1}{4} & \frac{-1}{4} & \frac{3}{4} \end{bmatrix}$

7. $\begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{5}{8} & \frac{1}{8} & \frac{-3}{8} \\ 0 & 1 & 0 & \frac{-1}{4} & \frac{3}{4} & \frac{-1}{4} \\ 0 & 0 & 1 & \frac{-1}{4} & \frac{-1}{4} & \frac{3}{4} \end{bmatrix} \xrightarrow{R_{12}(-1/2)} \begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} & \frac{-1}{4} & \frac{-1}{4} \\ 0 & 1 & 0 & \frac{-1}{4} & \frac{3}{4} & \frac{-1}{4} \\ 0 & 0 & 1 & \frac{-1}{4} & \frac{-1}{4} & \frac{3}{4} \end{bmatrix}$.

8. Thus, the inverse of the given matrix is $\begin{bmatrix} 3/4 & -1/4 & -1/4 \\ -1/4 & 3/4 & -1/4 \\ -1/4 & -1/4 & 3/4 \end{bmatrix}$.

Exercise 2.7.14 Find the inverse of the following matrices using Gauss-Jordan method.

(i) $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \\ 2 & 4 & 7 \end{bmatrix}$, (ii) $\begin{bmatrix} 1 & 3 & 3 \\ 2 & 3 & 2 \\ 2 & 4 & 7 \end{bmatrix}$, (iii) $\begin{bmatrix} 2 & -1 & 3 \\ -1 & 3 & -2 \\ 2 & 4 & 1 \end{bmatrix}$.

2.8 Determinant

Notation: For an $n \times n$ matrix A, by $A(\alpha|\beta)$, we mean the submatrix B of A, which is obtained by deleting the αth row and β^{th} column.

Example 2.8.1 Consider a matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \\ 2 & 4 & 7 \end{bmatrix}$. Then $A(1|2) = \begin{bmatrix} 1 & 2 \\ 2 & 7 \end{bmatrix}$, $A(1|3) = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, and $A(1,2|1,3) = [4]$.

Definition 2.8.2 (Determinant of a Square Matrix) Let A be a square matrix of order n. With A, we associate inductively (on n) a number, called the determinant of A, written $\det(A)$ (or $|A|$) by

$$\det(A) = \begin{cases} a & \text{if } A = [a] \ (n = 1), \\ \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det\big(A(1|j)\big), & \text{otherwise.} \end{cases}$$

Definition 2.8.3 (Minor, Cofactor of a Matrix) The number $\det(A(i|j))$ is called the $(i,j)^{\text{th}}$ minor of A. We write $A_{ij} = \det(A(i|j))$. The $(i,j)^{\text{th}}$ cofactor of A, denoted C_{ij}, is the number $(-1)^{i+j}A_{ij}$.

Example 2.8.4 1. Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. Then, $\det(A) = |A| = a_{11}A_{11} - a_{12}A_{12} = a_{11}a_{22} - a_{12}a_{21}$.

For example, for $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ $\det(A) = |A| = 1 - 2 \cdot 2 = -3$.

2. Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$. Then,

$$\begin{aligned} \det(A) &= |A| = a_{11}A_{11} - a_{12}A_{12} + a_{13}A_{13} \\ &= a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22}) \\ &= a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} \quad (2.8.1) \end{aligned}$$

For example, if $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ then

$$\det(A) = |A| = 1 \cdot \begin{vmatrix} 3 & 1 \\ 2 & 2 \end{vmatrix} - 2 \cdot \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + 3 \cdot \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 4 - 2(3) + 3(1) = 1.$$

Exercise 2.8.5 1. Find the determinant of the following matrices.

$i)$ $\begin{bmatrix} 1 & 2 & 7 & 8 \\ 0 & 4 & 3 & 2 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$, $ii)$ $\begin{bmatrix} 3 & 5 & 2 & 1 \\ 0 & 2 & 0 & 5 \\ 6 & -7 & 1 & 0 \\ 2 & 0 & 3 & 0 \end{bmatrix}$, $iii)$ $\begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$.

2. Show that the determinant of a triangular matrix is the product of its diagonal entries.

Definition 2.8.6 A matrix A is said to be a **singular** matrix if $\det(A) = 0$. It is called **non-singular** if $\det(A) \neq 0$.

The proof of the next theorem is omitted. The interested reader is advised to go through Appendix 14.3.

Theorem 2.8.7 Let A be an $n \times n$ matrix. Then

1. if B is obtained from A by interchanging two rows, then $\det(B) = -\det(A)$,
2. if B is obtained from A by multiplying a row by c then $\det(B) = c\det(A)$,
3. if all the elements of one row or column are 0 then $\det(A) = 0$,
4. if B is obtained from A by replacing the jth row by itself plus k times the ith row, where $i \neq j$ then $\det(B) = \det(A)$,
5. if A is a square matrix having two rows equal then $\det(A) = 0$.

Remark 2.8.8 *1. Many authors define the determinant using "Permutations." It turns out that* THE WAY WE HAVE DEFINED DETERMINANT *is usually called the expansion of the determinant along the first row.*

2. Part 1 of Lemma 2.8.7 implies that "one can also calculate the determinant by expanding along any row." Hence, for an $n \times n$ matrix A, for every k, $1 \le k \le n$, one also has

$$\det(A) = \sum_{j=1}^{n} (-1)^{k+j} a_{kj} \det\big(A(k|j)\big).$$

Remark 2.8.9 *1. Let $\mathbf{u}^t = (u_1, u_2)$ and $\mathbf{v}^t = (v_1, v_2)$ be two vectors in $\mathbb{R}^2$. Then consider the parallelogram, $PQRS$, formed by the vertices $\{P = (0,0)^t, Q = \mathbf{u}, S = \mathbf{v}, R = \mathbf{u} + \mathbf{v}\}$. We*

$$\textit{Claim:} \qquad \textit{Area } (PQRS) = \left|\det\left(\begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix}\right)\right| = |u_1v_2 - u_2v_1|.$$

Recall that the dot product, $\mathbf{u} \bullet \mathbf{v} = u_1v_1 + u_2v_2$, and $\sqrt{\mathbf{u} \bullet \mathbf{u}} = \sqrt{(u_1^2 + u_2^2)}$, is the length of the vector $\mathbf{u}$. We denote the length by $\ell(\mathbf{u})$. With the above notation, if θ is the angle between the vectors $\mathbf{u}$ and $\mathbf{v}$, then

$$\cos(\theta) = \frac{\mathbf{u} \bullet \mathbf{v}}{\ell(\mathbf{u})\ell(\mathbf{v})}.$$

Which tells us,

$$\begin{aligned} \textit{Area}(PQRS) &= \ell(\mathbf{u})\ell(\mathbf{v})\sin(\theta) = \ell(\mathbf{u})\ell(\mathbf{v})\sqrt{1 - \left(\frac{\mathbf{u} \bullet \mathbf{v}}{\ell(\mathbf{u})\ell(\mathbf{v})}\right)^2} \\ &= \sqrt{\ell(\mathbf{u})^2\ell(v)^2 - (\mathbf{u} \bullet \mathbf{v})^2} = \sqrt{(u_1v_2 - u_2v_1)^2} \\ &= |u_1v_2 - u_2v_1|. \end{aligned}$$

Hence, the claim holds. That is, in $\mathbb{R}^2$, the determinant is $\pm$ times the area of the parallelogram.

2. Let $\mathbf{u} = (u_1, u_2, u_3), \mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$ be three elements of $\mathbb{R}^3$. Recall that the cross product of two vectors in $\mathbb{R}^3$ is,

$$\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1).$$

Note here that if $A = [\mathbf{u}^t, \mathbf{v}^t, \mathbf{w}^t]$, then

$$\det(A) = \begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} = \mathbf{u} \bullet (\mathbf{v} \times \mathbf{w}) = \mathbf{v} \bullet (\mathbf{w} \times \mathbf{u}) = \mathbf{w} \bullet (\mathbf{u} \times \mathbf{v}).$$

Let P be the parallelopiped formed with $(0,0,0)$ as a vertex and the vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ as adjacent vertices. Then observe that $\mathbf{u} \times \mathbf{v}$ is a vector perpendicular to the plane that contains the parallelogram formed by the vectors $\mathbf{u}$ and $\mathbf{v}$. So, to compute the volume of the parallelopiped P, we need to look at $\cos(\theta)$, where θ is the angle between the vector $\mathbf{w}$ and the normal vector to the parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$. So,

$$\textit{volume } (P) = |\mathbf{w} \bullet (\mathbf{u} \times \mathbf{v})|.$$

Hence, $|\det(A)| = \textit{volume } (P)$.

3. Let $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n \in \mathbb{R}^{n \times 1}$ and let $A = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n]$ be an $n \times n$ matrix. Then the following properties of $\det(A)$ also hold for the volume of an n-dimensional parallelopiped formed with $\mathbf{0} \in \mathbb{R}^{n \times 1}$ as one vertex and the vectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ as adjacent vertices:

(a) If $\mathbf{u}_1 = (1,0,\ldots,0)^t, \mathbf{u}_2 = (0,1,0,\ldots,0)^t, \ldots,$ and $\mathbf{u}_n = (0,\ldots,0,1)^t$, then $\det(A) = 1$. *Also, volume of a unit n-dimensional cube is* 1.

(b) *If we replace the vector $\mathbf{u}_i$ by $\alpha\mathbf{u}_i$, for some $\alpha \in \mathbb{R}$, then the determinant of the new matrix is $\alpha \cdot \det(A)$. This is also true for the volume, as the original volume gets multiplied by α.*

(c) *If $\mathbf{u}_1 = \mathbf{u}_i$ for some i, $2 \le i \le n$, then the vectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ will give rise to an $(n-1)$-dimensional parallelopiped. So, this parallelopiped lies on an $(n-1)$-dimensional hyperplane. Thus, its n-dimensional volume will be zero. Also, $|\det(A)| = |0| = 0$.*

In general, for any $n \times n$ matrix A, it can be proved that $|\det(A)|$ is indeed equal to the volume of the n-dimensional parallelepiped. The actual proof is beyond the scope of this book.

2.8.1 Adjoint of a Matrix

Recall that for a square matrix A, the notations A_{ij} and $C_{ij} = (-1)^{i+j}A_{ij}$ were respectively used to denote the $(i,j)^{\text{th}}$ minor and the $(i,j)^{\text{th}}$ cofactor of A.

Definition 2.8.10 (Adjoint of a Matrix) Let A be an $n \times n$ matrix. The matrix $B = [b_{ij}]$ with $b_{ij} = C_{ji}$, for $1 \le i, j \le n$ is called the Adjoint of A, denoted $Adj(A)$.

Example 2.8.11 Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$. Then $Adj(A) = \begin{bmatrix} 4 & 2 & -7 \\ -3 & -1 & 5 \\ 1 & 0 & -1 \end{bmatrix}$; as $C_{11} = (-1)^{1+1}A_{11} = 4, C_{12} = (-1)^{1+2}A_{12} = -3, C_{13} = (-1)^{1+3}A_{13} = 1$, and so on.

Theorem 2.8.12 Let A be an $n \times n$ matrix. Then

1. for $1 \le i \le n$, $\sum_{j=1}^{n} a_{ij}\, C_{ij} = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\, A_{ij} = \det(A)$,
2. for $i \ne \ell$, $\sum_{j=1}^{n} a_{ij}\, C_{\ell j} = \sum_{j=1}^{n} a_{ij}(-1)^{\ell+j}\, A_{\ell j} = 0$, and
3. $A(Adj(A)) = \det(A) I_n$. Thus,

$$\det(A) \ne 0 \Rightarrow A^{-1} = \frac{1}{\det(A)} Adj(A). \tag{2.8.2}$$

PROOF. Let $B = [b_{ij}]$ be a square matrix with

- the ℓth row of B as the ith row of A,
- the other rows of B are the same as that of A.

By the construction of B, two rows (i^{th} and ℓ^{th}) are equal. By Part 5 of Lemma 2.8.7, $\det(B) = 0$. By construction again, $\det\big(A(\ell|j)\big) = \det\big(B(\ell|j)\big)$ for $1 \le j \le n$. Thus, by Remark 2.8.8, we have

$$\begin{aligned} 0 = \det(B) &= \sum_{j=1}^{n} (-1)^{\ell+j} b_{\ell j} \det\big(B(\ell|j)\big) = \sum_{j=1}^{n} (-1)^{\ell+j} a_{ij} \det\big(B(\ell|j)\big) \\ &= \sum_{j=1}^{n} (-1)^{\ell+j} a_{ij} \det\big(A(\ell|j)\big) = \sum_{j=1}^{n} a_{ij} C_{\ell j}. \end{aligned}$$

Now,

$$\begin{aligned} \Big(A\big(\text{Adj}(A)\big)\Big)_{ij} &= \sum_{k=1}^{n} a_{ik}\big(\text{Adj}(A)\big)_{kj} = \sum_{k=1}^{n} a_{ik} C_{jk} \\ &= \begin{cases} 0 & \text{if } i \ne j \\ \det(A) & \text{if } i = j \end{cases} \end{aligned}$$

Thus, $A(Adj(A)) = \det(A) I_n$. Since, $\det(A) \ne 0$, $A\frac{1}{\det(A)} Adj(A) = I_n$. Therefore, A has a right inverse. Hence, by Theorem 2.7.8 A has an inverse and

$$A^{-1} = \frac{1}{\det(A)} Adj(A).$$

□

Example 2.8.13 Let $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$. Then

$$Adj(A) = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -3 & 1 \end{bmatrix}$$

and $\det(A) = -2$. By Theorem 2.8.12.3, $A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -1/2 & -1/2 & 1/2 \\ 1/2 & 3/2 & -1/2 \end{bmatrix}$.

The next corollary is an easy consequence of Theorem 2.8.12 (recall Theorem 2.7.8).

Corollary 2.8.14 If A is a non-singular matrix, then
$\big(Adj(A)\big)A = \det(A) I_n$ and $\sum_{i=1}^{n} a_{ij}\, C_{ik} = \begin{cases} \det(A) & \text{if } j = k \\ 0 & \text{if } j \ne k \end{cases}$.

Theorem 2.8.15 Let A and B be square matrices of order n. Then $\det(AB) = \det(A)\det(B)$.

PROOF. **Step 1.** Let $\det(A) \ne 0$.
This means, A is invertible. Therefore, either A is an elementary matrix or is a product of elementary matrices (see Theorem 2.7.7). So, let $E_1, E_2, \ldots, E_k$ be elementary matrices such that $A = E_1 E_2 \cdots E_k$. Then, by using Parts 1, 2 and 4 of Lemma 2.8.7 repeatedly, we get

$$\begin{aligned} \det(AB) &= \det(E_1 E_2 \cdots E_k B) = \det(E_1)\det(E_2 \cdots E_k B) \\ &= \det(E_1)\det(E_2)\det(E_3 \cdots E_k B) \\ &= \det(E_1 E_2)\det(E_3 \cdots E_k B) \\ &= \vdots \\ &= \det(E_1 E_2 \cdots E_k)\det(B) \\ &= \det(A)\det(B). \end{aligned}$$

Thus, we get the required result in case A is non-singular.

Step 2. Suppose $\det(A) = 0$.

Then A is not invertible. Hence, there exists an invertible matrix P such that $PA = C$, where $C = \begin{bmatrix} C_1 \\ \mathbf{0} \end{bmatrix}$. So, $A = P^{-1}C$, and therefore

$$\begin{aligned} \det(AB) &= \det((P^{-1}C)B) = \det(P^{-1}(CB)) = \det\left(P^{-1}\begin{bmatrix} C_1 B \\ \mathbf{0} \end{bmatrix}\right) \\ &= \det(P^{-1}) \cdot \det\left(\begin{bmatrix} C_1 B \\ \mathbf{0} \end{bmatrix}\right) \quad \text{as } P^{-1} \text{ is non-singular} \\ &= \det(P) \cdot 0 = 0 = 0 \cdot \det(B) = \det(A)\det(B). \end{aligned}$$

Thus, the proof of the theorem is complete. □

Corollary 2.8.16 Let A be a square matrix. Then A is non-singular if and only if A has an inverse.

PROOF. Suppose A is non-singular. Then $\det(A) \ne 0$ and therefore, $A^{-1} = \frac{1}{\det(A)} Adj(A)$. Thus, A has an inverse.

Suppose A has an inverse. Then there exists a matrix B such that $AB = I = BA$. Taking determinant of both sides, we get

$$\det(A)\det(B) = \det(AB) = \det(I) = 1.$$

This implies that $\det(A) \ne 0$. Thus, A is non-singular. □

Theorem 2.8.17 Let A be a square matrix. Then $\det(A) = \det(A^t)$.

PROOF. If A is a non-singular Corollary 2.8.14 gives $\det(A) = \det(A^t)$.

If A is singular, then $\det(A) = 0$. Hence, by Corollary 2.8.16, A doesn't have an inverse. Therefore, A^t also doesn't have an inverse (for if A^t has an inverse then $A^{-1} = \big((A^t)^{-1}\big)^t$). Thus again by Corollary 2.8.16, $\det(A^t) = 0$. Therefore, we again have $\det(A) = 0 = \det(A^t)$.

Hence, we have $\det(A) = \det(A^t)$. □

2.8.2 Cramer's Rule

Recall the following:

- The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$ if and only if A^{-1} exists.
- A has an inverse if and only if $\det(A) \ne 0$.

Thus, $A\mathbf{x} = \mathbf{b}$ has a unique solution FOR EVERY $\mathbf{b}$ if and only if $\det(A) \ne 0$.

The following theorem gives a direct method of finding the solution of the linear system $A\mathbf{x} = \mathbf{b}$ when $\det(A) \ne 0$.

Theorem 2.8.18 (Cramer's Rule) Let $A\mathbf{x} = \mathbf{b}$ be a linear system with n equations in n unknowns. If $\det(A) \ne 0$, then the unique solution to this system is

$$x_j = \frac{\det(A_j)}{\det(A)}, \quad \text{for } j = 1, 2, \ldots, n,$$

where A_j is the matrix obtained from A by replacing the jth column of A by the column vector $\mathbf{b}$.

PROOF. Since $\det(A) \ne 0$, $A^{-1} = \frac{1}{\det(A)} Adj(A)$. Thus, the linear system $A\mathbf{x} = \mathbf{b}$ has the solution $\mathbf{x} = \frac{1}{\det(A)} Adj(A)\mathbf{b}$. Hence, x_j, the jth coordinate of $\mathbf{x}$ is given by

$$x_j = \frac{b_1 C_{1j} + b_2 C_{2j} + \cdots + b_n C_{nj}}{\det(A)} = \frac{\det(A_j)}{\det(A)}.$$

□

The theorem implies that

$$x_1 = \frac{1}{\det(A)} \begin{vmatrix} b_1 & a_{12} & \cdots & a_{1n} \\ b_2 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_n & a_{n2} & \cdots & a_{nn} \end{vmatrix},$$

and in general

$$x_j = \frac{1}{\det(A)} \begin{vmatrix} a_{11} & \cdots & a_{1j-1} & b_1 & a_{1j+1} & \cdots & a_{1n} \\ a_{12} & \cdots & a_{2j-1} & b_2 & a_{2j+1} & \cdots & a_{2n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{nj-1} & b_n & a_{nj+1} & \cdots & a_{nn} \end{vmatrix}$$

for $j = 2, 3, \ldots, n$.

Example 2.8.19 Suppose that $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Use Cramer's rule to find a vector $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{b}$.

Solution: Check that $\det(A) = 1$. Therefore $x_1 = \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{vmatrix} = -1$,

$x_2 = \begin{vmatrix} 1 & 1 & 3 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 1$, and $x_3 = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 1 \end{vmatrix} = 0$. That is, $\mathbf{x}^t = (-1, 1, 0)$.

2.9 Miscellaneous Exercises

Exercise 2.9.1

1. Let A be an orthogonal matrix. Show that $\det A = \pm 1$.
2. If A and B are two $n \times n$ non-singular matrices, are the matrices $A + B$ and $A - B$ non-singular? Justify your answer.
3. For an $n \times n$ matrix A, prove that the following conditions are equivalent:
   (a) A is singular (A^{-1} doesn't exist).
   (b) $\text{rank}(A) \ne n$.
   (c) $\det(A) = 0$.
   (d) A is not row-equivalent to I_n, the identity matrix of order n.
   (e) $A\mathbf{x} = \mathbf{0}$ has a non-trivial solution for $\mathbf{x}$.
   (f) $A\mathbf{x} = b$ doesn't have a unique solution, i.e., it has no solutions or it has infinitely many solutions.

4. Let $A = \begin{bmatrix} 2 & 0 & 6 & 0 & 4 \\ 5 & 3 & 2 & 2 & 7 \\ 2 & 5 & 7 & 5 & 5 \\ 2 & 0 & 9 & 2 & 7 \\ 7 & 8 & 4 & 2 & 1 \end{bmatrix}$. We know that the numbers $20604, 53227, 25755, 20927$ and 78421 are all divisible by 17. Does this imply 17 divides $\det(A)$?

5. Let $A = [a_{ij}]_{n\times n}$ where $a_{ij} = x_i^{j-1}$. Show that $\det(A) = \prod_{1\le i<j\le n}(x_j - x_i)$. [The matrix A is usually called the Van-dermonde matrix.]

6. Let $A = [a_{ij}]$ with $a_{ij} = \max\{i, j\}$ be an $n \times n$ matrix. Compute $\det A$.

7. Let $A = [a_{ij}]$ with $a_{ij} = 1/(i + j)$ be an $n \times n$ matrix. Show that A is invertible.

8. Solve the following system of equations by Cramer's rule.
   i) $x + y + z - w = 1,\ x + y - z + w = 2,\ 2x + y + z - w = 7,\ x + y + z + w = 3.$
   ii) $x - y + z - w = 1,\ x + y - z + w = 2,\ 2x + y - z - w = 7,\ x - y - z + w = 3.$

9. Suppose $A = [a_{ij}]$ and $B = [b_{ij}]$ are two $n \times n$ matrices such that $b_{ij} = p^{i-j}a_{ij}$ for $1 \le i, j \le n$ for some non-zero real number p. Then compute $\det(B)$ in terms of $\det(A)$.

10. The position of an element a_{ij} of a determinant is called even or odd according as $i + j$ is even or odd. Show that

    (a) If all the entries in odd positions are multiplied with -1 then the value of the determinant doesn't change.

    (b) If all entries in even positions are multiplied with -1 then the determinant

        i. does not change if the matrix is of even order.

        ii. is multiplied by -1 if the matrix is of odd order.

11. Let A be an $n \times n$ Hermitian matrix, that is, $A^* = A$. Show that $\det A$ is a real number. [A is a matrix with complex entries and $A^* = \overline{A^t}$.]

12. Let A be an $n \times n$ matrix. Then show that

$$A \text{ is invertible} \iff Adj(A) \text{ is invertible.}$$

13. Let A and B be invertible matrices. Prove that $\mathrm{Adj}(AB) = \mathrm{Adj}(B)\mathrm{Adj}(A)$.

14. Let $P = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ be a rectangular matrix with A a square matrix of order n and $|A| \neq 0$. Then show that rank $(P) = n$ if and only if $D = CA^{-1}B$.

Chapter 3

Finite Dimensional Vector Spaces

Consider the problem of finding the set of points of intersection of the two planes $2x + 3y + z + u = 0$ and $3x + y + 2z + u = 0$.
Let V be the set of points of intersection of the two planes. Then V has the following properties:

1. The point $(0, 0, 0, 0)$ is an element of V.
2. For the points $(-1, 0, 1, 1)$ and $(-5, 1, 7, 0)$ which belong to V; the point $(-6, 1, 8, 1) = (-1, 0, 1, 1) + (-5, 1, 7, 0) \in V$.
3. Let $\alpha \in \mathbb{R}$. Then the point $\alpha(-1, 0, 1, 1) = (-\alpha, 0, \alpha, \alpha)$ also belongs to V.

Similarly, for an $m \times n$ real matrix A, consider the set V, of solutions of the homogeneous linear system $A\mathbf{x} = \mathbf{0}$. This set satisfies the following properties:

1. If $A\mathbf{x} = \mathbf{0}$ and $A\mathbf{y} = \mathbf{0}$, then $\mathbf{x}, \mathbf{y} \in V$. Then $\mathbf{x} + \mathbf{y} \in V$ as $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0}$. Also, $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$.
2. It is clear that if $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ then $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$.
3. The vector $\mathbf{0} \in V$ as $A\mathbf{0} = \mathbf{0}$.
4. If $A\mathbf{x} = \mathbf{0}$ then $A(-\mathbf{x}) = -A\mathbf{x} = \mathbf{0}$. Hence, $-\mathbf{x} \in V$.
5. Let $\alpha \in \mathbb{R}$ and $\mathbf{x} \in V$. Then $\alpha\mathbf{x} \in V$ as $A(\alpha\mathbf{x}) = \alpha A\mathbf{x} = \mathbf{0}$.

Thus we are lead to the following.

3.1 Vector Spaces

3.1.1 Definition

Definition 3.1.1 (Vector Space) A vector space over $\mathbb{F}$, denoted $V(\mathbb{F})$, is a non-empty set, satisfying the following axioms:

1. Vector Addition: To every pair $\mathbf{u}, \mathbf{v} \in V$ there corresponds a unique element $\mathbf{u} \oplus \mathbf{v}$ in V such that

   (a) $\mathbf{u} \oplus \mathbf{v} = \mathbf{v} \oplus \mathbf{u}$ (Commutative law).

   (b) $(\mathbf{u} \oplus \mathbf{v}) \oplus \mathbf{w} = \mathbf{u} \oplus (\mathbf{v} \oplus \mathbf{w})$ (Associative law).

   (c) There is a unique element $\mathbf{0}$ in V (the zero vector) such that $\mathbf{u} \oplus \mathbf{0} = \mathbf{u}$, for every $\mathbf{u} \in V$ (called **the additive identity**).

   (d) For every $\mathbf{u} \in V$ there is a unique element $-\mathbf{u} \in V$ such that $\mathbf{u} \oplus (-\mathbf{u}) = \mathbf{0}$ (called **the additive inverse**).

   $\oplus$ is called vector addition.

2. Scalar Multiplication: For each $\mathbf{u} \in V$ and $\alpha \in \mathbb{F}$, there corresponds a unique element $\alpha \odot \mathbf{u}$ in V such that

   (a) $\alpha \cdot (\beta \odot \mathbf{u}) = (\alpha\beta) \odot \mathbf{u}$ for every $\alpha, \beta \in \mathbb{F}$ and $\mathbf{u} \in V$.

   (b) $1 \odot \mathbf{u} = \mathbf{u}$ for every $\mathbf{u} \in V$, where $1 \in \mathbb{R}$.

3. Distributive Laws: relating vector addition with scalar multiplication
   For any $\alpha, \beta \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in V$, the following distributive laws hold:

   (a) $\alpha \odot (\mathbf{u} \oplus \mathbf{v}) = (\alpha \odot \mathbf{u}) \ \oplus \ (\alpha \odot \mathbf{v})$.

   (b) $(\alpha + \beta) \odot \mathbf{u} = (\alpha \odot \mathbf{u}) \ \oplus \ (\beta \odot \mathbf{u})$.

Note: the number 0 is the element of $\mathbb{F}$ whereas $\mathbf{0}$ is the zero vector.

Remark 3.1.2 *The elements of $\mathbb{F}$ are called* scalars, *and that of V are called* vectors. *If $\mathbb{F} = \mathbb{R}$, the vector space is called a* real vector space. *If $\mathbb{F} = \mathbb{C}$, the vector space is called a* complex vector space.

We may sometimes write V for a vector space if $\mathbb{F}$ is understood from the context.

Some interesting consequences of Definition 3.1.1 is the following useful result. Intuitively, these results seem to be obvious but for better understanding of the axioms it is desirable to go through the proof.

Theorem 3.1.3 Let V be a vector space over $\mathbb{F}$. Then

1. $\mathbf{u} \oplus \mathbf{v} = \mathbf{u}$ implies $\mathbf{v} = \mathbf{0}$.
2. $\alpha \odot \mathbf{u} = \mathbf{0}$ if and only if either $\mathbf{u}$ is the zero vector or $\alpha = 0$.
3. $(-1) \odot \mathbf{u} = -\mathbf{u}$ for every $\mathbf{u} \in V$.

Proof. Proof of Part 1.
For $\mathbf{u} \in V$, by Axiom 1d there exists $-\mathbf{u} \in V$ such that $-\mathbf{u} \oplus \mathbf{u} = \mathbf{0}$.
Hence, $\mathbf{u} \oplus \mathbf{v} = \mathbf{u}$ is equivalent to

$$-\mathbf{u} \oplus (\mathbf{u} \oplus \mathbf{v}) = -\mathbf{u} \oplus \mathbf{u} \iff (-\mathbf{u} \oplus \mathbf{u}) \oplus \mathbf{v} = \mathbf{0} \iff \mathbf{0} \oplus \mathbf{v} = \mathbf{0} \iff \mathbf{v} = \mathbf{0}.$$

Proof of Part 2.
As $\mathbf{0} = \mathbf{0} \oplus \mathbf{0}$, using the distributive law, we have

$$\alpha \odot \mathbf{0} = \alpha \odot (\mathbf{0} \oplus \mathbf{0}) = (\alpha \odot \mathbf{0}) \ \oplus \ (\alpha \odot \mathbf{0}).$$

Thus, for any $\alpha \in \mathbb{F}$, the first part implies $\alpha \odot \mathbf{0} = \mathbf{0}$. In the same way,

$$0 \odot \mathbf{u} = (0 + 0) \odot \mathbf{u} = (0 \odot \mathbf{u}) \ \oplus (0 \odot \mathbf{u}).$$

Hence, using the first part, one has $0 \odot \mathbf{u} = \mathbf{0}$ for any $\mathbf{u} \in V$.
Now suppose $\alpha \odot \mathbf{u} = \mathbf{0}$. If $\alpha = 0$ then the proof is over. Therefore, let us assume $\alpha \neq 0$ (note that α is a real or complex number, hence $\frac{1}{\alpha}$ exists and

$$\mathbf{0} = \frac{1}{\alpha} \odot \mathbf{0} = \frac{1}{\alpha} \odot (\alpha \odot \mathbf{u}) = (\frac{1}{\alpha}\ \alpha) \odot \mathbf{u} = 1 \odot \mathbf{u} = \mathbf{u}$$

as $1 \odot \mathbf{u} = \mathbf{u}$ for every vector $\mathbf{u} \in V$.

Thus we have shown that if $\alpha \neq 0$ and $\alpha \odot \mathbf{u} = \mathbf{0}$ then $\mathbf{u} = \mathbf{0}$.

Proof of Part 3.

We have $\mathbf{0} = \mathbf{0u} = (1 + (-1))\mathbf{u} = \mathbf{u} + (-1)\mathbf{u}$ and hence $(-1)\mathbf{u} = -\mathbf{u}$. □

3.1.2 Examples

Example 3.1.4 1. The set $\mathbb{R}$ of real numbers, with the usual addition and multiplication (*i.e.,* $\oplus \equiv +$ and $\odot \equiv \cdot$) forms a vector space over $\mathbb{R}$.

2. Consider the set $\mathbb{R}^2 = \{(x_1, x_2) : x_1, x_2 \in \mathbb{R}\}$. For $x_1, x_2, y_1, y_2 \in \mathbb{R}$ and $\alpha \in \mathbb{R}$, define,

$$(x_1, x_2) \oplus (y_1, y_2) = (x_1 + y_1, x_2 + y_2) \text{ and } \alpha \odot (x_1, x_2) = (\alpha x_1, \alpha x_2).$$

Then $\mathbb{R}^2$ is a real vector space.

3. Let $\mathbb{R}^n = \{(a_1, a_2, \ldots, a_n) : a_i \in \mathbb{R}, 1 \leq i \leq n\}$, be the set of n-tuples of real numbers. For $\mathbf{u} = (a_1, \ldots, a_n)$, $\mathbf{v} = (b_1, \ldots, b_n)$ in V and $\alpha \in \mathbb{R}$, we define

$$\mathbf{u} \oplus \mathbf{v} = (a_1 + b_1, \ldots, a_n + b_n) \text{ and } \alpha \odot \mathbf{u} = (\alpha a_1, \ldots, \alpha a_n)$$

(called component wise or coordinate wise operations). Then V is a real vector space with addition and scalar multiplication defined as above. This vector space is denoted by $\mathbb{R}^n$, called **the real vector space of n-tuples**.

4. Let $V = \mathbb{R}^+$ (the set of positive real numbers). This is NOT A VECTOR SPACE under usual operations of addition and scalar multiplication (why?). We now define a new vector addition and scalar multiplication as

$$\mathbf{v}_1 \oplus \mathbf{v}_2 = \mathbf{v}_1 \cdot \mathbf{v}_2 \text{ and } \alpha \odot \mathbf{v} = \mathbf{v}^\alpha$$

for all $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v} \in \mathbb{R}^+$ and $\alpha \in \mathbb{R}$. Then $\mathbb{R}^+$ is a real vector space with 1 as the additive identity.

5. Let $V = \mathbb{R}^2$. Define $(x_1, x_2) \oplus (y_1, y_2) = (x_1 + y_1 + 1, x_2 + y_2 - 3)$, $\alpha \odot (x_1, x_2) = (\alpha x_1 + \alpha - 1, \alpha x_2 - 3\alpha + 3)$ for $(x_1, x_2), (y_1, y_2) \in \mathbb{R}^2$ and $\alpha \in \mathbb{R}$. Then it can be easily verified that the vector $(-1, 3)$ is the additive identity and V is indeed a real vector space.

Recall $\sqrt{-1}$ is denoted i.

6. Consider the set $\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\}$ of complex numbers.

(a) For $x_1 + iy_1, x_2 + iy_2 \in \mathbb{C}$ and $\alpha \in \mathbb{R}$, define,

$$\begin{aligned}(x_1 + iy_1) \oplus (x_2 + iy_2) &= (x_1 + x_2) + i(y_1 + y_2) \text{ and} \\ \alpha \odot (x_1 + iy_1) &= (\alpha x_1) + i(\alpha y_1).\end{aligned}$$

Then $\mathbb{C}$ is a real vector space.

(b) For $x_1 + iy_1, x_2 + iy_2 \in \mathbb{C}$ and $\alpha + i\beta \in \mathbb{C}$, define,

$$\begin{aligned}(x_1 + iy_1) \oplus (x_2 + iy_2) &= (x_1 + x_2) + i(y_1 + y_2) \text{ and} \\ (\alpha + i\beta) \odot (x_1 + iy_1) &= (\alpha x_1 - \beta y_1) + i(\alpha y_1 + \beta x_1).\end{aligned}$$

Then $\mathbb{C}$ forms a complex vector space.

7. Consider the set $\mathbb{C}^n = \{(z_1, z_2, \ldots, z_n) : z_i \in \mathbb{C} \text{ for } 1 \leq i \leq n\}$. For $(z_1, \ldots, z_n), (w_1, \ldots, w_n) \in \mathbb{C}^n$ and $\alpha \in \mathbb{F}$, define,

$$\begin{aligned}(z_1, \ldots, z_n) \oplus (w_1, \ldots, w_n) &= (z_1 + w_1, \ldots, z_n + w_n) \text{ and} \\ \alpha \odot (z_1, \ldots, z_n) &= (\alpha z_1, \ldots, \alpha z_n).\end{aligned}$$

(a) If the set $\mathbb{F}$ is the set $\mathbb{C}$ of complex numbers, then $\mathbb{C}^n$ is a complex vector space having n-tuple of complex numbers as its vectors.

(b) If the set $\mathbb{F}$ is the set $\mathbb{R}$ of real numbers, then $\mathbb{C}^n$ is a real vector space having n-tuple of complex numbers as its vectors.

Remark 3.1.5 *In Example 7a, the scalars are Complex numbers and hence* $i(1, 0) = (i, 0)$. *Whereas, in Example 7b, the scalars are Real Numbers and hence* WE CANNOT WRITE $i(1, 0) = (i, 0)$.

8. Fix a positive integer n and let $M_n(\mathbb{R})$ denote the set of all $n \times n$ matrices with real entries. Then $M_n(\mathbb{R})$ is a real vector space with vector addition and scalar multiplication defined by

$$A \oplus B = [a_{ij}] \oplus [b_{ij}] = [a_{ij} + b_{ij}], \qquad \alpha \odot A = \alpha \odot [a_{ij}] = [\alpha a_{ij}].$$

9. Fix a positive integer n. Consider the set, $\mathcal{P}_n(\mathbb{R})$, of all polynomials of degree $\leq n$ with coefficients from $\mathbb{R}$ in the indeterminate x. Algebraically,

$$\mathcal{P}_n(\mathbb{R}) = \{a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n : a_i \in \mathbb{R}, 0 \leq i \leq n\}.$$

Let $f(x), g(x) \in \mathcal{P}_n(\mathbb{R})$. Then $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$ and $g(x) = b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n$ for some $a_i, b_i \in \mathbb{R}$, $0 \leq i \leq n$. It can be verified that $\mathcal{P}_n(\mathbb{R})$ is a real vector space with the addition and scalar multiplication defined by:

$$\begin{aligned}f(x) \oplus g(x) &= (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x^n, \text{ and} \\ \alpha \odot f(x) &= \alpha a_0 + \alpha a_1 x + \cdots + \alpha a_n x^n \text{ for } \alpha \in \mathbb{R}.\end{aligned}$$

10. Consider the set $\mathcal{P}(\mathbb{R})$, of all polynomials with real coefficients. Let $f(x), g(x) \in \mathcal{P}(\mathbb{R})$. Observe that a polynomial of the form $a_0 + a_1 x + \cdots + a_m x^m$ can be written as $a_0 + a_1 x + \cdots + a_m x^m + 0 \cdot x^{m+1} + \cdots + 0 \cdot x^p$ for any $p > m$. Hence, we can assume $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_p x^p$ and $g(x) = b_0 + b_1 x + b_2 x^2 + \cdots + b_p x^p$ for some $a_i, b_i \in \mathbb{R}$, $0 \leq i \leq p$, for some large positive integer p. We now define the vector addition and scalar multiplication as

$$\begin{aligned}f(x) \oplus g(x) &= (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_p + b_p)x^p, \text{ and} \\ \alpha \odot f(x) &= \alpha a_0 + \alpha a_1 x + \cdots + \alpha a_p x^p \text{ for } \alpha \in \mathbb{R}.\end{aligned}$$

Then $\mathcal{P}(\mathbb{R})$ forms a real vector space.

11. Let $C([-1, 1])$ be the set of all real valued continuous functions on the interval $[-1, 1]$. For $f, g \in C([-1, 1])$ and $\alpha \in \mathbb{R}$, define

$$\begin{aligned}(f \oplus g)(x) &= f(x) + g(x), \text{ and} \\ (\alpha \odot f)(x) &= \alpha f(x), \text{ for all } x \in [-1, 1].\end{aligned}$$

Then $C([-1, 1])$ forms a real vector space. The operations defined above are called POINT WISE ADDITION AND SCALAR MULTIPLICATION.

12. Let V and W be real vector spaces with binary operations $(+, \bullet)$ and $(\oplus, \odot)$, respectively. Consider the following operations on the set $V \times W$: for $(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2) \in V \times W$ and $\alpha \in \mathbb{R}$, define

$$\begin{aligned}(\mathbf{x}_1, \mathbf{y}_1) \oplus' (\mathbf{x}_2, \mathbf{y}_2) &= (\mathbf{x}_1 + \mathbf{x}_2, \mathbf{y}_1 \oplus \mathbf{y}_2), \text{ and} \\ \alpha \circ (\mathbf{x}_1, \mathbf{y}_1) &= (\alpha \bullet \mathbf{x}_1, \alpha \odot \mathbf{y}_1).\end{aligned}$$

On the right hand side, we write $\mathbf{x}_1 + \mathbf{x}_2$ to mean the addition in V, while $\mathbf{y}_1 \oplus \mathbf{y}_2$ is the addition in W. Similarly, $\alpha \bullet \mathbf{x}_1$ and $\alpha \odot \mathbf{y}_1$ come from scalar multiplication in V and W, respectively. With the above definitions, $V \times W$ also forms a real vector space.

The readers are advised to justify the statements made in the above examples.

From now on, we will use '$\mathbf{u} + \mathbf{v}$' in place of '$\mathbf{u} \oplus \mathbf{v}$' and '$\alpha \cdot \mathbf{u}$ or $\alpha\mathbf{u}$' in place of '$\alpha \odot \mathbf{u}$'.

3.1.3 Subspaces

Definition 3.1.6 (Vector Subspace) Let S be a NON-EMPTY SUBSET of V. $S(\mathbb{F})$ is said to be a subspace of $V(\mathbb{F})$ if $\alpha\mathbf{u} + \beta\mathbf{v} \in S$ whenever $\alpha, \beta \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in S$; where the vector addition and scalar multiplication are the same as that of $V(\mathbb{F})$.

Remark 3.1.7 *Any subspace is a vector space in its own right with respect to the vector addition and scalar multiplication that is defined for* $V(\mathbb{F})$.

Example 3.1.8 1. Let $V(\mathbb{F})$ be a vector space. Then

(a) $S = \{\mathbf{0}\}$, the set consisting of the zero vector $\mathbf{0}$,

(b) $S = V$

are vector subspaces of V. These are called **trivial subspaces**.

2. Let $S = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 0\}$. Then S is a subspace of $\mathbb{R}^3$. (S is a plane in $\mathbb{R}^3$ passing through the origin.)

3. Let $S = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 3\}$. Then S is not a subspace of $\mathbb{R}^3$. (S is again a plane in $\mathbb{R}^3$ but it doesn't pass through the origin.)

4. Let $S = \{(x, y, z) \in \mathbb{R}^3 : z = x\}$. Then S is a subspace of $\mathbb{R}^3$.

5. The vector space $\mathcal{P}_n(\mathbb{R})$ is a subspace of the vector space $\mathcal{P}(\mathbb{R})$.

Exercise 3.1.9 1. Which of the following are correct statements?

(a) Let $S = \{(x, y, z) \in \mathbb{R}^3 : z = x^2\}$. Then S is a subspace of $\mathbb{R}^3$.

(b) Let $V(\mathbb{F})$ be a vector space. Let $\mathbf{x} \in V$. Then the set $\{\alpha\mathbf{x} : \alpha \in \mathbb{F}\}$ forms a vector subspace of V.

(c) Let $W = \{f \in C([-1, 1]) : f(1/2) = 0\}$. Then W is a subspace of the real vector space, $C([-1, 1])$.

2. Which of the following are subspaces of $\mathbb{R}^n(\mathbb{R})$?

(a) $\{(x_1, x_2, \ldots, x_n) : x_1 \geq 0\}$.

(b) $\{(x_1, x_2, \ldots, x_n) : x_1 + 2x_2 = 4x_3\}$.

(c) $\{(x_1, x_2, \ldots, x_n) : x_1 \text{is rational }\}$.

(d) $\{(x_1, x_2, \ldots, x_n) : x_1 = x_3^2\}$.

(e) $\{(x_1, x_2, \ldots, x_n) : \text{either } x_1 \text{ or } x_2 \text{ or both is} 0\}$.

(f) $\{(x_1, x_2, \ldots, x_n) : |x_1| \leq 1\}$.

3. Which of the following are subspaces of $i)\mathbb{C}^n(\mathbb{R})$ $ii)\mathbb{C}^n(\mathbb{C})$?

(a) $\{(z_1, z_2, \ldots, z_n) : z_1 \text{is real }\}$.

(b) $\{(z_1, z_2, \ldots, z_n) : z_1 + z_2 = \overline{z_3}\}$.

(c) $\{(z_1, z_2, \ldots, z_n) : |\, z_1 \,| = |\, z_2 \,|\}$.

3.1.4 Linear Combinations

Definition 3.1.10 (Linear Span) Let $V(\mathbb{F})$ be a vector space and let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be a non-empty subset of V. The linear span of S is the set defined by

$$L(S) = \{\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n : \alpha_i \in \mathbb{F}, 1 \leq i \leq n\}$$

If S is an empty set we define $L(S) = \{\mathbf{0}\}$.

Example 3.1.11 1. Note that $(4, 5, 5)$ is a linear combination of $(1, 0, 0), (1, 1, 0)$, and $(1, 1, 1)$ as $(4, 5, 5) = 5(1, 1, 1) - 1(1, 0, 0) + 0(1, 1, 0)$.

For each vector, the LINEAR COMBINATION IN TERMS OF THE VECTORS $(1, 0, 0), (1, 1, 0)$, AND $(1, 1, 1)$ IS UNIQUE.

2. Is $(4, 5, 5)$ a linear combination of $(1, 2, 3), (-1, 1, 4)$ and $(3, 3, 2)$?
Solution: We want to find $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ such that

$$\alpha_1(1, 2, 3) + \alpha_2(-1, 1, 4) + \alpha_3(3, 3, 2) = (4, 5, 5). \tag{3.1.1}$$

Check that $3(1, 2, 3) + (-1)(-1, 1, 4) + 0(3, 3, 2) = (4, 5, 5)$. Also, in this case, the vector $(4, 5, 5)$ DOES NOT HAVE A UNIQUE EXPRESSION AS LINEAR COMBINATION OF VECTORS $(1, 2, 3), (-1, 1, 4)$ AND $(3, 3, 2)$.

3. Verify that $(4, 5, 5)$ is not a linear combination of the vectors $(1, 2, 1)$ and $(1, 1, 0)$?

4. The linear span of $S = \{(1, 1, 1), (2, 1, 3)\}$ over $\mathbb{R}$ is

$$\begin{aligned}L(S) &= \{\alpha(1, 1, 1) + \beta(2, 1, 3) : \alpha, \beta \in \mathbb{R}\} \\ &= \{(\alpha + 2\beta, \alpha + \beta, \alpha + 3\beta) : \alpha, \beta \in \mathbb{R}\} \\ &= \{(x, y, z) \in \mathbb{R}^3 : 2x - y = z\}.\end{aligned}$$

as $2(\alpha + 2\beta) - (\alpha + \beta) = \alpha + 3\beta$, and if $z = 2x - y$, take $\alpha = 2y - x$ and $\beta = x - y$.

Lemma 3.1.12 (Linear Span is a subspace) Let $V(\mathbb{F})$ be a vector space and let S be a non-empty subset of V. Then $L(S)$ is a subspace of $V(\mathbb{F})$.

PROOF. By definition, $S \subset L(S)$ and hence $L(S)$ is non-empty subset of V. Let $\mathbf{u}, \mathbf{v} \in L(S)$. Then, for $1 \leq i \leq n$ there exist vectors $\mathbf{w}_i \in S$, and scalars $\alpha_i, \beta_i \in \mathbb{F}$ such that $\mathbf{u} = \alpha_1\mathbf{w}_1 + \alpha_2\mathbf{w}_2 + \cdots + \alpha_n\mathbf{w}_n$ and $\mathbf{v} = \beta_1\mathbf{w}_1 + \beta_2\mathbf{w}_2 + \cdots + \beta_n\mathbf{w}_n$. Hence,

$$\mathbf{u} + \mathbf{v} = (\alpha_1 + \beta)\mathbf{w}_1 + \cdots + (\alpha_n + \beta_n)\mathbf{w}_n \in L(S).$$

Thus, $L(S)$ is a vector subspace of $V(\mathbb{F})$. □

Remark 3.1.13 *Let $V(\mathbb{F})$ be a vector space and $W \subset V$ be a subspace. If $S \subset W$, then $L(S) \subset W$ is a subspace of W as W is a vector space in its own right.*

Theorem 3.1.14 Let S be a non-empty subset of a vector space V. Then $L(S)$ is the smallest subspace of V containing S.

PROOF. For every $\mathbf{u} \in S$, $\mathbf{u} = 1.\mathbf{u} \in L(S)$ and therefore, $S \subseteq L(S)$. To show $L(S)$ is the smallest subspace of V containing S, consider any subspace W of V containing S. Then by Proposition 3.1.13, $L(S) \subseteq W$ and hence the result follows. □

Definition 3.1.15 Let A be an $m \times n$ matrix with real entries. Then using the rows $\mathbf{a}_1^t, \mathbf{a}_2^t, \ldots, \mathbf{a}_m^t \in \mathbb{R}^n$ and columns $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n \in \mathbb{R}^m$, we define

1. $\mathbf{R}ow Space(A) = L(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_m)$,
2. $\mathbf{C}olumn Space(A) = L(\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$,
3. $\mathbf{N}ull Space(A)$, denoted $\mathcal{N}(A)$ as $\{\mathbf{x}^t \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$.
4. $\mathbf{R}ange(A)$, denoted Im $(A) = \{\mathbf{y} : A\mathbf{x} = \mathbf{y}$ for some $\mathbf{x}^t \in \mathbb{R}^n\}$.

Note that the "column space" of a matrix A consists of all $\mathbf{b}$ such that $A\mathbf{x} = \mathbf{b}$ has a solution. Hence, $\mathbf{C}olumn Space(A) = \mathbf{R}ange(A)$.

Lemma 3.1.16 Let A be a real $m \times n$ matrix. Suppose $B = EA$ for some elementary matrix E. Then Row Space(A) = Row Space(B).

PROOF. We prove the result for the elementary matrix $E_{ij}(c)$, where $c \neq 0$ and $i < j$. Let $\mathbf{a}_1^t, \mathbf{a}_2^t, \ldots, \mathbf{a}_m^t$ be the rows of the matrix A. Then $B = E_{ij}(c)A$ gives us

$$\begin{aligned}
\text{Row Space}(B) &= L(\mathbf{a}_1, \ldots, \mathbf{a}_{i-1}, \mathbf{a}_i + c\mathbf{a}_j, \ldots, \mathbf{a}_m) \\
&= \{\alpha_1\mathbf{a}_1 + \cdots + \alpha_{i-1}\mathbf{a}_{i-1} + \alpha_i(\mathbf{a}_i + c\mathbf{a}_j) + \cdots \\
&\qquad + \alpha_m\mathbf{a}_m : \ \alpha_\ell \in \mathbb{R}, 1 \leq \ell \leq m\} \\
&= \left\{\sum_{\ell=1}^{m} \alpha_\ell\mathbf{a}_\ell + \alpha_i \cdot c\mathbf{a}_j : \alpha_\ell \in \mathbb{R}, 1 \leq \ell \leq m\right\} \\
&= \left\{\sum_{\ell=1}^{m} \beta_\ell\mathbf{a}_\ell : \beta_\ell \in \mathbb{R}, 1 \leq \ell \leq m\right\} \\
&= L(\mathbf{a}_1, \ldots, \mathbf{a}_{i-1}, \mathbf{a}_i, \ldots, \mathbf{a}_m) \\
&= \text{Row Space}(A)
\end{aligned}$$

□

Theorem 3.1.17 Let A be an $m \times n$ matrix with real entries. Then

1. $\mathcal{N}(A)$ is a subspace of $\mathbb{R}^n$;
2. the non-zero row vectors of a matrix in row-reduced form, forms a basis for the row-space. Hence dim(Row Space(A)) = row rank of (A).

PROOF. Part 1) can be easily proved. Let A be an $m \times n$ matrix. For part 2), let D be the row-reduced form of A with non-zero rows $\mathbf{d}_1^t, \mathbf{d}_2^t, \ldots, \mathbf{d}_r^t$. Then $B = E_kE_{k-1}\cdots E_2E_1A$ for some elementary matrices $E_1, E_2, \ldots, E_k$. Then, a repeated application of Lemma 3.1.16 implies Row Space(A) = Row Space(B). That is, if the rows of the matrix A are $\mathbf{a}_1^t, \mathbf{a}_2^t, \ldots, \mathbf{a}_m^t$, then

$$L(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_m) = L(\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_r).$$

Hence the required result follows. □

Exercise 3.1.18

1. Show that any two row-equivalent matrices have the same row space. Give examples to show that the column space of two row-equivalent matrices need not be same.
2. Find all the vector subspaces of $\mathbb{R}^2$.
3. Let P and Q be two subspaces of a vector space V. Show that $P \cap Q$ is a subspace of V. Also show that $P \cup Q$ need not be a subspace of V. When is $P \cup Q$ a subspace of V?
4. Let P and Q be two subspaces of a vector space V. Define $P + Q = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in P, \mathbf{v} \in Q\}$. Show that $P + Q$ is a subspace of V. Also show that $L(P \cup Q) = P + Q$.
5. Let $S = \{x_1, x_2, x_3, x_4\}$ where $x_1 = (1,0,0,0)$, $x_2 = (1,1,0,0)$, $x_3 = (1,2,0,0)$, $x_4 = (1,1,1,0)$. Determine all x_i such that $L(S) = L(S \setminus \{x_i\})$.
6. Let $C([-1,1])$ be the set of all continuous functions on the interval $[-1,1]$ (cf. Example 3.1.4.11). Let
$$\begin{aligned} W_1 &= \{f \in C([-1,1]) : f(0.2) = 0\}, \text{ and} \\ W_2 &= \{f \in C([-1,1]) : f'(\tfrac{1}{4}) \text{exists }\}. \end{aligned}$$
Are W_1, W_2 subspaces of $C([-1,1])$?
7. Let $V = \{(x,y) : x, y \in \mathbb{R}\}$ over $\mathbb{R}$. Define $(x,y) \oplus (x_1,y_1) = (x + x_1, 0)$ and $\alpha \odot (x,y) = (\alpha x, 0)$. Show that V is not a vector space over $\mathbb{R}$.
8. Recall that $M_n(\mathbb{R})$ is the real vector space of all $n \times n$ real matrices. Prove that the following subsets are subspaces of $M_n(\mathbb{R})$.
   (a) $\mathsf{sl}_n = \{A \in M_n(\mathbb{R}) : \text{trace}(A) = 0\}$
   (b) $\mathsf{Sym}_n = \{A \in M_n(\mathbb{R}) : A = A^t\}$
   (c) $\mathsf{Skew}_n = \{A \in M_n(\mathbb{R}) : A + A^t = \mathbf{0}\}$
9. Let $V = \mathbb{R}$. Define $x \oplus y = x - y$ and $\alpha \odot x = -\alpha x$. Which vector space axioms are not satisfied here?

In this section, we saw that a vector space has infinite number of vectors. Hence, one can start with any finite collection of vectors and obtain their span. It means that any vector space contains infinite number of other vector subspaces. Therefore, the following questions arise:

1. What are the conditions under which, the linear span of two distinct sets the same?
2. Is it possible to find/choose vectors so that the linear span of the chosen vectors is the whole vector space itself?
3. Suppose we are able to choose certain vectors whose linear span is the whole space. Can we find the minimum number of such vectors?

We try to answer these questions in the subsequent sections.

3.2 Linear Independence

Definition 3.2.1 (Linear Independence and Dependence) Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ be any non-empty subset of V. If there exist some non-zero α_i's $1 \leq i \leq m$, such that

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_m\mathbf{u}_m = \mathbf{0},$$

then the set S is called a linearly dependent set. Otherwise, the set S is called linearly independent.

Example 3.2.2

1. Let $S = \{(1,2,1), (2,1,4), (3,3,5)\}$. Then check that $1(1,2,1)+1(2,1,4)+(-1)(3,3,5) = (0,0,0)$. Since $\alpha_1 = 1, \alpha_2 = 1$ and $\alpha_3 = -1$ is a solution of (3.2.1), so the set S is a linearly dependent subset of $\mathbb{R}^3$.
2. Let $S = \{(1,1,1), (1,1,0), (1,0,1)\}$. Suppose there exists $\alpha, \beta, \gamma \in \mathbb{R}$ such that $\alpha(1,1,1)+\beta(1,1,0)+\gamma(1,0,1) = (0,0,0)$. Then check that in this case we necessarily have $\alpha = \beta = \gamma = 0$ which shows that the set $S = \{(1,1,1), (1,1,0), (1,0,1)\}$ is a linearly independent subset of $\mathbb{R}^3$.

In other words, if $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ is a non-empty subset of a vector space V, then to check whether the set S is linearly dependent or independent, one needs to consider the equation

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_m\mathbf{u}_m = \mathbf{0}. \tag{3.2.1}$$

In case $\alpha_1 = \alpha_2 = \cdots = \alpha_m = 0$ is THE ONLY SOLUTION of (3.2.1), the set S becomes a linearly independent subset of V. Otherwise, the set S becomes a linearly dependent subset of V.

Proposition 3.2.3 Let V be a vector space.

1. Then the zero-vector cannot belong to a linearly independent set.
2. If S is a linearly independent subset of V, then every subset of S is also linearly independent.
3. If S is a linearly dependent subset of V then every set containing S is also linearly dependent.

PROOF. We give the proof of the first part. The reader is required to supply the proof of other parts. Let $S = \{\mathbf{0} = \mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be a set consisting of the zero vector. Then for any $\gamma \neq o$, $\gamma\mathbf{u}_1 + o\mathbf{u}_2 + \cdots + 0\mathbf{u}_n = \mathbf{0}$. Hence, for the system $\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_m\mathbf{u}_m = \mathbf{0}$, we have a non-zero solution $\alpha_1 = \gamma$ and $o = \alpha_2 = \cdots = \alpha_n$. Therefore, the set S is linearly dependent. □

Theorem 3.2.4 Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ be a linearly independent subset of a vector space V. Suppose there exists a vector $\mathbf{v}_{p+1} \in V$, such that the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p, \mathbf{v}_{p+1}\}$ is linearly dependent, then $\mathbf{v}_{p+1}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$.

PROOF. Since the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p, \mathbf{v}_{p+1}\}$ is linearly dependent, there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_{p+1}$, NOT ALL ZERO such that

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_p\mathbf{v}_p + \alpha_{p+1}\mathbf{v}_{p+1} = \mathbf{0}. \tag{3.2.2}$$

CLAIM: $\alpha_{p+1} \neq 0$.
Let if possible $\alpha_{p+1} = 0$. Then equation (3.2.2) gives $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_p\mathbf{v}_p = \mathbf{0}$ with not all α_i, $1 \leq i \leq p$ zero. Hence, by the definition of linear independence, the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ is linearly dependent which is contradictory to our hypothesis. Thus, $\alpha_{p+1} \neq 0$ and we get

$$\mathbf{v}_{p+1} = -\frac{1}{\alpha_{p+1}}(\alpha_1\mathbf{v}_1 + \cdots + \alpha_p\mathbf{v}_p).$$

Note that $\alpha_i \in \mathbb{F}$ for every i, $1 \leq i \leq p+1$ and hence $-\frac{\alpha_i}{\alpha_{p+1}} \in \mathbb{F}$ for $1 \leq i \leq p$. Hence the result follows. □

We now state two important corollaries of the above theorem. We don't give their proofs as they are easy consequence of the above theorem.

Corollary 3.2.5 Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be a linearly dependent subset of a vector space V. Then there exists a smallest k, $2 \leq k \leq n$ such that

$$L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k) = L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{k-1}).$$

The next corollary follows immediately from Theorem 3.2.4 and Corollary 3.2.5.

Corollary 3.2.6 Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ be a linearly independent subset of a vector space V. Suppose there exists a vector $\mathbf{v} \in V$, such that $\mathbf{v} \notin L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p)$. Then the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p, \mathbf{v}\}$ is also linearly independent subset of V.

Exercise 3.2.7

1. Consider the vector space $\mathbb{R}^2$. Let $\mathbf{u}_1 = (1,0)$. Find all choices for the vector $\mathbf{u}_2$ such that the set $\{\mathbf{u}_1, \mathbf{u}_2\}$ is linear independent subset of $\mathbb{R}^2$. Does there exist choices for vectors $\mathbf{u}_2$ and $\mathbf{u}_3$ such that the set $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is linearly independent subset of $\mathbb{R}^2$?
2. If none of the elements appearing along the principal diagonal of a lower triangular matrix is zero, show that the row vectors are linearly independent in $\mathbb{R}^n$. The same is true for column vectors.
3. Let $S = \{(1,1,1,1), (1,-1,1,2), (1,1,-1,1)\} \subset \mathbb{R}^4$. Determine whether or not the vector $(1,1,2,1) \in L(S)$?
4. Show that $S = \{(1,2,3), (-2,1,1), (8,6,10)\}$ is linearly dependent in $\mathbb{R}^3$.
5. Show that $S = \{(1,0,0), (1,1,0), (1,1,1)\}$ is a linearly independent set in $\mathbb{R}^3$. In general if $\{f_1, f_2, f_3\}$ is a linearly independent set then $\{f_1, f_1 + f_2, f_1 + f_2 + f_3\}$ is also a linearly independent set.
6. In $\mathbb{R}^3$, give an example of 3 vectors $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ such that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly dependent but any set of 2 vectors from $\mathbf{u}, \mathbf{v}, \mathbf{w}$ is linearly independent.
7. What is the maximum number of linearly independent vectors in $\mathbb{R}^3$?
8. Show that any set of k vectors in $\mathbb{R}^3$ is linearly dependent if $k \geq 4$.
9. Is the set of vectors $(1,0), (i,0)$ linearly independent subset of $\mathbb{C}^2$ $(\mathbb{R})$?
10. Under what conditions on α are the vectors $(1+\alpha, 1-\alpha)$ and $(\alpha-1, 1+\alpha)$ in $\mathbb{C}^2(\mathbb{R})$ linearly independent?
11. Let $\mathbf{u}, \mathbf{v} \in V$ and M be a subspace of V. Further, let K be the subspace spanned by M and $\mathbf{u}$ and H be the subspace spanned by M and $\mathbf{v}$. Show that if $\mathbf{v} \in K$ and $\mathbf{v} \notin M$ then $\mathbf{u} \in H$.

3.3 Bases

Definition 3.3.1 (Basis of a Vector Space) 1. A non-empty subset $\mathcal{B}$ of a vector space V is called a basis of V if

(a) $\mathcal{B}$ is a linearly independent set, and

(b) $L(\mathcal{B}) = V$, i.e., every vector in V can be expressed as a linear combination of the elements of $\mathcal{B}$.

2. A vector in $\mathcal{B}$ is called a basis vector.

Remark 3.3.2 *Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ be a basis of a vector space $V(\mathbb{F})$. Then any $\mathbf{v} \in V$ is a* UNIQUE LINEAR COMBINATION *of the basis vectors, $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$.*

Observe that if there exists a $\mathbf{v} \in W$ such that $\mathbf{v} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_p\mathbf{v}_p$ and $\mathbf{v} = \beta_1\mathbf{v}_1 + \beta_2\mathbf{v}_2 + \cdots + \beta_p\mathbf{v}_p$ then

$$\mathbf{0} = \mathbf{v} - \mathbf{v} = (\alpha_1 - \beta_1)\mathbf{v}_1 + (\alpha_2 - \beta_2)\mathbf{v}_2 + \cdots + (\alpha_p - \beta_p)\mathbf{v}_p.$$

But then the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ is linearly independent and therefore the scalars $\alpha_i - \beta_i$ for $1 \le i \le p$ must all be equal to zero. Hence, for $1 \le i \le p$, $\alpha_i = \beta_i$ and we have the uniqueness.

By convention, the linear span of an empty set is $\{\mathbf{0}\}$. Hence, the empty set is a basis of the vector space $\{\mathbf{0}\}$.

Example 3.3.3

1. Check that if $V = \{(x, y, 0) : x, y \in \mathbb{R}\} \subset \mathbb{R}^3$, then $\mathcal{B} = \{(1, 0, 0), (0, 1, 0)\}$ or $\mathcal{B} = \{(1, 0, 0), (1, 1, 0)\}$ or $\mathcal{B} = \{(2, 0, 0), (1, 3, 0)\}$ or $\cdots$ are bases of V.
2. For $1 \le i \le n$, let $\mathbf{e}_i = (0, \ldots, 0, \underbrace{1}_{i \text{ th place}}, 0, \ldots, 0) \in \mathbb{R}^n$. Then, the set $\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ forms a basis of $\mathbb{R}^n$. This set is called the **standard basis** of $\mathbb{R}^n$.

   That is, if $n = 3$, then the set $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ forms an standard basis of $\mathbb{R}^3$.
3. Let $V = \{(x, y, z) : x+y-z = 0,\ x, y, z \in \mathbb{R}\}$ be a vector subspace of $\mathbb{R}^3$. Then $S = \{(1, 1, 2), (2, 1, 3), (1, 2, 3)\} \subset V$. It can be easily verified that the vector $(3, 2, 5) \in V$ and

   $$(3, 2, 5) = (1, 1, 2) + (2, 1, 3) = 4(1, 1, 2) - (1, 2, 3).$$

   Then by Remark 3.3.2, S cannot be a basis of V.

   A basis of V can be obtained by the following method:
   The condition $x + y - z = 0$ is equivalent to $z = x + y$. we replace the value of z with $x + y$ to get

   $$(x, y, z) = (x, y, x + y) = (x, 0, x) + (0, y, y) = x(1, 0, 1) + y(0, 1, 1).$$

   Hence, $\{(1, 0, 1), (0, 1, 1)\}$ forms a basis of V.
4. Let $V = \{a + ib : a, b \in \mathbb{R}\}$ and $\mathbb{F} = \mathbb{C}$. That is, V is a complex vector space. Note that any element $a + ib \in V$ can be written as $a + ib = (a + ib)\mathbf{1}$. Hence, a basis of V is $\{\mathbf{1}\}$.
5. Let $V = \{a + ib : a, b \in \mathbb{R}\}$ and $\mathbb{F} = \mathbb{R}$. That is, V is a real vector space. Any element $a + ib \in V$ is expressible as $a \cdot \mathbf{1} + b \cdot \mathbf{i}$. Hence a basis of V is $\{\mathbf{1}, \mathbf{i}\}$.

   Observe that i is a vector in $\mathbb{C}$. Also, $i \notin \mathbb{R}$ and hence $i \cdot (1 + 0 \cdot i)$ is not defined.
6. Recall the vector space $\mathcal{P}(\mathbb{R})$, the vector space of all polynomials with real coefficients. A basis of this vector space is the set

   $$\{1, x, x^2, \ldots, x^n, \ldots\}.$$

   This basis has infinite number of vectors as the degree of the polynomial can be any positive integer.

Definition 3.3.4 (Finite Dimensional Vector Space) A vector space V is said to be finite dimensional if there exists a basis consisting of finite number of elements. Otherwise, the vector space V is called infinite dimensional.

In Example 3.3.3, the vector space of all polynomials is an example of an infinite dimensional vector space. All the other vector spaces are finite dimensional.

Remark 3.3.5 *We can use the above results to obtain a basis of any finite dimensional vector space V as follows:*

Step 1: *Choose a non-zero vector, say, $\mathbf{v}_1 \in V$. Then the set $\{\mathbf{v}_1\}$ is linearly independent.*

Step 2: *If $V = L(\mathbf{v}_1)$, we have got a basis of V. Else there exists a vector, say, $\mathbf{v}_2 \in V$ such that $\mathbf{v}_2 \notin L(\mathbf{v}_1)$. Then by Corollary 3.2.6, the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent.*

Step 3: *If $V = L(\mathbf{v}_1, \mathbf{v}_2)$, then $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis of V. Else there exists a vector, say, $\mathbf{v}_3 \in V$ such that $\mathbf{v}_3 \notin L(\mathbf{v}_1, \mathbf{v}_2)$. So, by Corollary 3.2.6, the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.*

At the ith step, either $V = L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_i)$, or $L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_i) \neq V$.

In the first case, we have $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_i\}$ as a basis of V.

In the second case, $L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_i) \subset V$. So, we choose a vector, say, $\mathbf{v}_{i+1} \in V$ such that $\mathbf{v}_{i+1} \notin L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_i)$. Therefore, by Corollary 3.2.6, the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{i+1}\}$ is linearly independent.

This process will finally end as V is a finite dimensional vector space.

Exercise 3.3.6

1. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ be a subset of a vector space $V(\mathbb{F})$. Suppose $L(S) = V$ but S is not a linearly independent set. Then prove that each vector in V can be expressed in more than one way as a linear combination of vectors from S.
2. Show that the set $\{(1, 0, 1), (1, i, 0), (1, 1, 1 - i)\}$ is a basis of $\mathbb{C}^3(\mathbb{C})$.
3. Let A be a matrix of rank r. Then show that the r non-zero rows in the row-reduced echelon form of A are linearly independent and they form a basis of the row space of A.

3.3.1 Important Results

Theorem 3.3.7 Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis of a given vector space V. If $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ is a set of vectors from V with $m > n$ then this set is linearly dependent.

PROOF. Since we want to find whether the set $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ is linearly independent or not, we consider the linear system

$$\alpha_1\mathbf{w}_1 + \alpha_2\mathbf{w}_2 + \cdots + \alpha_m\mathbf{w}_m = \mathbf{0} \tag{3.3.1}$$

with $\alpha_1, \alpha_2, \ldots, \alpha_m$ as the m unknowns. If the solution set of this linear system of equations has more than one solution, then this set will be linearly dependent.

As $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis of V and $\mathbf{w}_i \in V$, for each i, $1 \le i \le m$, there exist scalars a_{ij}, $1 \le i \le n$, $1 \le j \le m$, such that

$$\begin{aligned}
\mathbf{w}_1 &= a_{11}\mathbf{v}_1 + a_{21}\mathbf{v}_2 + \cdots + a_{n1}\mathbf{v}_n \\
\mathbf{w}_2 &= a_{12}\mathbf{v}_1 + a_{22}\mathbf{v}_2 + \cdots + a_{n2}\mathbf{v}_n \\
&\vdots = \vdots \\
\mathbf{w}_m &= a_{1m}\mathbf{v}_1 + a_{2m}\mathbf{v}_2 + \cdots + a_{nm}\mathbf{v}_n.
\end{aligned}$$

The set of equations (3.3.1) can be rewritten as

$$\alpha_1\left(\sum_{j=1}^{n} a_{j1}\mathbf{v}_j\right) + \alpha_2\left(\sum_{j=1}^{n} a_{j2}\mathbf{v}_j\right) + \cdots + \alpha_m\left(\sum_{j=1}^{n} a_{jm}\mathbf{v}_j\right) = \mathbf{0}$$

$$\text{i.e.,}\quad \left(\sum_{i=1}^{m} \alpha_i a_{1i}\right)\mathbf{v}_1 + \left(\sum_{i=1}^{m} \alpha_i a_{2i}\right)\mathbf{v}_2 + \cdots + \left(\sum_{i=1}^{m} \alpha_i a_{ni}\right)\mathbf{v}_n = \mathbf{0}.$$

Since the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is linearly independent, we have

$$\sum_{i=1}^{m} \alpha_i a_{1i} = \sum_{i=1}^{m} \alpha_i a_{2i} = \cdots = \sum_{i=1}^{m} \alpha_i a_{ni} = 0.$$

Therefore, finding α_i's satisfying equation (3.3.1) reduces to solving the system of homogeneous equations $A\alpha = \mathbf{0}$ where $\alpha^t = (\alpha_1, \alpha_2, \ldots, \alpha_m)$ and $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}$. Since $n < m$, i.e., THE NUMBER OF EQUATIONS is strictly less than THE NUMBER OF UNKNOWNS, Corollary 2.6.3 implies that the solution set consists of infinite number of elements. Therefore, the equation (3.3.1) has a solution with not all α_i, $1 \le i \le m$, zero. Hence, the set $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ is a linearly dependent set. □

Remark 3.3.8 *Let V be a vector subspace of $\mathbb{R}^n$ with spanning set S. We give a method of finding a basis of V from S.*

1. *Construct a matrix A whose rows are the vectors in S.*
2. *Use only the elementary row operations $R_i(c)$ and $R_{ij}(c)$ to get the row-reduced form B of A (in fact we just need to make as many zero-rows as possible).*
3. *Let $\mathcal{B}$ be the set of vectors in S corresponding to the non-zero rows of B.*

Then the set $\mathcal{B}$ is a basis of $L(S) = V$.

Example 3.3.9 Let $S = \{(1, 1, 1, 1), (1, 1, -1, 1), (1, 1, 0, 1), (1, -1, 1, 1)\}$ be a subset of $\mathbb{R}^4$. Find a basis of $L(S)$.

Solution: Here $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & -1 & 1 & 1 \end{bmatrix}$. Applying row-reduction to A, we have

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & -1 & 1 & 1 \end{bmatrix} \xrightarrow[R_{12}(-1), R_{13}(-1), R_{14}(-1)]{} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -2 & 0 & 0 \end{bmatrix} \xrightarrow[R_{32}(-2)]{} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -2 & 0 & 0 \end{bmatrix}.$$

Observe that the rows 1, 3 and 4 are non-zero. Hence, a basis of $L(S)$ consists of the first, third and fourth vectors of the set S. Thus, $\mathcal{B} = \{(1, 1, 1, 1),\ (1, 1, 0, 1),\ (1, -1, 1, 1)\}$ is a basis of $L(S)$.

Observe that at the last step, in place of the elementary row operation $R_{32}(-2)$, we can apply $R_{23}(-\frac{1}{2})$ to make the third row as the zero-row. In this case, we get $\{(1, 1, 1, 1),\ (1, 1, -1, 1),\ (1, -1, 1, 1)\}$ as a basis of $L(S)$.

Corollary 3.3.10 Let V be a finite dimensional vector space. Then any two bases of V have the same number of vectors.

PROOF. Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ and $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\}$ be two bases of V with $m > n$. Then by the above theorem the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\}$ is linearly dependent if we take $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ as the basis of V. This contradicts the assumption that $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\}$ is also a basis of V. Hence, we get $m = n$. □

Definition 3.3.11 (Dimension of a Vector Space) The dimension of a finite dimensional vector space V is the number of vectors in a basis of V, denoted $\dim(V)$.

Note that the Corollary 3.2.6 can be used to generate a basis of ANY NON-TRIVIAL FINITE DIMENSIONAL VECTOR SPACE.

Example 3.3.12

1. Consider the complex vector space $\mathbb{C}^2(\mathbb{C})$. Then,

   $$(a + ib, c + id) = (a + ib)(1, 0) + (c + id)(0, 1).$$

   So, $\{(1, 0), (0, 1)\}$ is a basis of $\mathbb{C}^2(\mathbb{C})$ and thus $\dim(V) = 2$.
2. Consider the real vector space $\mathbb{C}^2(\mathbb{R})$. In this case, any vector

   $$(a + ib, c + id) = a(1, 0) + b(i, 0) + c(0, 1) + d(0, i).$$

   Hence, the set $\{(1, 0), (i, 0), (0, 1), (0, i)\}$ is a basis and $\dim(V) = 4$.

Remark 3.3.13 *It is important to note that the dimension of a vector space may change if the underlying field (the set of scalars) is changed.*

Example 3.3.14 Let V be the set of all functions $f : \mathbb{R}^n \longrightarrow \mathbb{R}$ with the property that $f(\mathbf{x}+\mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$ and $f(\alpha\mathbf{x}) = \alpha f(\mathbf{x})$. For $f, g \in V$, and $t \in \mathbb{R}$, define

$$\begin{aligned}
(f \oplus g)(\mathbf{x}) &= f(\mathbf{x}) + g(\mathbf{x}) \quad \text{and} \\
(t \odot f)(\mathbf{x}) &= f(t\mathbf{x}).
\end{aligned}$$

Then V is a real vector space.

For $1 \le i \le n$, consider the functions

$$\mathbf{e}_i(\mathbf{x}) = \mathbf{e}_i\big((x_1, x_2, \ldots, x_n)\big) = x_i.$$

Then it can be easily verified that the set $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ is a basis of V and hence $\dim(V) = n$.

The next theorem follows directly from Corollary 3.2.6 and Theorem 3.3.7. Hence, the proof is omitted.

Theorem 3.3.15 Let S be a linearly independent subset of a finite dimensional vector space V. Then the set S can be extended to form a basis of V.

Theorem 3.3.15 is equivalent to the following statement:
Let V be a vector space of dimension n. Suppose, we have found a linearly independent set $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r\} \subset V$. Then there exist vectors $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n$ in V such that $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis of V.

Corollary 3.3.16 Let V be a vector space of dimension n. Then any set of n linearly independent vectors forms a basis of V. Also, every set of m vectors, $m > n$, is linearly dependent.

Example 3.3.17 Let $V = \{(v, w, x, y, z) \in \mathbb{R}^5 \ : v + x - 3y + z = 0\}$ and $W = \{(v, w, x, y, z) \in \mathbb{R}^5 \ : w - x - z = 0, v = y\}$ be two subspaces of $\mathbb{R}^5$. Find bases of V and W containing a basis of $V \cap W$.
Solution: Let us find a basis of $V \cap W$. The solution set of the linear equations

$$v + x - 3y + z = 0, \quad w - x - z = 0 \quad \text{and} \quad v = y$$

is given by

$$(v, w, x, y, z)^t = (y, 2y, x, y, 2y - x)^t = y(1, 2, 0, 1, 2)^t + x(0, 0, 1, 0, -1)^t.$$

Thus, a basis of $V \cap W$ is

$$\{(1, 2, 0, 1, 2), (0, 0, 1, 0, -1)\}.$$

To find a basis of W containing a basis of $V \cap W$, we can proceed as follows:

1. Find a basis of W.
2. Take the basis of $V \cap W$ found above as the first two vectors and that of W as the next set of vectors.

   Now use Remark 3.3.8 to get the required basis.

Heuristically, we can also find the basis in the following way:
A vector of W has the form $(y, x+z, x, y, z)$ for $x, y, z \in \mathbb{R}$. Substituting $y = 1, x = 1$, and $z = 0$ in $(y, x+z, x, y, z)$ gives us the vector $(1, 1, 1, 1, 0) \in W$. It can be easily verified that a basis of W is

$$\{(1, 2, 0, 1, 2), (0, 0, 1, 0, -1), (1, 1, 1, 1, 0)\}.$$

Similarly, a vector of V has the form $(v, w, x, y, 3y-v-x)$ for $v, w, x, y \in \mathbb{R}$. Substituting $v = 0, w = 1, x = 0$ and $y = 0$, gives a vector $(0, 1, 0, 0, 0) \in V$. Also, substituting $v = 0, w = 1, x = 1$ and $y = 1$, gives another vector $(0, 1, 1, 1, 2) \in V$. So, a basis of V can be taken as

$$\{(1, 2, 0, 1, 2), (0, 0, 1, 0, -1), (0, 1, 0, 0, 0), (0, 1, 1, 1, 2)\}.$$

Recall that for two vector subspaces M and N of a vector space $V(\mathbb{F})$, the vector subspace $M + N$ is defined by

$$M + N = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in M,\ \mathbf{v} \in N\}.$$

With this definition, we have the following very important theorem (for a proof, see Appendix 14.4.1).

Theorem 3.3.18 Let $V(\mathbb{F})$ be a finite dimensional vector space and let M and N be two subspaces of V. Then

$$\dim(M) + \dim(N) = \dim(M + N) + \dim(M \cap N). \tag{3.3.2}$$

Exercise 3.3.19

1. Find a basis of the vector space $\mathcal{P}_n(\mathbb{R})$. Also, find $\dim(\mathcal{P}_n(\mathbb{R}))$. What can you say about the dimension of $\mathcal{P}(\mathbb{R})$?
2. Consider the real vector space, $C([0, 2\pi])$, of all real valued continuous functions. For each n consider the vector $\mathbf{e}_n$ defined by $\mathbf{e}_n(x) = \sin(nx)$. Prove that the collection of vectors $\{\mathbf{e}_n : 1 \le n < \infty\}$ is a linearly independent set.
   [*Hint: On the contrary, assume that the set is linearly dependent. Then we have a finite set of vectors, say* $\{\mathbf{e}_{k_1}, \mathbf{e}_{k_2}, \ldots, \mathbf{e}_{k_\ell}\}$ *that are linearly dependent. That is, there exist scalars* $\alpha_i \in \mathbb{R}$ *for* $1 \le i \le \ell$ *not all zero such that*

   $$\alpha_1 \sin(k_1 x) + \alpha_2 \sin(k_2 x) + \cdots + \alpha_\ell \sin(k_\ell x) = \mathbf{0} \quad \textit{for all} \quad x \in [0, 2\pi].$$

   Now for different values of m *integrate the function*

   $$\int_0^{2\pi} \sin(mx)\,(\alpha_1 \sin(k_1 x) + \alpha_2 \sin(k_2 x) + \cdots + \alpha_\ell \sin(k_\ell x))\ dx$$

   to get the required result.]
3. Show that the set $\{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ is a basis of $\mathbb{C}^3(\mathbb{C})$. Is it a basis of $\mathbb{C}^3(\mathbb{R})$ also?
4. Let $W = \{(x, y, z, w) \in \mathbb{R}^4 : x + y - z + w = 0\}$ be a subspace of $\mathbb{R}^4$. Find its basis and dimension.
5. Let $V = \{(x, y, z, w) \in \mathbb{R}^4 : x + y - z + w = 0, x + y + z + w = 0\}$ and $W = \{(x, y, z, w) \in \mathbb{R}^4 : x - y - z + w = 0, x + 2y - w = 0\}$ be two subspaces of $\mathbb{R}^4$. Find bases and dimensions of V, W, $V \cap W$ and $V + W$.
6. Let V be the set of all real symmetric $n \times n$ matrices. Find its basis and dimension. What if V is the complex vector space of all $n \times n$ Hermitian matrices?

7. If M and N are 4-dimensional subspaces of a vector space V of dimension 7 then show that M and N have at least one vector in common other than the zero vector.
8. Let $P = L\{(1, 0, 0), (1, 1, 0)\}$ and $Q = L\{(1, 1, 1)\}$ be vector subspaces of $\mathbb{R}^3$. Show that $P + Q = \mathbb{R}^3$ and $P \cap Q = \{\mathbf{0}\}$. If $\mathbf{u} \in \mathbb{R}^3$, determine $\mathbf{u}_P, \mathbf{u}_Q$ such that $\mathbf{u} = \mathbf{u}_P + \mathbf{u}_Q$ where $\mathbf{u}_P \in P$ and $\mathbf{u}_Q \in Q$. Is it necessary that $\mathbf{u}_P$ and $\mathbf{u}_Q$ are unique?
9. Let W_1 be a k-dimensional subspace of an n-dimensional vector space $V(\mathbb{F})$ where $k \ge 1$. Prove that there exists an $(n-k)$-dimensional subspace W_2 of V such that $W_1 \cap W_2 = \{\mathbf{0}\}$ and $W_1 + W_2 = V$.
10. Let P and Q be subspaces of $\mathbb{R}^n$ such that $P + Q = \mathbb{R}^n$ and $P \cap Q = \{\mathbf{0}\}$. Then show that each $\mathbf{u} \in \mathbb{R}^n$ can be uniquely expressed as $\mathbf{u} = \mathbf{u}_P + \mathbf{u}_Q$ where $\mathbf{u}_P \in P$ and $\mathbf{u}_Q \in Q$.
11. Let $P = L\{(1, -1, 0), (1, 1, 0)\}$ and $Q = L\{(1, 1, 1), (1, 2, 1)\}$ be vector subspaces of $\mathbb{R}^3$. Show that $P + Q = \mathbb{R}^3$ and $P \cap Q \ne \{\mathbf{0}\}$. Show that there exists a vector $\mathbf{u} \in \mathbb{R}^3$ such that $\mathbf{u}$ cannot be written uniquely in the form $\mathbf{u} = \mathbf{u}_P + \mathbf{u}_Q$ where $\mathbf{u}_P \in P$ and $\mathbf{u}_Q \in Q$.
12. Recall the vector space $\mathcal{P}_4(\mathbb{R})$. Is the set,

    $$W = \{p(x) \in \mathcal{P}_4(\mathbb{R})\ :\ p(-1) = p(1) = 0\}$$

    a subspace of $\mathcal{P}_4(\mathbb{R})$? If yes, find its dimension.
13. Let V be the set of all 2×2 matrices with complex entries and $a_{11} + a_{22} = 0$. Show that V is a real vector space. Find its basis. Also let $W = \{A \in V : a_{21} = \overline{-a_{12}}\}$. Show W is a vector subspace of V, and find its dimension.
14. Let $A = \begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 0 & 2 & 2 & 2 & 4 \\ 2 & -2 & 4 & 0 & 8 \\ 4 & 2 & 5 & 6 & 10 \end{bmatrix}$, and $B = \begin{bmatrix} 2 & 4 & 0 & 6 \\ -1 & 0 & -2 & 5 \\ -3 & -5 & 1 & -4 \\ -1 & -1 & 1 & 2 \end{bmatrix}$ be two matrices. For A and B find the following:
    (a) their row-reduced echelon forms.
    (b) the matrices P_1 and P_2 such that $P_1 A$ and $P_2 B$ are in row-reduced form.
    (c) a basis each for the row spaces of A and B.
    (d) a basis each for the range spaces of A and B.
    (e) bases of the null spaces of A and B.
    (f) the dimensions of all the vector subspaces so obtained.
15. Let $M(n, \mathbb{R})$ denote the space of all $n \times n$ real matrices. For the sets given below, check that they are subspaces of $M(n, \mathbb{R})$ and also find their dimension.
    (a) $sl(n, \mathbb{R}) = \{A \in M(n, \mathbb{R})\ :\ \mathrm{tr}(A) = 0\}$, where recall that $\mathrm{tr}(A)$ stands for trace of A.
    (b) $S(n, \mathbb{R}) = \{A \in M(n, \mathbb{R})\ :\ A = A^t\}$.
    (c) $A(n, \mathbb{R}) = \{A \in M(n, \mathbb{R})\ :\ A + A^t = \mathbf{0}\}$.

Before going to the next section, we prove that for any matrix A of order $m \times n$

$$\text{Row rank}(A) = \text{Column rank}(A).$$

Proposition 3.3.20 Let A be an $m \times n$ real matrix. Then

$$\text{Row rank}(A) = \text{Column rank}(A).$$

PROOF. Let $R_1, R_2, \ldots, R_m$ be the rows of A and $C_1, C_2, \ldots, C_n$ be the columns of A. Note that Row rank$(A) = r$, means that

$$\dim\big(L(R_1, R_2, \ldots, R_m)\big) = r.$$

Hence, there exists vectors

$$\mathbf{u}_1 = (u_{11}, \ldots, u_{1n}), \mathbf{u}_2 = (u_{21}, \ldots, u_{2n}), \ldots, \mathbf{u}_r = (u_{r1}, \ldots, u_{rn}) \in \mathbb{R}^n$$

with

$$R_i \in L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_r) \in \mathbb{R}^n, \ \text{ for all } \ i, 1 \le i \le m.$$

Therefore, there exist real numbers $\alpha_{ij},\ 1 \le i \le m,\ 1 \le j \le r$ such that

$$R_1 = \alpha_{11}\mathbf{u}_1 + \alpha_{12}\mathbf{u}_2 + \cdots + \alpha_{1r}\mathbf{u}_r = (\sum_{i=1}^{r} \alpha_{1i}u_{i1}, \sum_{i=1}^{r} \alpha_{1i}u_{i2}, \ldots, \sum_{i=1}^{r} \alpha_{1i}u_{in}),$$

$$R_2 = \alpha_{21}\mathbf{u}_1 + \alpha_{22}\mathbf{u}_2 + \cdots + \alpha_{2r}\mathbf{u}_r = (\sum_{i=1}^{r} \alpha_{2i}u_{i1}, \sum_{i=1}^{r} \alpha_{2i}u_{i2}, \ldots, \sum_{i=1}^{r} \alpha_{2i}u_{in}),$$

and so on, till

$$R_m = \alpha_{m1}\mathbf{u}_1 + \cdots + \alpha_{mr}\mathbf{u}_r = (\sum_{i=1}^{r} \alpha_{mi}u_{i1}, \sum_{i=1}^{r} \alpha_{mi}u_{i2}, \ldots, \sum_{i=1}^{r} \alpha_{mi}u_{in}).$$

So,

$$C_1 = \begin{bmatrix} \sum_{i=1}^{r} \alpha_{1i}u_{i1} \\ \sum_{i=1}^{r} \alpha_{2i}u_{i1} \\ \vdots \\ \sum_{i=1}^{r} \alpha_{mi}u_{i1} \end{bmatrix} = u_{11} \begin{bmatrix} \alpha_{11} \\ \alpha_{21} \\ \vdots \\ \alpha_{m1} \end{bmatrix} + u_{21} \begin{bmatrix} \alpha_{12} \\ \alpha_{22} \\ \vdots \\ \alpha_{m2} \end{bmatrix} + \cdots + u_{r1} \begin{bmatrix} \alpha_{1r} \\ \alpha_{2r} \\ \vdots \\ \alpha_{mr} \end{bmatrix}.$$

In general, for $1 \le j \le n$, we have

$$C_j = \begin{bmatrix} \sum_{i=1}^{r} \alpha_{1i}u_{ij} \\ \sum_{i=1}^{r} \alpha_{2i}u_{ij} \\ \vdots \\ \sum_{i=1}^{r} \alpha_{mi}u_{ij} \end{bmatrix} = u_{1j} \begin{bmatrix} \alpha_{11} \\ \alpha_{21} \\ \vdots \\ \alpha_{m1} \end{bmatrix} + u_{2j} \begin{bmatrix} \alpha_{12} \\ \alpha_{22} \\ \vdots \\ \alpha_{m2} \end{bmatrix} + \cdots + u_{rj} \begin{bmatrix} \alpha_{1r} \\ \alpha_{2r} \\ \vdots \\ \alpha_{mr} \end{bmatrix}.$$

Therefore, we observe that the columns $C_1, C_2, \ldots, C_n$ are linear combination of the r vectors

$$(\alpha_{11}, \alpha_{21}, \ldots, \alpha_{m1})^t, (\alpha_{12}, \alpha_{22}, \ldots, \alpha_{m2})^t, \ldots, (\alpha_{1r}, \alpha_{2r}, \ldots, \alpha_{mr})^t.$$

Therefore,

$$\text{Column rank}(A) = \dim\big(L(C_1, C_2, \ldots, C_n)\big) =\le r = \text{Row rank}(A).$$

A similar argument gives

$$\text{Row rank}(A) \le \text{Column rank}(A).$$

Thus, we have the required result. □

3.4 Ordered Bases

Let $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be a basis of a vector space $V(\mathbb{F})$. As $\mathcal{B}$ is a set, there is no ordering of its elements. In this section, we want to associate an order among the vectors in any basis of V.

Definition 3.4.1 (Ordered Basis) An ordered basis for a vector space $V(\mathbb{F})$ of dimension n, is a basis $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ together with a one-to-one correspondence between the sets $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ and $\{1, 2, 3, \ldots, n\}$.

If the ordered basis has $\mathbf{u}_1$ as the first vector, $\mathbf{u}_2$ as the second vector and so on, then we denote this ordered basis by

$$(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n).$$

Example 3.4.2 Consider $\mathcal{P}_2(\mathbb{R})$, the vector space of all polynomials of degree less than or equal to 2 with coefficients from $\mathbb{R}$. The set $\{1 - x, 1 + x, x^2\}$ is a basis of $\mathcal{P}_2(\mathbb{R})$.

For any element $a_0 + a_1 x + a_2 x^2 \in \mathcal{P}_2(\mathbb{R})$, we have

$$a_0 + a_1 x + a_2 x^2 = \frac{a_0 - a_1}{2}(1 - x) + \frac{a_0 + a_1}{2}(1 + x) + a_2 x^2.$$

If $(1-x, 1+x, x^2)$ is an ordered basis, then $\frac{a_0 - a_1}{2}$ is the first component, $\frac{a_0 + a_1}{2}$ is the second component, and a_2 is the third component of the vector $a_0 + a_1 x + a_2 x^2$.

If we take $(1 + x, 1 - x, x^2)$ as an ordered basis, then $\frac{a_0 + a_1}{2}$ is the first component, $\frac{a_0 - a_1}{2}$ is the second component, and a_2 is the third component of the vector $a_0 + a_1 x + a_2 x^2$.

That is, as ordered bases $(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n)$, $(\mathbf{u}_2, \mathbf{u}_3, \ldots, \mathbf{u}_n, \mathbf{u}_1)$, and $(\mathbf{u}_n, \mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{n-1})$ are different even though they have the same set of vectors as elements.

Definition 3.4.3 (Coordinates of a Vector) Let $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ be an ordered basis of a vector space $V(\mathbb{F})$ and let $\mathbf{v} \in V$. If

$$\mathbf{v} = \beta_1 \mathbf{v}_1 + \beta_2 \mathbf{v}_2 + \cdots + \beta_n \mathbf{v}_n$$

then the tuple $(\beta_1, \beta_2, \ldots, \beta_n)$ is called the coordinate of the vector $\mathbf{v}$ with respect to the ordered basis $\mathcal{B}$.

Mathematically, we denote it by $[\mathbf{v}]_{\mathcal{B}} = (\beta_1, \ldots, \beta_n)^t$, A COLUMN VECTOR.

Suppose $\mathcal{B}_1 = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n)$ and $\mathcal{B}_2 = (\mathbf{u}_n, \mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{n-1})$ are two ordered bases of V. Then for any $\mathbf{x} \in V$ there exists unique scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that

$$\mathbf{x} = \alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2 + \cdots + \alpha_n \mathbf{u}_n = \alpha_n \mathbf{u}_n + \alpha_1 \mathbf{u}_1 + \cdots + \alpha_{n-1} \mathbf{u}_{n-1}.$$

Therefore,

$$[\mathbf{x}]_{\mathcal{B}_1} = (\alpha_1, \alpha_2, \ldots, \alpha_n)^t \quad \text{and} \quad [\mathbf{x}]_{\mathcal{B}_2} = (\alpha_n, \alpha_1, \alpha_2, \ldots, \alpha_{n-1})^t.$$

Note that $\mathbf{x}$ is uniquely written as $\sum_{i=1}^{n} \alpha_i \mathbf{u}_i$ and hence the coordinates with respect to an ordered basis are unique.

Suppose that the ordered basis $\mathcal{B}_1$ is changed to the ordered basis $\mathcal{B}_3 = (\mathbf{u}_2, \mathbf{u}_1, \mathbf{u}_3, \ldots, \mathbf{u}_n)$. Then $[\mathbf{x}]_{\mathcal{B}_3} = (\alpha_2, \alpha_1, \alpha_3, \ldots, \alpha_n)^t$. So, the coordinates of a vector depend on the ordered basis chosen.

Example 3.4.4 Let $V = \mathbb{R}^3$. Consider the ordered bases
$\mathcal{B}_1 = \big((1, 0, 0), (0, 1, 0), (0, 0, 1)\big)$, $\mathcal{B}_2 = \big((1, 0, 0), (1, 1, 0), (1, 1, 1)\big)$ and $\mathcal{B}_3 = \big((1, 1, 1), (1, 1, 0), (1, 0, 0)\big)$ of V. Then, with respect to the above bases we have

$$\begin{aligned} (1, -1, 1) &= 1 \cdot (1, 0, 0) + (-1) \cdot (0, 1, 0) + 1 \cdot (0, 0, 1). \\ &= 2 \cdot (1, 0, 0) + (-2) \cdot (1, 1, 0) + 1 \cdot (1, 1, 1). \\ &= 1 \cdot (1, 1, 1) + (-2) \cdot (1, 1, 0) + 2 \cdot (1, 0, 0). \end{aligned}$$

Therefore, if we write $\mathbf{u} = (1,-1,1)$, then

$$[\mathbf{u}]_{\mathcal{B}_1} = (1,-1,1)^t,\ [\mathbf{u}]_{\mathcal{B}_2} = (2,-2,1)^t,\ [\mathbf{u}]_{\mathcal{B}_3} = (1,-2,2)^t.$$

In general, let V be an n-dimensional vector space with ordered bases $\mathcal{B}_1 = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n)$ and $\mathcal{B}_2 = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$. Since, $\mathcal{B}_1$ is a basis of V, there exists unique scalars $a_{ij},\ 1 \le i, j \le n$ such that

$$\mathbf{v}_i = \sum_{l=1}^{n} a_{li}\mathbf{u}_l \qquad \text{for } 1 \le i \le n.$$

That is, for each $i,\ 1 \le i \le n,\ [\mathbf{v}_i]_{\mathcal{B}_1} = (a_{1i}, a_{2i}, \ldots, a_{ni})^t$.

Let $\mathbf{v} \in V$ with $[\mathbf{v}]_{\mathcal{B}_2} = (\alpha_1, \alpha_2, \ldots, \alpha_n)^t$. As $\mathcal{B}_2$ as ordered basis $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$, we have

$$\mathbf{v} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i = \sum_{i=1}^{n} \alpha_i \left(\sum_{j=1}^{n} a_{ji}\mathbf{u}_j \right) = \sum_{j=1}^{n} \left(\sum_{i=1}^{n} a_{ji}\alpha_i \right) \mathbf{u}_j.$$

Since $\mathcal{B}_1$ is a basis this representation of $\mathbf{v}$ in terms of $\mathbf{u}_i$'s is unique. So,

$$\begin{aligned} [\mathbf{v}]_{\mathcal{B}_1} &= \left(\sum_{i=1}^{n} a_{1i}\alpha_i, \sum_{i=1}^{n} a_{2i}\alpha_i, \ldots, \sum_{i=1}^{n} a_{ni}\alpha_i \right)^t \\ &= \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} \\ &= A[\mathbf{v}]_{\mathcal{B}_2}. \end{aligned}$$

Note that the ith column of the matrix A is equal to $[\mathbf{v}_i]_{\mathcal{B}_1}$, i.e., the ith column of A is the coordinate of the ith vector $\mathbf{v}_i$ of $\mathcal{B}_2$ with respect to the ordered basis $\mathcal{B}_1$. Hence, we have proved the following theorem.

Theorem 3.4.5 Let V be an n-dimensional vector space with ordered bases $\mathcal{B}_1 = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n)$ and $\mathcal{B}_2 = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$. Let

$$A = [[\mathbf{v}_1]_{\mathcal{B}_1}, [\mathbf{v}_2]_{\mathcal{B}_1}, \ldots, [\mathbf{v}_n]_{\mathcal{B}_1}].$$

Then for any $\mathbf{v} \in V$,

$$[v]_{\mathcal{B}_1} = A[\mathbf{v}]_{\mathcal{B}_2}.$$

Example 3.4.6 Consider two bases $\mathcal{B}_1 = \big((1,0,0),(1,1,0),(1,1,1)\big)$ and $\mathcal{B}_2 = \big((1,1,1),(1,-1,1),(1,1,0)\big)$ of $\mathbb{R}^3$.

1. Then

$$\begin{aligned} [(x,y,z)]_{\mathcal{B}_1} &= (x-y)\cdot(1,0,0) + (y-z)\cdot(1,1,0) + z\cdot(1,1,1) \\ &= (x-y, y-z, z)^t \end{aligned}$$

and

$$\begin{aligned} [(x,y,z)]_{\mathcal{B}_2} &= (\frac{y-x}{2} + z)\cdot(1,1,1) + \frac{x-y}{2}\cdot(1,-1,1) \\ &\quad +(x-z)\cdot(1,1,0) \\ &= (\frac{y-x}{2} + z, \frac{x-y}{2}, x-z)^t. \end{aligned}$$

2. Let $A = [a_{ij}] = \begin{bmatrix} 0 & 2 & 0 \\ 0 & -2 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. The columns of the matrix A are obtained by the following rule:

$$[(1,1,1)]_{\mathcal{B}_1} = 0\cdot(1,0,0) + 0\cdot(1,1,0) + 1\cdot(1,1,1) = (0,0,1)^t,$$

$$[(1,-1,1)]_{\mathcal{B}_1} = 2\cdot(1,0,0) + (-2)\cdot(1,1,0) + 1\cdot(1,1,1) = (2,-2,1)^t$$

and

$$[(1,1,0)]_{\mathcal{B}_1} = 0\cdot(1,0,0) + 1\cdot(1,1,0) + 0\cdot(1,1,1) = (0,1,0)^t.$$

That is, the elements of $\mathcal{B}_2 = \big((1,1,1),(1,-1,1),(1,1,0)\big)$ are expressed in terms of the ordered basis $\mathcal{B}_1$.

3. Note that for any $(x,y,z) \in \mathbb{R}^3$,

$$[(x,y,z)]_{\mathcal{B}_1} = \begin{bmatrix} x-y \\ y-z \\ z \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 0 & -2 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{y-x}{2} + z \\ \frac{x-y}{2} \\ x-z \end{bmatrix} = A\ [(x,y,z)]_{\mathcal{B}_2}.$$

4. The matrix A is invertible and hence $[(x,y,z)]_{\mathcal{B}_2} = A^{-1}\ [(x,y,z)]_{\mathcal{B}_1}$.

In the next chapter, we try to understand Theorem 3.4.5 again using the ideas of 'linear transformations / functions'.

Exercise 3.4.7 1. Determine the coordinates of the vectors $(1,2,1)$ and $(4,-2,2)$ with respect to the basis $\mathcal{B} = \big((2,1,0),(2,1,1),(2,2,1)\big)$ of $\mathbb{R}^3$.

2. Consider the vector space $\mathcal{P}_3(\mathbb{R})$.
   (a) Show that $\mathcal{B}_1 = (1-x, 1+x^2, 1-x^3, 3+x^2-x^3)$ and $\mathcal{B}_2 = (1, 1-x, 1+x^2, 1-x^3)$ are bases of $\mathcal{P}_3(\mathbb{R})$.
   (b) Find the coordinates of the vector $\mathbf{u} = 1+x+x^2+x^3$ with respect to the ordered basis $\mathcal{B}_1$ and $\mathcal{B}_2$.
   (c) Find the matrix A such that $[\mathbf{u}]_{\mathcal{B}_2} = A[\mathbf{u}]_{\mathcal{B}_1}$.
   (d) Let $\mathbf{v} = a_0 + a_1x + a_2x^2 + a_3x^3$. Then verify the following:

$$\begin{aligned} [\mathbf{v}]_{\mathcal{B}_1} &= \begin{bmatrix} -a_1 \\ -a_0 - a_1 + 2a_2 - a_3 \\ -a_0 - a_1 + a_2 - 2a_3 \\ a_0 + a_1 - a_2 + a_3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 + a_1 - a_2 + a_3 \\ -a_1 \\ a_2 \\ -a_3 \end{bmatrix} \\ &= [\mathbf{v}]_{\mathcal{B}_2}. \end{aligned}$$

Chapter 4

Linear Transformations

4.1 Definitions and Basic Properties

Throughout this chapter, the scalar field $\mathbb{F}$ is either always the set $\mathbb{R}$ or always the set $\mathbb{C}$.

Definition 4.1.1 (Linear Transformation) Let V and W be vector spaces over $\mathbb{F}$. A map $T : V \longrightarrow W$ is called a linear transformation if

$$T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}), \qquad \text{for all } \alpha, \beta \in \mathbb{F}, \text{ and } \mathbf{u}, \mathbf{v} \in V.$$

We now give a few examples of linear transformations.

Example 4.1.2 1. Define $T : \mathbb{R} \longrightarrow \mathbb{R}^2$ by $T(x) = (x, 3x)$ for all $x \in \mathbb{R}$. Then T is a linear transformation as

$$T(x+y) = (x+y, 3(x+y)) = (x, 3x) + (y, 3y) = T(x) + T(y).$$

2. Verify that the maps given below from $\mathbb{R}^n$ to $\mathbb{R}$ are linear transformations. Let $\mathbf{x} = (x_1, x_2, \ldots, x_n)$.
   (a) Define $T(\mathbf{x}) = \sum_{i=1}^{n} x_i$.
   (b) For any $i,\ 1 \le i \le n$, define $T_i(\mathbf{x}) = x_i$.
   (c) For a fixed vector $\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \mathbb{R}^n$, define $T(\mathbf{x}) = \sum_{i=1}^{n} a_i x_i$. Note that examples (a) and (b) can be obtained by assigning particular values for the vector $\mathbf{a}$.

3. Define $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ by $T((x,y)) = (x+y, 2x-y, x+3y)$.
   Then T is a linear transformation with $T((1,0)) = (1,2,1)$ and $T((0,1)) = (1,-1,3)$.

4. Let A be an $m \times n$ real matrix. Define a map $T_A : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ by

$$T_A(\mathbf{x}) = A\mathbf{x} \quad \text{for every} \quad \mathbf{x}^t = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n.$$

   Then T_A is a linear transformation. That is, every $m \times n$ real matrix defines a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

5. Recall that $\mathcal{P}_n(\mathbb{R})$ is the set of all polynomials of degree less than or equal to n with real coefficients. Define $T : \mathbb{R}^{n+1} \longrightarrow \mathcal{P}_n(\mathbb{R})$ by

$$T((a_1, a_2, \ldots, a_{n+1})) = a_1 + a_2x + \cdots + a_{n+1}x^n$$

for $(a_1, a_2, \ldots, a_{n+1}) \in \mathbb{R}^{n+1}$. Then T is a linear transformation.

Proposition 4.1.3 Let $T : V \longrightarrow W$ be a linear transformation. Suppose that $\mathbf{0}_V$ is the zero vector in V and $\mathbf{0}_W$ is the zero vector of W. Then $T(\mathbf{0}_V) = \mathbf{0}_W$.

PROOF. Since $\mathbf{0}_V = \mathbf{0}_V + \mathbf{0}_V$, we have

$$T(\mathbf{0}_V) = T(\mathbf{0}_V + \mathbf{0}_V) = T(\mathbf{0}_V) + T(\mathbf{0}_V).$$

So, $T(\mathbf{0}_V) = \mathbf{0}_W$ as $T(\mathbf{0}_V) \in W$. □

From now on, we write $\mathbf{0}$ for both the zero vector of the domain space and the zero vector of the range space.

Definition 4.1.4 (Zero Transformation) Let V be a vector space and let $T : V \longrightarrow W$ be the map defined by

$$T(\mathbf{v}) = \mathbf{0} \ \text{ for every } \ \mathbf{v} \in V.$$

Then T is a linear transformation. Such a linear transformation is called the zero transformation and is denoted by $\mathbf{0}$.

Definition 4.1.5 (Identity Transformation) Let V be a vector space and let $T : V \longrightarrow V$ be the map defined by

$$T(\mathbf{v}) = \mathbf{v} \ \text{ for every } \ \mathbf{v} \in V.$$

Then T is a linear transformation. Such a linear transformation is called the Identity transformation and is denoted by I.

We now prove a result that relates a linear transformation T with its value on a basis of the domain space.

Theorem 4.1.6 Let $T : V \longrightarrow W$ be a linear transformation and $\mathcal{B} = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n)$ be an ordered basis of V. Then the linear transformation T is a linear combination of the vectors $T(\mathbf{u}_1), T(\mathbf{u}_2), \ldots, T(\mathbf{u}_n)$.
In other words, T is determined by $T(\mathbf{u}_1), T(\mathbf{u}_2), \ldots, T(\mathbf{u}_n)$.

PROOF. Since $\mathcal{B}$ is a basis of V, for any $\mathbf{x} \in V$, there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that $\mathbf{x} = \alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n$. So, by the definition of a linear transformation

$$T(\mathbf{x}) = T(\alpha_1\mathbf{u}_1 + \cdots + \alpha_n\mathbf{u}_n) = \alpha_1 T(\mathbf{u}_1) + \cdots + \alpha_n T(\mathbf{u}_n).$$

Observe that, given $\mathbf{x} \in V$, we know the scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$. Therefore, to know $T(\mathbf{x})$, we just need to know the vectors $T(\mathbf{u}_1), T(\mathbf{u}_2), \ldots, T(\mathbf{u}_n)$ in W.
That is, for every $\mathbf{x} \in V$, $T(\mathbf{x})$ is determined by the coordinates $(\alpha_1, \alpha_2, \ldots, \alpha_n)$ of $\mathbf{x}$ with respect to the ordered basis $\mathcal{B}$ and the vectors $T(\mathbf{u}_1), T(\mathbf{u}_2), \ldots, T(\mathbf{u}_n) \in W$. □

Exercise 4.1.7 1. Which of the following are linear transformations $T : V \longrightarrow W$? Justify your answers.
   (a) Let $V = \mathbb{R}^2$ and $W = \mathbb{R}^3$ with $T\big((x,y)\big) = (x+y+1, 2x-y, x+3y)$
   (b) Let $V = W = \mathbb{R}^2$ with $T\big((x,y)\big) = (x-y, x^2-y^2)$
   (c) Let $V = W = \mathbb{R}^2$ with $T\big((x,y)\big) = (x-y, |x|)$
   (d) Let $V = \mathbb{R}^2$ and $W = \longrightarrow \mathbb{R}^4$ with $T\big((x,y)\big) = (x+y, x-y, 2x+y, 3x-4y)$
   (e) Let $V = W = \mathbb{R}^4$ with $T\big((x,y,z,w)\big) = (z,x,w,y)$

2. Recall that $M_2(\mathbb{R})$ is the space of all 2×2 matrices with real entries. Then, which of the following are linear transformations $T : M_2(\mathbb{R}) \longrightarrow M_2(\mathbb{R})$?

(a) $T(A) = A^t$ (b) $T(A) = I + A$ (c) $T(A) = A^2$

(d) $T(A) = BAB^{-1}$, where B is some fixed 2×2 matrix.

3. Let $T : \mathbb{R} \longrightarrow \mathbb{R}$ be a map. Then T is a linear transformation if and only if there exists a unique $c \in \mathbb{R}$ such that $T(\mathbf{x}) = c\mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}$.

4. Let A be an $n \times n$ real matrix. Consider the linear transformation

$$T_A(\mathbf{x}) = A\mathbf{x} \quad \text{for every} \quad \mathbf{x} \in \mathbb{R}^n.$$

Then prove that $T^2(\mathbf{x}) := T(T(\mathbf{x})) = A^2\mathbf{x}$. In general, for $k \in \mathbb{N}$, prove that $T^k(\mathbf{x}) = A^k\mathbf{x}$.

5. Use the ideas of matrices to give examples of linear transformations $T, S : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ that satisfy:

(a) $T \neq \mathbf{0}$, $T^2 \neq \mathbf{0}$, $T^3 = \mathbf{0}$.

(b) $T \neq \mathbf{0}$, $S \neq \mathbf{0}$, $S \circ T \neq \mathbf{0}$, $T \circ S = \mathbf{0}$; where $T \circ S(\mathbf{x}) = T\big(S(\mathbf{x})\big)$.

(c) $S^2 = T^2$, $S \neq T$.

(d) $T^2 = I$, $T \neq I$.

6. Let $T : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be a linear transformation such that $T \neq \mathbf{0}$ and $T^2 = \mathbf{0}$. Let $\mathbf{x} \in \mathbb{R}^n$ such that $T(\mathbf{x}) \neq \mathbf{0}$. Then prove that the set $\{\mathbf{x}, T(\mathbf{x})\}$ is linearly independent. In general, if $T^k \neq \mathbf{0}$ for $1 \leq k \leq p$ and $T^{p+1} = \mathbf{0}$, then for any vector $\mathbf{x} \in \mathbb{R}^n$ with $T^p(\mathbf{x}) \neq \mathbf{0}$ prove that the set $\{\mathbf{x}, T(\mathbf{x}), \ldots, T^p(\mathbf{x})\}$ is linearly independent.

7. Let $T : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be a linear transformation, and let $\mathbf{x}_0 \in \mathbb{R}^n$ with $T(\mathbf{x}_0) = y$. Consider the sets

$$S = \{\mathbf{x} \in \mathbb{R}^n : T(\mathbf{x}) = \mathbf{y}\} \quad \text{and} \quad N = \{\mathbf{x} \in \mathbb{R}^n : T(\mathbf{x}) = \mathbf{0}\}.$$

Show that for every $\mathbf{x} \in S$ there exists $\mathbf{z} \in N$ such that $\mathbf{x} = \mathbf{x}_0 + z$.

8. Define a map $T : \mathbb{C} \longrightarrow \mathbb{C}$ by $T(z) = \overline{z}$, the complex conjugate of z. Is T linear on

(a) $\mathbb{C}$ over $\mathcal{R}$ (b) $\mathbb{C}$ over $\mathbb{C}$.

9. Find all functions $f : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ that satisfy the conditions

(a) $f(\ (x, x)\ ) = (x, x)$ and

(b) $f(\ (x, y)\ ) = (y, x)$ for all $(x, y) \in \mathbb{R}^2$.

That is, f fixes the line $y = x$ and sends the point (x_1, y_1) for $x_1 \neq y_1$ to its mirror image along the line $y = x$.

Is this function a linear transformation? Justify your answer.

Theorem 4.1.8 Let $T : V \longrightarrow W$ be a linear transformation. For $\mathbf{w} \in W$, define the set

$$T^{-1}(\mathbf{w}) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{w}\}.$$

Suppose that the map T is one-one and onto.

1. Then for each $\mathbf{w} \in W$, the set $T^{-1}(\mathbf{w})$ is a set consisting of a single element.

2. The map $T^{-1} : W \longrightarrow V$ defined by

$$T^{-1}(\mathbf{w}) = \mathbf{v} \ \text{ whenever } \ T(\mathbf{v}) = \mathbf{w}.$$

is a linear transformation.

PROOF. Since T is onto, for each $\mathbf{w} \in W$ there exists a vector $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{w}$. So, the set $T^{-1}(\mathbf{w})$ is non-empty.

Suppose there exist vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$ such that $T(\mathbf{v}_1) = T(\mathbf{v}_2)$. But by assumption, T is one-one and therefore $\mathbf{v}_1 = \mathbf{v}_2$. This completes the proof of Part 1.

We now show that T^{-1} as defined above is a linear transformation. Let $\mathbf{w}_1, \mathbf{w}_2 \in W$. Then by Part 1, there exist unique vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$ such that $T^{-1}(\mathbf{w}_1) = \mathbf{v}_1$ and $T^{-1}(\mathbf{w}_2) = \mathbf{v}_2$. Or equivalently, $T(\mathbf{v}_1) = \mathbf{w}_1$ and $T(\mathbf{v}_2) = \mathbf{w}_2$. So, for any $\alpha_1, \alpha_2 \in \mathbb{F}$, we have $T(\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2) = \alpha_1\mathbf{w}_1 + \alpha_2\mathbf{w}_2$.

Thus for any $\alpha_1, \alpha_2 \in \mathbb{F}$,

$$T^{-1}(\alpha_1\mathbf{w}_1 + \alpha_2\mathbf{w}_2) = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 = \alpha_1 T^{-1}(\mathbf{w}_1) + \alpha_2 T^{-1}(\mathbf{w}_2).$$

Hence $T^{-1} : W \longrightarrow V$, defined as above, is a linear transformation. □

Definition 4.1.9 (Inverse Linear Transformation) Let $T : V \longrightarrow W$ be a linear transformation. If the map T is one-one and onto, then the map $T^{-1} : W \longrightarrow V$ defined by

$$T^{-1}(\mathbf{w}) = \mathbf{v} \quad \text{whenever } T(\mathbf{v}) = \mathbf{w}$$

is called the inverse of the linear transformation T.

Example 4.1.10 1. Define $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ by $T((x, y)) = (x + y, x - y)$. Then $T^{-1} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ is defined by

$$T^{-1}((x, y)) = (\frac{x+y}{2}, \frac{x-y}{2}).$$

Note that

$$\begin{aligned} T \circ T^{-1}((x, y)) &= T(T^{-1}((x, y))) = T((\frac{x+y}{2}, \frac{x-y}{2})) \\ &= (\frac{x+y}{2} + \frac{x-y}{2}, \frac{x+y}{2} - \frac{x-y}{2}) \\ &= (x, y). \end{aligned}$$

Hence, $T \circ T^{-1} = I$, the identity transformation. Verify that $T^{-1} \circ T = I$. Thus, the map T^{-1} is indeed the inverse of the linear transformation T.

2. Recall the vector space $\mathcal{P}_n(\mathbb{R})$ and the linear transformation $T : \mathbb{R}^{n+1} \longrightarrow \mathcal{P}_n(\mathbb{R})$ defined by

$$T((a_1, a_2, \ldots, a_{n+1})) = a_1 + a_2x + \cdots + a_{n+1}x^n$$

for $(a_1, a_2, \ldots, a_{n+1}) \in \mathbb{R}^{n+1}$. Then $T^{-1} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathbb{R}^{n+1}$ is defined as

$$T^{-1}(a_1 + a_2x + \cdots + a_{n+1}x^n) = (a_1, a_2, \ldots, a_{n+1})$$

for $a_1 + a_2x + \cdots + a_{n+1}x^n \in \mathcal{P}_n(\mathbb{R})$. Verify that $T \circ T^{-1} = T^{-1} \circ T = I$. Hence, conclude that the map T^{-1} is indeed the inverse of the linear transformation T.

4.2 Matrix of a linear transformation

In this section, we relate linear transformation over finite dimensional vector spaces with matrices. For this, we ask the reader to recall the results on ordered basis, studied in Section 3.4.

Let V and W be finite dimensional vector spaces over the set $\mathbb{F}$ with respective dimensions m and n. Also, let $T : V \longrightarrow W$ be a linear transformation. Suppose $\mathcal{B}_1 = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ is an ORDERED BASIS of

V. In the last section, we saw that a linear transformation is determined by its image on a basis of the domain space. We therefore look at the images of the vectors $\mathbf{v}_j \in \mathcal{B}_1$ for $1 \leq j \leq n$.

Now for each j, $1 \leq j \leq n$, the vectors $T(\mathbf{v}_j) \in W$. We now express these vectors in terms of an ordered basis $\mathcal{B}_2 = (\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m)$ of W. So, for each j, $1 \leq j \leq n$, there exist unique scalars $a_{1j}, a_{2j}, \ldots, a_{mj} \in \mathbb{F}$ such that

$$\begin{aligned} T(\mathbf{v}_1) &= a_{11}\mathbf{w}_1 + a_{21}\mathbf{w}_2 + \cdots + a_{m1}\mathbf{w}_m \\ T(\mathbf{v}_2) &= a_{12}\mathbf{w}_1 + a_{22}\mathbf{w}_2 + \cdots + a_{m2}\mathbf{w}_m \\ &\vdots \\ T(\mathbf{v}_n) &= a_{1n}\mathbf{w}_1 + a_{2n}\mathbf{w}_2 + \cdots + a_{mn}\mathbf{w}_m. \end{aligned}$$

Or in short, $T(\mathbf{v}_j) = \sum_{i=1}^{m} a_{ij}\mathbf{w}_i$ for $1 \leq j \leq n$. In other words, for each j, $1 \leq j \leq n$, the coordinates of $T(\mathbf{v}_j)$ with respect to the ordered basis $\mathcal{B}_2$ is the column vector $[a_{1j}, a_{2j}, \ldots, a_{mj}]^t$. Equivalently,

$$[T(\mathbf{v}_j)]_{\mathcal{B}_2} = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}.$$

Let $[\mathbf{x}]_{\mathcal{B}_1} = [x_1, x_2, \ldots, x_n]^t$ be the coordinates of a vector $\mathbf{x} \in V$. Then

$$\begin{aligned} T(\mathbf{x}) &= T(\sum_{j=1}^{n} x_j\mathbf{v}_j) = \sum_{j=1}^{n} x_j T(\mathbf{v}_j) \\ &= \sum_{j=1}^{n} x_j (\sum_{i=1}^{m} a_{ij}\mathbf{w}_i) \\ &= \sum_{i=1}^{m} (\sum_{j=1}^{n} a_{ij}x_j)\mathbf{w}_i. \end{aligned}$$

Define a matrix A by $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$. Then the coordinates of the vector $T(\mathbf{x})$ with respect to the ordered basis $\mathcal{B}_2$ is

$$\begin{aligned} [T(\mathbf{x})]_{\mathcal{B}_2} &= \begin{bmatrix} \sum_{j=1}^{n} a_{1j}x_j \\ \sum_{j=1}^{n} a_{2j}x_j \\ \vdots \\ \sum_{j=1}^{n} a_{mj}x_j \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \\ &= A\ [\mathbf{x}]_{\mathcal{B}_1}. \end{aligned}$$

The matrix A is called the matrix of the linear transformation T with respect to the ordered bases $\mathcal{B}_1$ and $\mathcal{B}_2$, and is denoted by $T[\mathcal{B}_1, \mathcal{B}_2]$.

We thus have the following theorem.

Theorem 4.2.1 Let V and W be finite dimensional vector spaces with dimensions n and m, respectively. Let $T : V \longrightarrow W$ be a linear transformation. If $\mathcal{B}_1$ is an ordered basis of V and $\mathcal{B}_2$ is an ordered basis of W, then there exists an $m \times n$ matrix $A = T[\mathcal{B}_1, \mathcal{B}_2]$ such that

$$[T(\mathbf{x})]_{\mathcal{B}_2} = A\ [x]_{\mathcal{B}_1}.$$

Remark 4.2.2 *Let $\mathcal{B}_1 = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ be an ordered basis of V and $\mathcal{B}_2 = (\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m)$ be an ordered basis of W. Let $T : V \longrightarrow W$ be a linear transformation with $A = T[\mathcal{B}_1, \mathcal{B}_2]$. Then the first column of A is the coordinate of the vector $T(\mathbf{v}_1)$ in the basis $\mathcal{B}_2$. In general, the ith column of A is the coordinate of the vector $T(\mathbf{v}_i)$ in the basis $\mathcal{B}_2$.*

We now give a few examples to understand the above discussion and the theorem.

Example 4.2.3 1. Let $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be a linear transformation, given by

$$T(\ (x, y)\ ) = (x + y, x - y).$$

We obtain $T[\mathcal{B}_1, \mathcal{B}_2]$, the matrix of the linear transformation T with respect to the ordered bases

$$\mathcal{B}_1 = \big((1, 0), (0, 1)\big) \quad \text{and} \quad \mathcal{B}_2 = \big((1, 1), (1, -1)\big) \quad \text{of} \quad \mathbb{R}^2.$$

For any vector

$$(x, y) \in \mathbb{R}^2, \quad [(x, y)]_{\mathcal{B}_1} = \begin{bmatrix} x \\ y \end{bmatrix}$$

as $(x, y) = x(1, 0) + y(0, 1)$. Also, by definition of the linear transformation T, we have

$$T(\ (1, 0)\ ) = (1, 1) = 1 \cdot (1, 1) + 0 \cdot (1, -1). \ \text{ So, } [T(\ (1, 0)\ )]_{\mathcal{B}_2} = (1, 0)^t$$

and

$$T(\ (0, 1)\ ) = (1, -1) = 0 \cdot (1, 1) + 1 \cdot (1, -1).$$

That is, $[T(\ (0, 1)\ )]_{\mathcal{B}_2} = (0, 1)^t$. So the $T[\mathcal{B}_1, \mathcal{B}_2] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Observe that in this case,

$$[T(\ (x, y)\ )]_{\mathcal{B}_2} = [(x + y, x - y)]_{\mathcal{B}_2} = x(1, 1) + y(1, -1) = \begin{bmatrix} x \\ y \end{bmatrix}, \ \text{ and}$$

$$T[\mathcal{B}_1, \mathcal{B}_2]\ [(x, y)]_{\mathcal{B}_1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = [T(\ (x, y)\ )]_{\mathcal{B}_2}.$$

2. Let $\mathcal{B}_1 = \big((1, 0, 0), (0, 1, 0), (0, 0, 1)\big)$, $\mathcal{B}_2 = \big((1, 0, 0), (1, 1, 0), (1, 1, 1)\big)$ be two ordered bases of $\mathbb{R}^3$. Define

$$T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3 \quad \text{by} \quad T(\mathbf{x}) = \mathbf{x}.$$

Then

$$\begin{aligned} T((1, 0, 0)) &= 1 \cdot (1, 0, 0) + 0 \cdot (1, 1, 0) + 0 \cdot (1, 1, 1), \\ T((0, 1, 0)) &= -1 \cdot (1, 0, 0) + 1 \cdot (1, 1, 0) + 0 \cdot (1, 1, 1), \ \text{and} \\ T((0, 0, 1)) &= 0 \cdot (1, 0, 0) + (-1) \cdot (1, 1, 0) + 1 \cdot (1, 1, 1). \end{aligned}$$

Thus, we have

$$\begin{aligned} T[\mathcal{B}_1, \mathcal{B}_2] &= [[T((1, 0, 0))]_{\mathcal{B}_2},\ [T((0, 1, 0))]_{\mathcal{B}_2},\ [T((0, 0, 1))]_{\mathcal{B}_2}] \\ &= [(1, 0, 0)^t,\ (-1, 1, 0)^t,\ (0, -1, 1)^t] \\ &= \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

Similarly check that $T[\mathcal{B}_1, \mathcal{B}_1] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

3. Let $T:\mathbb{R}^3\longrightarrow\mathbb{R}^2$ be define by $T((x,y,z))=(x+y-z,x+z)$. Let $\mathcal{B}_1=\big((1,0,0),(0,1,0),(0,0,1)\big)$ and $\mathcal{B}_2=\big((1,0),(0,1)\big)$ be the ordered bases of the domain and range space, respectively. Then

$$T[\mathcal{B}_1,\mathcal{B}_2]=\begin{bmatrix}1&1&-1\\1&0&1\end{bmatrix}.$$

Check that that $[T(x,y,z)]_{\mathcal{B}_2}=T[\mathcal{B}_1,\mathcal{B}_2]\ \ [(x,y,z)]_{\mathcal{B}_1}$.

Exercise 4.2.4 Recall the space $\mathcal{P}_n(\mathbb{R})$ (the vector space of all polynomials of degree less than or equal to n). We define a linear transformation $D:\mathcal{P}_n(\mathbb{R})\longrightarrow\mathcal{P}_n(\mathbb{R})$ by

$$D(a_0+a_1x+a_2x^2+\cdots+a_nx^n)=a_1+2a_2x+\cdots+na_nx^{n-1}.$$

Find the matrix of the linear transformation D.

However, note that the image of the linear transformation is contained in $\mathcal{P}_{n-1}(\mathbb{R})$.

Remark 4.2.5 *1. Observe that*

$$T[\mathcal{B}_1,\mathcal{B}_2]=[[T(\mathbf{v}_1)]_{\mathcal{B}_2},[T(\mathbf{v}_2)]_{\mathcal{B}_2},\ldots,[T(\mathbf{v}_n)]_{\mathcal{B}_2}].$$

2. It is important to note that

$$[T(\mathbf{x})]_{\mathcal{B}_2}=T[\mathcal{B}_1,\mathcal{B}_2]\ [\mathbf{x}]_{\mathcal{B}_1}.$$

That is, we multiply the matrix of the linear transformation with the coordinates $[\mathbf{x}]_{\mathcal{B}_1}$, of the vector $\mathbf{x}\in V$ to obtain the coordinates of the vector $T(\mathbf{x})\in W$.

3. If A is an $m\times n$ matrix, then A induces a linear transformation $T_A:\mathbb{R}^n\longrightarrow\mathbb{R}^m$, defined by

$$T_A(\mathbf{x})=A\mathbf{x}.$$

We sometimes write A for T_A. Suppose that the standard bases for $\mathbb{R}^n$ and $\mathbb{R}^m$ are the ordered bases $\mathcal{B}_1$ and $\mathcal{B}_2$, respectively. Then observe that

$$T[\mathcal{B}_1,\mathcal{B}_2]=A.$$

4.3 Rank-Nullity Theorem

Definition 4.3.1 (Range and Null Space) Let V,W be finite dimensional vector spaces over the same set of scalars and $T:V\longrightarrow W$ be a linear transformation. We define

1. $\mathcal{R}(T)=\{T(\mathbf{x}):\mathbf{x}\in V\}$, and
2. $\mathcal{N}(T)=\{\mathbf{x}\in V:T(\mathbf{x})=\mathbf{0}\}$.

Proposition 4.3.2 Let V and W be finite dimensional vector spaces and let $T:V\longrightarrow W$ be a linear transformation. Suppose that $(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_n)$ is an ordered basis of V. Then

1. (a) $\mathcal{R}(T)$ is a subspace of W.
   (b) $\mathcal{R}(T)=L(T(\mathbf{v}_1),T(\mathbf{v}_2),\ldots,T(\mathbf{v}_n))$.
   (c) $\dim(\mathcal{R}(T))\le\dim(W)$.
2. (a) $\mathcal{N}(T)$ is a subspace of V.
   (b) $\dim(\mathcal{N}(T))\le\dim(V)$.

3. T is one-one $\iff$ $\mathcal{N}(T)=\{\mathbf{0}\}$ is the zero subspace of V $\iff$ $\{T(u_i):1\le i\le n\}$ is a basis of $\mathcal{R}(T)$.
4. $\dim(\mathcal{R}(T))=\dim(V)$ if and only if $\mathcal{N}(T)=\{\mathbf{0}\}$.

PROOF. The results about $\mathcal{R}(T)$ and $\mathcal{N}(T)$ can be easily proved. We thus leave the proof for the readers.

We now assume that T is one-one. We need to show that $\mathcal{N}(T)=\{\mathbf{0}\}$.

Let $\mathbf{u}\in\mathcal{N}(T)$. Then by definition, $T(\mathbf{u})=\mathbf{0}$. Also for any linear transformation (see Proposition 4.1.3), $T(\mathbf{0})=\mathbf{0}$. Thus $T(\mathbf{u})=T(\mathbf{0})$. So, T is one-one implies $\mathbf{u}=\mathbf{0}$. That is, $\mathcal{N}(T)=\{\mathbf{0}\}$.

Let $\mathcal{N}(T)=\{\mathbf{0}\}$. We need to show that T is one-one. So, let us assume that for some $\mathbf{u},\mathbf{v}\in V$, $T(\mathbf{u})=T(\mathbf{v})$. Then, by linearity of T, $T(\mathbf{u}-\mathbf{v})=\mathbf{0}$. This implies, $\mathbf{u}-\mathbf{v}\in\mathcal{N}(T)=\{\mathbf{0}\}$. This in turn implies $\mathbf{u}=\mathbf{v}$. Hence, T is one-one.

The other parts can be similarly proved. □

Remark 4.3.3 *1. The space $\mathcal{R}(T)$ is called the* RANGE SPACE *of T and $\mathcal{N}(T)$ is called the* NULL SPACE *of T.*

2. We write $\rho(T)=\dim(\mathcal{R}(T))$ and $\nu(T)=\dim(\mathcal{N}(T))$.

3. $\rho(T)$ is called the rank of the linear transformation T and $\nu(T)$ is called the nullity of T.

Example 4.3.4 Determine the range and null space of the linear transformation

$$T:\mathbb{R}^3\longrightarrow\mathbb{R}^4\quad\text{with}\quad T(x,y,z)=(x-y+z,y-z,x,2x-5y+5z).$$

Solution: By Definition $\mathcal{R}(T)=L(T(1,0,0),T(0,1,0),T(0,0,1))$. We therefore have

$$\begin{aligned}\mathcal{R}(T)&=L\big((1,0,1,2),(-1,1,0,-5),(1,-1,0,5)\big)\\&=L\big((1,0,1,2),(1,-1,0,5)\big)\\&=\{\alpha(1,0,1,2)+\beta(1,-1,0,5)\ :\alpha,\beta\in\mathbb{R}\}\\&=\{(\alpha+\beta,-\beta,\alpha,2\alpha+5\beta)\ :\alpha,\beta\in\mathbb{R}\}\\&=\{(x,y,z,w)\in\mathbb{R}^4\ :x+y-z=0,5y-2z+w=0\}.\end{aligned}$$

Also, by definition

$$\begin{aligned}\mathcal{N}(T)&=\{(x,y,z)\in\mathbb{R}^3\ :T(x,y,z)=\mathbf{0}\}\\&=\{(x,y,z)\in\mathbb{R}^3\ :(x-y+z,y-z,x,2x-5y+5z)=\mathbf{0}\}\\&=\{(x,y,z)\in\mathbb{R}^3\ :x-y+z=0,y-z=0,\\&\qquad\qquad x=0,2x-5y+5z=0\}\\&=\{(x,y,z)\in\mathbb{R}^3\ :y-z=0,x=0\}\\&=\{(x,y,z)\in\mathbb{R}^3\ :y=z,x=0\}\\&=\{(0,y,y)\in\mathbb{R}^3\ :y\ \text{arbitrary}\}\\&=L((0,1,1))\end{aligned}$$

Exercise 4.3.5 1. Let $T:V\longrightarrow W$ be a linear transformation and let $\{T(\mathbf{v}_1),T(\mathbf{v}_2),\ldots,T(\mathbf{v}_n)\}$ be linearly independent in $\mathcal{R}(T)$. Prove that $\{\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_n\}\subset V$ is linearly independent.

2. Let $T:\mathbb{R}^2\longrightarrow\mathbb{R}^3$ be defined by

$$T\big((1,0)\big)=(1,0,0),\ T\big((0,1)\big)=(1,0,0).$$

Then the vectors $(1,0)$ and $(0,1)$ are linearly independent whereas $T\big((1,0)\big)$ and $T\big((0,1)\big)$ are linearly dependent.

3. Is there a linear transformation

$$T:\mathbb{R}^3\longrightarrow\mathbb{R}^2\ \text{ such that }\ T(1,-1,1)=(1,2),\quad\text{and}\quad T(-1,1,2)=(1,0)?$$

4. Recall the vector space $\mathcal{P}_n(\mathbb{R})$. Define a linear transformation

$$D:\mathcal{P}_n(\mathbb{R})\longrightarrow\mathcal{P}_n(\mathbb{R})$$

by

$$D(a_0+a_1x+a_2x^2+\cdots+a_nx^n)=a_1+2a_2x+\cdots+na_nx^{n-1}.$$

Describe the null space and range space of D. Note that the range space is contained in the space $\mathcal{P}_{n-1}(\mathbb{R})$.

5. Let $T:\mathbb{R}^3\longrightarrow\mathbb{R}^3$ be defined by

$$T(1,0,0)=(0,0,1),\ \ T(1,1,0)=(1,1,1)\ \text{ and }\ T(1,1,1)=(1,1,0).$$

   (a) Find $T(x,y,z)$ for $x,y,z\in\mathbb{R}$,
   (b) Find $\mathcal{R}(T)$ and $\mathcal{N}(T)$. Also calculate $\rho(T)$ and $\nu(T)$.
   (c) Show that $T^3=T$ and find the matrix of the linear transformation with respect to the standard basis.

6. Let $T:\mathbb{R}^2\longrightarrow\mathbb{R}^2$ be a linear transformation with

$$T((3,4))=(0,1),\ T((-1,1))=(2,3).$$

Find the matrix representation $T[\mathcal{B},\mathcal{B}]$ of T with respect to the ordered basis $\mathcal{B}=\big((1,0),(1,1)\big)$ of $\mathbb{R}^2$.

7. Determine a linear transformation $T:\mathbb{R}^3\longrightarrow\mathbb{R}^3$ whose range space is $L\{(1,2,0),(0,1,1),(1,3,1)\}$.

8. Suppose the following chain of matrices is given.

$$A\longrightarrow B_1\longrightarrow B_1\longrightarrow B_2\cdots\longrightarrow B_{k-1}\longrightarrow B_k\longrightarrow B.$$

If row space of B is in the row space of B_k and the row space of B_l is in the row space of B_{l-1} for $2\le l\le k$ then show that the row space of B is in the row space of A.

We now state and prove the rank-nullity Theorem. This result also follows from Proposition 4.3.2.

Theorem 4.3.6 (Rank Nullity Theorem) Let $T:V\longrightarrow W$ be a linear transformation and V be a finite dimensional vector space. Then

$$\dim(\mathcal{R}(T))+\dim(\mathcal{N}(T))=\dim(V),$$

or equivalently $\rho(T)+\nu(T)=\dim(V)$.

PROOF. Let $\dim(V)=n$ and $\dim(\mathcal{N}(T))=r$. Suppose $\{u_1,u_2,\ldots,u_r\}$ is a basis of $\mathcal{N}(T)$. Since $\{u_1,u_2,\ldots,u_r\}$ is a linearly independent set in V, we can extend it to form a basis of V (see Corollary 3.3.15). So, there exist vectors $\{u_{r+1},u_{r+2},\ldots,u_n\}$ such that $\{u_1,\ldots,u_r,u_{r+1},\ldots,u_n\}$ is a basis of V. Therefore, by Proposition 4.3.2

$$\begin{aligned}\mathcal{R}(T)&=L(T(u_1),T(u_2),\ldots,T(u_n))\\&=L(\mathbf{0},\ldots,\mathbf{0},T(u_{r+1}),T(u_{r+2}),\ldots,T(u_n))\\&=L(T(u_{r+1}),T(u_{r+2}),\ldots,T(u_n)).\end{aligned}$$

We now prove that the set $\{T(u_{r+1}),T(u_{r+2}),\ldots,T(u_n)\}$ is linearly independent. Suppose the set is not linearly independent. Then, there exists scalars, $\alpha_{r+1},\alpha_{r+2},\ldots,\alpha_n$, not all zero such that

$$\alpha_{r+1}T(u_{r+1})+\alpha_{r+2}T(u_{r+2})+\cdots+\alpha_nT(u_n)=\mathbf{0}.$$

That is,

$$T(\alpha_{r+1}u_{r+1}+\alpha_{r+2}u_{r+2}+\cdots+\alpha_nu_n)=\mathbf{0}.$$

So, by definition of $\mathcal{N}(T)$,

$$\alpha_{r+1}u_{r+1}+\alpha_{r+2}u_{r+2}+\cdots+\alpha_nu_n\in\mathcal{N}(T)=L(u_1,\ldots,u_r).$$

Hence, there exists scalars α_i, $1\le i\le r$ such that

$$\alpha_{r+1}u_{r+1}+\alpha_{r+2}u_{r+2}+\cdots+\alpha_nu_n=\alpha_1u_1+\alpha_2u_2+\cdots+\alpha_ru_r.$$

That is,

$$\alpha_1u_1++\cdots+\alpha_ru_r-\alpha_{r+1}u_{r+1}-\cdots-\alpha_nu_n=\mathbf{0}.$$

But the set $\{u_1,u_2,\ldots,u_n\}$ is a basis of V and so linearly independent. Thus by definition of linear independence

$$\alpha_i=0\ \text{ for all }\ i,\ 1\le i\le n.$$

In other words, we have shown that $\{T(u_{r+1}),T(u_{r+2}),\ldots,T(u_n)\}$ is a basis of $\mathcal{R}(T)$. Hence,

$$\dim(\mathcal{R}(T))+\dim(\mathcal{N}(T))=(n-r)+r=n=\dim(V).$$

□

Using the Rank-nullity theorem, we give a short proof of the following result.

Corollary 4.3.7 Let $T:V\longrightarrow V$ be a linear transformation on a finite dimensional vector space V. Then

$$T\text{ is one-one }\iff T\text{ is onto}\iff T\text{ is invertible.}$$

PROOF. By Proposition 4.3.2, T is one-one if and only if $\mathcal{N}(T)=\{\mathbf{0}\}$. By the rank-nullity Theorem 4.3.6 $\mathcal{N}(T)=\{\mathbf{0}\}$ is equivalent to the condition $\dim(\mathcal{R}(T))=\dim(V)$. Or equivalently T is onto.

By definition, T is invertible if T is one-one and onto. But we have shown that T is one-one if and only if T is onto. Thus, we have the last equivalent condition. □

Remark 4.3.8 *Let V be a finite dimensional vector space and let $T:V\longrightarrow V$ be a linear transformation. If either T is one-one or T is onto, then T is invertible.*

The following are some of the consequences of the rank-nullity theorem. The proof is left as an exercise for the reader.

Corollary 4.3.9 The following are equivalent for an $m \times n$ real matrix A.

1. Rank $(A) = k$.
2. There exist exactly k rows of A that are linearly independent.
3. There exist exactly k columns of A that are linearly independent.
4. There is a $k \times k$ submatrix of A with non-zero determinant and every $(k+1) \times (k+1)$ submatrix of A has zero determinant.
5. The dimension of the range space of A is k.
6. There is a subset of $\mathbb{R}^m$ consisting of exactly k linearly independent vectors $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k$ such that the system $A\mathbf{x} = \mathbf{b}_i$ for $1 \leq i \leq k$ is consistent.
7. The dimension of the null space of $A = n - k$.

Exercise 4.3.10 1. Let $T : V \longrightarrow W$ be a linear transformation.

(a) If V is finite dimensional then show that the null space and the range space of T are also finite dimensional.

(b) If V and W are both finite dimensional then show that

i. if $\dim(V) < \dim(W)$ then T is onto.

ii. if $\dim(V) > \dim(W)$ then T is not one-one.

2. Let A be an $m \times n$ real matrix. Then

(a) if $n > m$, then the system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions,

(b) if $n < m$, then there exists a non-zero vector $\mathbf{b} = (b_1, b_2, \ldots, b_m)^t$ such that the system $A\mathbf{x} = \mathbf{b}$ does not have any solution.

3. Let A be an $m \times n$ matrix. Prove that
Row Rank (A) = Column Rank (A).
[Hint: Define $T_A : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ by $T_A(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$. Let Row Rank $(A) = r$. Use Theorem 2.6.1 to show, $A\mathbf{x} = \mathbf{0}$ has $n - r$ linearly independent solutions. This implies, $\nu(T_A) = \dim(\{\mathbf{v} \in \mathbb{R}^n : T_A(\mathbf{v}) = \mathbf{0}\}) = \dim(\{\mathbf{v} \in \mathbb{R}^n : A\mathbf{v} = \mathbf{0}\}) = n - r$. Now observe that $\mathcal{R}(T_A)$ is the linear span of columns of A and use the rank-nullity Theorem 4.3.6 to get the required result.]

4. Prove Theorem 2.6.1.
[Hint: Consider the linear system of equation $A\mathbf{x} = \mathbf{b}$ with the orders of $A, \mathbf{x}$ and $\mathbf{b}$, respectively as $m \times n, n \times 1$ and $m \times 1$. Define a linear transformation $T : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ by $T(\mathbf{v}) = A\mathbf{v}$. First observe that if the solution exists then $\mathbf{b}$ is a linear combination of the columns of A and the linear span of the columns of A give us $\mathcal{R}(T)$. Note that $\rho(A) =$ column rank$(A) = \dim(\mathcal{R}(T)) = \ell$(say). Then for part i) one can proceed as follows.
i) Let $C_{i_1}, C_{i_2}, \ldots, C_{i_t}$ be the linearly independent columns of A. Then rank(A) < rank$([A\ \mathbf{b}])$ implies that $\{C_{i_1}, C_{i_2}, \ldots, C_{i_t}, \mathbf{b}\}$ is linearly independent. Hence $\mathbf{b} \notin L(C_{i_1}, C_{i_2}, \ldots, C_{i_t})$. Hence, the system doesn't have any solution.
On similar lines prove the other two parts.]

5. Let $T, S : V \longrightarrow V$ be linear transformations with $\dim(V) = n$.

(a) Show that $\mathcal{R}(T+S) \subset \mathcal{R}(T) + \mathcal{R}(S)$. Deduce that $\rho(T+S) \leq \rho(T) + \rho(S)$.
Hint: For two subspaces M, N of a vector space V, recall the definition of the vector subspace $M + N$.

(b) Use the above and the rank-nullity Theorem 4.3.6 to prove $\nu(T+S) \geq \nu(T) + \nu(S) - n$.

6. Let V be the complex vector space of all complex polynomials of degree at most n. Given k distinct complex numbers $z_1, z_2, \ldots, z_k$, we define a linear transformation

$$T : V \longrightarrow \mathbb{C}^k \quad \text{by} \quad T\big(P(z)\big) = \big(P(z_1), P(z_2), \ldots, P(z_k)\big).$$

For each $k \geq 1$, determine the dimension of the range space of T.

7. Let A be an $n \times n$ real matrix with $A^2 = A$. Consider the linear transformation $T_A : \mathbb{R}^n \longrightarrow \mathbb{R}^n$, defined by $T_A(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$. Prove that

(a) $T_A \circ T_A = T_A$ (use the condition $A^2 = A$).

(b) $\mathcal{N}(T_A) \cap \mathcal{R}(T_A) = \{\mathbf{0}\}$.
Hint: Let $\mathbf{x} \in \mathcal{N}(T_A) \cap \mathcal{R}(T_A)$. This implies $T_A(\mathbf{x}) = \mathbf{0}$ and $\mathbf{x} = T_A(\mathbf{y})$ for some $\mathbf{y} \in \mathbb{R}^n$. So,

$$\mathbf{x} = T_A(\mathbf{y}) = (T_A \circ T_A)(\mathbf{y}) = T_A\big(T_A(\mathbf{y})\big) = T_A(\mathbf{x}) = \mathbf{0}.$$

(c) $\mathbb{R}^n = \mathcal{N}(T_A) + \mathcal{R}(T_A)$.
Hint: Let $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ be a basis of $\mathcal{N}(T_A)$. Extend it to get a basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_k, \mathbf{v}_{k+1}, \ldots, \mathbf{v}_n\}$ of $\mathbb{R}^n$. Then by Rank-nullity Theorem 4.3.6, $\{T_A(\mathbf{v}_{k+1}), \ldots, T_A(\mathbf{v}_n)\}$ is a basis of $\mathcal{R}(T_A)$.

4.4 Similarity of Matrices

In the last few sections, the following has been discussed in detail:
Given a finite dimensional vector space V of dimension n, we fixed an ordered basis $\mathcal{B}$. For any $\mathbf{v} \in V$, we calculated the column vector $[\mathbf{v}]_{\mathcal{B}}$, to obtain the coordinates of $\mathbf{v}$ with respect to the ordered basis $\mathcal{B}$. Also, for any linear transformation $T : V \longrightarrow V$, we got an $n \times n$ matrix $T[\mathcal{B}, \mathcal{B}]$, the matrix of T with respect to the ordered basis $\mathcal{B}$. That is, once an ordered basis of V is fixed, every linear transformation is represented by a matrix with entries from the scalars.

In this section, we understand the matrix representation of T in terms of different bases $\mathcal{B}_1$ and $\mathcal{B}_2$ of V. That is, we relate the two $n \times n$ matrices $T[\mathcal{B}_1, \mathcal{B}_1]$ and $T[\mathcal{B}_2, \mathcal{B}_2]$. We start with the following important theorem. This theorem also enables us to understand WHY THE MATRIX PRODUCT IS DEFINED SOMEWHAT DIFFERENTLY.

Theorem 4.4.1 (Composition of Linear Transformations) Let V, W and Z be finite dimensional vector spaces with ordered bases $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$, respectively. Also, let $T : V \longrightarrow W$ and $S : W \longrightarrow Z$ be linear transformations. Then the composition map $S \circ T : V \longrightarrow Z$ is a linear transformation and

$$(S \circ T)\,[\mathcal{B}_1, \mathcal{B}_3] = S[\mathcal{B}_2, \mathcal{B}_3]\ T[\mathcal{B}_1, \mathcal{B}_2].$$

PROOF. Let $\mathcal{B}_1 = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n)$, $\mathcal{B}_2 = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m)$ and $\mathcal{B}_3 = (\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_p)$ be ordered bases of V, W and Z, respectively. Then

$$(S \circ T)\,[\mathcal{B}_1, \mathcal{B}_3] = [[S \circ T(\mathbf{u}_1)]_{\mathcal{B}_3},\ [S \circ T(\mathbf{u}_2)]_{\mathcal{B}_3}, \ldots, [S \circ T(\mathbf{u}_n)]_{\mathcal{B}_3}].$$

Now for $1 \leq t \leq n$,

$$\begin{aligned}
(S \circ T)\,(\mathbf{u}_t) &= S(T(\mathbf{u}_t)) = S\Big(\sum_{j=1}^{m} (T[\mathcal{B}_1, \mathcal{B}_2])_{jt} \mathbf{v}_j\Big) = \sum_{j=1}^{m} (T[\mathcal{B}_1, \mathcal{B}_2])_{jt} S(\mathbf{v}_j) \\
&= \sum_{j=1}^{m} (T[\mathcal{B}_1, \mathcal{B}_2])_{jt} \sum_{k=1}^{p} (S[\mathcal{B}_2, \mathcal{B}_3])_{kj} \mathbf{w}_k \\
&= \sum_{k=1}^{p} \Big(\sum_{j=1}^{m} (S[\mathcal{B}_2, \mathcal{B}_3])_{kj} (T[\mathcal{B}_1, \mathcal{B}_2])_{jt}\Big) \mathbf{w}_k \\
&= \sum_{k=1}^{p} (S[\mathcal{B}_2, \mathcal{B}_3]\ T[\mathcal{B}_1, \mathcal{B}_2])_{kt} \mathbf{w}_k.
\end{aligned}$$

So,

$$[(S \circ T)\,(\mathbf{u}_t)]_{\mathcal{B}_3} = ((S[\mathcal{B}_2, \mathcal{B}_3]\ T[\mathcal{B}_1, \mathcal{B}_2])_{1t}, \ldots, (S[\mathcal{B}_2, \mathcal{B}_3]\ T[\mathcal{B}_1, \mathcal{B}_2])_{pt})^t.$$

Hence,

$$(S \circ T)\,[\mathcal{B}_1, \mathcal{B}_3] = \big[\,[(S \circ T)\,(u_1)]_{\mathcal{B}_3}, \ldots, [(S \circ T)\,(u_n)]_{\mathcal{B}_3}\big] = S[\mathcal{B}_2, \mathcal{B}_3]\ T[\mathcal{B}_1, \mathcal{B}_2].$$

This completes the proof. □

Proposition 4.4.2 Let V be a finite dimensional vector space and let $T, S : V \longrightarrow V$ be a linear transformations. Then

$$\nu(T) + \nu(S) \geq \nu(T \circ S) \geq \max\{\nu(T), \nu(S)\}.$$

PROOF. We first prove the second inequality.
Suppose that $\mathbf{v} \in \mathcal{N}(S)$. Then $T \circ S(\mathbf{v}) = T(S(\mathbf{v})) = T(\mathbf{0}) = \mathbf{0}$. So, $\mathcal{N}(S) \subset \mathcal{N}(T \circ S)$. Therefore, $\nu(S) \leq \nu(T \circ S)$.

Suppose $\dim(V) = n$. Then using the rank-nullity theorem, observe that

$$\nu(T \circ S) \geq \nu(T) \Longleftrightarrow n - \nu(T \circ S) \leq n - \nu(T) \Longleftrightarrow \rho(T \circ S) \leq \rho(T).$$

So, to complete the proof of the second inequality, we need to show that $\mathcal{R}(T \circ S) \subset \mathcal{R}(T)$. This is true as $\mathcal{R}(S) \subset V$.

We now prove the first inequality.
Let $k = \nu(S)$ and let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ be a basis of $\mathcal{N}(S)$. Clearly, $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\} \subset \mathcal{N}(T \circ S)$ as $T(\mathbf{0}) = \mathbf{0}$. We extend it to get a basis $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k, \mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_\ell\}$ of $\mathcal{N}(T \circ S)$.

Claim: The set $\{S(\mathbf{u}_1), S(\mathbf{u}_2), \ldots, S(\mathbf{u}_\ell)\}$ is linearly independent subset of $\mathcal{N}(T)$.

As $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_\ell \in \mathcal{N}(T \circ S)$, the set $\{S(\mathbf{u}_1), S(\mathbf{u}_2), \ldots, S(\mathbf{u}_\ell)\}$ is a subset of $\mathcal{N}(T)$. Let if possible the given set be linearly dependent. Then there exist non-zero scalars $c_1, c_2, \ldots, c_\ell$ such that

$$c_1 S(\mathbf{u}_1) + c_2 S(\mathbf{u}_2) + \cdots + c_\ell S(\mathbf{u}_\ell) = \mathbf{0}.$$

So, the vector $\sum_{i=1}^{\ell} c_i \mathbf{u}_i \in \mathcal{N}(S)$ and is a linear combination of the basis vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ of $\mathcal{N}(S)$. Therefore, there exist scalars $\alpha_1, \alpha_2, \alpha_k$ such that

$$\sum_{i=1}^{\ell} c_i \mathbf{u}_i = \sum_{i=1}^{k} \alpha_i \mathbf{v}_i.$$

Or equivalently

$$\sum_{i=1}^{\ell} c_i \mathbf{u}_i + \sum_{i=1}^{k} (-\alpha_i) \mathbf{v}_i = \mathbf{0}.$$

That is, the $\mathbf{0}$ vector is a non-trivial linear combination of the basis vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k, \mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_\ell$ of $\mathcal{N}(T \circ S)$. A contradiction.

Thus, the set $\{S(\mathbf{u}_1), S(\mathbf{u}_2), \ldots, S(\mathbf{u}_\ell)\}$ is a linearly independent subset of $\mathcal{N}(T)$ and so $\nu(T) \geq \ell$. Hence,

$$\nu(T \circ S) = k + \ell \leq \nu(S) + \nu(T).$$

□

Recall from Theorem 4.1.8 that if T is an invertible linear Transformation, then $T^{-1} : V \longrightarrow V$ is a linear transformation defined by $T^{-1}(\mathbf{u}) = \mathbf{v}$ whenever $T(\mathbf{v}) = \mathbf{u}$. We now state an important result about inverse of a linear transformation. The reader is required to supply the proof (use Theorem 4.4.1).

Theorem 4.4.3 (Inverse of a Linear Transformation) Let V be a finite dimensional vector space with ordered bases $\mathcal{B}_1$ and $\mathcal{B}_2$. Also let $T : V \longrightarrow V$ be an invertible linear transformation. Then the matrix of T and T^{-1} are related by

$$T[\mathcal{B}_1, \mathcal{B}_2]^{-1} = T^{-1}[\mathcal{B}_2, \mathcal{B}_1].$$

Exercise 4.4.4 For the linear transformations given below, find the matrix $T[\mathcal{B}, \mathcal{B}]$.

1. Let $\mathcal{B} = \big((1,1,1), (1,-1,1), (1,1,-1)\big)$ be an ordered basis of $\mathbb{R}^3$. Define $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ by $T(1,1,1) = (1,-1,1)$, $T(1,-1,1) = (1,1,-1)$, and $T(1,1,-1) = (1,1,1)$. Is T an invertible linear transformation? Give reasons.

2. Let $\mathcal{B} = \big(1, x, x^2, x^3\big)$ be an ordered basis of $\mathcal{P}_3(\mathbb{R})$. Define $T : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathcal{P}_3(\mathbb{R})$ by
$$T(1) = 1, T(x) = 1 + x, T(x^2) = (1+x)^2, \text{ and } T(x^3) = (1+x)^3.$$
Prove that T is an invertible linear transformation. Also, find $T^{-1}[\mathcal{B}, \mathcal{B}]$.

Let V be a vector space with $\dim(V) = n$. Let $\mathcal{B}_1 = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n)$ and $\mathcal{B}_2 = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be two ordered bases of V. Recall from Definition 4.1.5 that $I : V \longrightarrow V$ is the identity linear transformation defined by $I(\mathbf{x}) = \mathbf{x}$ for every $\mathbf{x} \in V$. Suppose $\mathbf{x} \in V$ with $[\mathbf{x}]_{\mathcal{B}_1} = (\alpha_1, \alpha_2, \ldots, \alpha_n)^t$ and $[\mathbf{x}]_{\mathcal{B}_2} = (\beta_1, \beta_2, \ldots, \beta_n)^t$.

We now express each vector in $\mathcal{B}_2$ as a linear combination of the vectors from $\mathcal{B}_1$. Since $\mathbf{v}_i \in V$, for $1 \leq i \leq n$, and $\mathcal{B}_1$ is a basis of V, we can find scalars $a_{ij}, 1 \leq i, j \leq n$ such that

$$\mathbf{v}_i = I(\mathbf{v}_i) = \sum_{j=1}^{n} a_{ji} \mathbf{u}_j \quad \text{for all } i, 1 \leq i \leq n.$$

Hence, $[I(\mathbf{v}_i)]_{\mathcal{B}_1} = [\mathbf{v}_i]_{\mathcal{B}_1} = (a_{1i}, a_{2i}, \cdots, a_{ni})^t$ and

$$\begin{aligned}
I[\mathcal{B}_2, \mathcal{B}_1] &= [[I(\mathbf{v}_1)]_{\mathcal{B}_1}, [I(\mathbf{v}_2)]_{\mathcal{B}_1}, \ldots, [I(\mathbf{v}_n)]_{\mathcal{B}_1}] \\
&= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}.
\end{aligned}$$

Thus, we have proved the following result.

Theorem 4.4.5 (Change of Basis Theorem) Let V be a finite dimensional vector space with ordered bases $\mathcal{B}_1 = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ and $\mathcal{B}_2 = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$. Suppose $\mathbf{x} \in V$ with $[\mathbf{x}]_{\mathcal{B}_1} = (\alpha_1, \alpha_2, \ldots, \alpha_n)^t$ and $[\mathbf{x}]_{\mathcal{B}_2} = (\beta_1, \beta_2, \ldots, \beta_n)^t$. Then

$$[\mathbf{x}]_{\mathcal{B}_1} = I[\mathcal{B}_2, \mathcal{B}_1]\ [\mathbf{x}]_{\mathcal{B}_2}.$$

Equivalently,

$$\begin{bmatrix}\alpha_1\\ \alpha_2\\ \vdots\\ \alpha_n\end{bmatrix} = \begin{bmatrix}a_{11} & a_{12} & \cdots & a_{1n}\\ a_{21} & a_{22} & \cdots & a_{2n}\\ \vdots & \vdots & \ddots & \vdots\\ a_{n1} & a_{n2} & \cdots & a_{nn}\end{bmatrix}\begin{bmatrix}\beta_1\\ \beta_2\\ \vdots\\ \beta_n\end{bmatrix}.$$

Note: Observe that the identity linear transformation $I : V \longrightarrow V$ defined by $I(\mathbf{x}) = \mathbf{x}$ for every $\mathbf{x} \in V$ is invertible and

$$I[\mathcal{B}_2, \mathcal{B}_1]^{-1} = I^{-1}[\mathcal{B}_1, \mathcal{B}_2] = I[\mathcal{B}_1, \mathcal{B}_2].$$

Therefore, we also have

$$[\mathbf{x}]_{\mathcal{B}_2} = I[\mathcal{B}_1, \mathcal{B}_2]\,[\mathbf{x}]_{\mathcal{B}_1}.$$

Let V be a finite dimensional vector space and let $\mathcal{B}_1$ and $\mathcal{B}_2$ be two ordered bases of V. Let $T : V \longrightarrow V$ be a linear transformation. We are now in a position to relate the two matrices $T[\mathcal{B}_1, \mathcal{B}_1]$ and $T[\mathcal{B}_2, \mathcal{B}_2]$.

Theorem 4.4.6 Let V be a finite dimensional vector space and let $\mathcal{B}_1 = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n)$ and $\mathcal{B}_2 = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ be two ordered bases of V. Let $T : V \longrightarrow V$ be a linear transformation with $B = T[\mathcal{B}_1, \mathcal{B}_1]$ and $C = T[\mathcal{B}_2, \mathcal{B}_2]$ as matrix representations of T in bases $\mathcal{B}_1$ and $\mathcal{B}_2$.

Also, let $A = [a_{ij}] = I[\mathcal{B}_2, \mathcal{B}_1]$, be the matrix of the identity linear transformation with respect to the bases $\mathcal{B}_1$ and $\mathcal{B}_2$. Then $BA = AC$. Equivalently $B = ACA^{-1}$.

PROOF. For any $\mathbf{x} \in V$, we represent $[T(\mathbf{x})]_{\mathcal{B}_2}$ in two ways. Using Theorem 4.2.1, the first expression is

$$[T(\mathbf{x})]_{\mathcal{B}_2} = T[\mathcal{B}_2, \mathcal{B}_2]\,[\mathbf{x}]_{\mathcal{B}_2}. \tag{4.4.1}$$

Using Theorem 4.4.5, the other expression is

$$\begin{aligned}[T(\mathbf{x})]_{\mathcal{B}_2} &= I[\mathcal{B}_1, \mathcal{B}_2]\,[T(\mathbf{x})]_{\mathcal{B}_1}\\ &= I[\mathcal{B}_1, \mathcal{B}_2]\,T[\mathcal{B}_1, \mathcal{B}_1]\,[\mathbf{x}]_{\mathcal{B}_1}\\ &= I[\mathcal{B}_1, \mathcal{B}_2]\,T[\mathcal{B}_1, \mathcal{B}_1]\,I[\mathcal{B}_2, \mathcal{B}_1]\,[\mathbf{x}]_{\mathcal{B}_2}.\end{aligned} \tag{4.4.2}$$

Hence, using (4.4.1) and (4.4.2), we see that for every $\mathbf{x} \in V$,

$$I[\mathcal{B}_1, \mathcal{B}_2]\,T[\mathcal{B}_1, \mathcal{B}_1]\,I[\mathcal{B}_2, \mathcal{B}_1]\,[\mathbf{x}]_{\mathcal{B}_2} = T[\mathcal{B}_2, \mathcal{B}_2]\,[\mathbf{x}]_{\mathcal{B}_2}.$$

Since the result is true for all $\mathbf{x} \in V$, we get

$$I[\mathcal{B}_1, \mathcal{B}_2]\,T[\mathcal{B}_1, \mathcal{B}_1]\,I[\mathcal{B}_2, \mathcal{B}_1] = T[\mathcal{B}_2, \mathcal{B}_2]. \tag{4.4.3}$$

That is, $A^{-1}BA = C$ or equivalently $ACA^{-1} = B$. □

Another Proof:

Let $B = [b_{ij}]$ and $C = [c_{ij}]$. Then for $1 \le i \le n$,

$$T(\mathbf{u}_i) = \sum_{j=1}^{n} b_{ji}\mathbf{u}_j \;\text{ and }\; T(\mathbf{v}_i) = \sum_{j=1}^{n} c_{ji}\mathbf{v}_j.$$

So, for each $j, 1 \le j \le n$,

$$\begin{aligned}T(\mathbf{v}_j) &= T(I(\mathbf{v}_j)) = T(\sum_{k=1}^{n} a_{kj}\mathbf{u}_k) = \sum_{k=1}^{n} a_{kj}T(\mathbf{u}_k)\\ &= \sum_{k=1}^{n} a_{kj}(\sum_{\ell=1}^{n} b_{\ell k}\mathbf{u}_\ell) = \sum_{\ell=1}^{n}(\sum_{k=1}^{n} b_{\ell k}a_{kj})\mathbf{u}_\ell\end{aligned}$$

and therefore,

$$[T(\mathbf{v}_j)]_{\mathcal{B}_1} = \begin{bmatrix}\sum_{k=1}^{n} b_{1k}a_{kj}\\ \sum_{k=1}^{n} b_{2k}a_{kj}\\ \vdots\\ \sum_{k=1}^{n} b_{nk}a_{kj}\end{bmatrix} = B\begin{bmatrix}a_{1j}\\ a_{2j}\\ \vdots\\ a_{nj}\end{bmatrix}.$$

Hence $T[\mathcal{B}_2, \mathcal{B}_1] = BA$.

Also, for each $j, 1 \le j \le n$,

$$\begin{aligned}T(\mathbf{v}_j) &= \sum_{k=1}^{n} c_{kj}\mathbf{v}_k = \sum_{k=1}^{n} c_{kj}I(\mathbf{v}_k) = \sum_{k=1}^{n} c_{kj}(\sum_{\ell=1}^{n} a_{\ell k}\mathbf{u}_\ell)\\ &= \sum_{\ell=1}^{n}(\sum_{k=1}^{n} a_{\ell k}c_{kj})\mathbf{u}_\ell\end{aligned}$$

and so

$$[T(\mathbf{v}_j)]_{\mathcal{B}_1} = \begin{bmatrix}\sum_{k=1}^{n} a_{1k}c_{kj}\\ \sum_{k=1}^{n} a_{2k}c_{kj}\\ \vdots\\ \sum_{k=1}^{n} a_{nk}c_{kj}\end{bmatrix} = A\begin{bmatrix}c_{1j}\\ c_{2j}\\ \vdots\\ c_{nj}\end{bmatrix}.$$

This gives us $T[\mathcal{B}_2, \mathcal{B}_1] = AC$. We thus have $AC = T[\mathcal{B}_2, \mathcal{B}_1] = BA$. ■

Let V be a vector space with $\dim(V) = n$, and let $T : V \longrightarrow V$ be a linear transformation. Then for each ordered basis $\mathcal{B}$ of V, we get an $n \times n$ matrix $T[\mathcal{B}, \mathcal{B}]$. Also, we know that for any vector space we have infinite number of choices for an ordered basis. So, as we change an ordered basis, the matrix of the linear transformation changes. Theorem 4.4.6 tells us that all these matrices are related.

Now, let A and B be two $n \times n$ matrices such that $P^{-1}AP = B$ for some invertible matrix P. Recall the linear transformation $T_A : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. Then we have seen that if the standard basis of $\mathbb{R}^n$ is the ordered basis $\mathcal{B}$, then $A = T_A[\mathcal{B}, \mathcal{B}]$. Since P is an invertible matrix, its columns are linearly independent and hence we can take its columns as an ordered basis $\mathcal{B}_1$. Then note that $B = T_A[\mathcal{B}_1, \mathcal{B}_1]$. The above observations lead to the following remark and the definition.

Remark 4.4.7 *The identity (4.4.3) shows how the matrix representation of a linear transformation T changes if the ordered basis used to compute the matrix representation is changed. Hence, the matrix $I[\mathcal{B}_1, \mathcal{B}_2]$ is called the $\mathcal{B}_1 : \mathcal{B}_2$ change of basis matrix.*

Definition 4.4.8 (Similar Matrices) Two square matrices B and C of the same order are said to be similar if there exists a non-singular matrix P such that $B = PCP^{-1}$ or equivalently $BP = PC$.

Remark 4.4.9 *Observe that if $A = T[\mathcal{B}, \mathcal{B}]$ then*

$$\{S^{-1}AS : S \text{ is } n \times n \text{ invertible matrix }\}$$

is the set of all matrices that are similar to the given matrix A. Therefore, similar matrices are just different matrix representations of a single linear transformation.

Example 4.4.10 1. Consider $\mathcal{P}_2(\mathbb{R})$, with ordered bases

$$\mathcal{B}_1 = \big(1, 1+x, 1+x+x^2\big) \;\text{ and }\; \mathcal{B}_2 = \big(1+x-x^2, 1+2x+x^2, 2+x+x^2\big).$$

Then

$$[1+x-x^2]_{\mathcal{B}_1} = 0\cdot 1 + 2\cdot(1+x) + (-1)\cdot(1+x+x^2) = (0, 2, -1)^t,$$
$$[1+2x+x^2]_{\mathcal{B}_1} = (-1)\cdot 1 + 1\cdot(1+x) + 1\cdot(1+x+x^2) = (-1, 1, 1)^t, \text{ and}$$
$$[2+x+x^2]_{\mathcal{B}_1} = 1\cdot 1 + 0\cdot(1+x) + 1\cdot(1+x+x^2) = (1, 0, 1)^t.$$

Therefore,

$$\begin{aligned}I[\mathcal{B}_2, \mathcal{B}_1] &= [[I(1+x-x^2)]_{\mathcal{B}_1}, [I(1+2x+x^2)]_{\mathcal{B}_1}, [I(2+x+x^2)]_{\mathcal{B}_1}]\\ &= [[1+x-x^2]_{\mathcal{B}_1}, [1+2x+x^2]_{\mathcal{B}_1}, [2+x+x^2]_{\mathcal{B}_1}]\\ &= \begin{bmatrix}0 & -1 & 1\\ 2 & 1 & 0\\ -1 & 1 & 1\end{bmatrix}.\end{aligned}$$

Find the matrices $T[\mathcal{B}_1, \mathcal{B}_1]$ and $T[\mathcal{B}_2, \mathcal{B}_2]$. Also verify that

$$\begin{aligned}T[\mathcal{B}_2, \mathcal{B}_2] &= I[\mathcal{B}_1, \mathcal{B}_2]\,T[\mathcal{B}_1, \mathcal{B}_1]\,I[\mathcal{B}_2, \mathcal{B}_1]\\ &= I^{-1}[\mathcal{B}_2, \mathcal{B}_1]\,T[\mathcal{B}_1, \mathcal{B}_1]\,I[\mathcal{B}_2, \mathcal{B}_1].\end{aligned}$$

2. Consider two bases $\mathcal{B}_1 = \big((1,0,0), (1,1,0), (1,1,1)\big)$ and $\mathcal{B}_2 = \big((1,1,-1), (1,2,1), (2,1,1)\big)$ of $\mathbb{R}^3$. Suppose $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is a linear transformation defined by

$$T((x, y, z)) = (x+y, x+y+2z, y-z).$$

Then

$$T[\mathcal{B}_1, \mathcal{B}_1] = \begin{bmatrix}0 & 0 & -2\\ 1 & 1 & 4\\ 0 & 1 & 0\end{bmatrix}, \text{ and } T[\mathcal{B}_2, \mathcal{B}_2] = \begin{bmatrix}-4/5 & 1 & 8/5\\ -2/5 & 2 & 9/5\\ 8/5 & 0 & -1/5\end{bmatrix}.$$

Find $I[\mathcal{B}_1, \mathcal{B}_2]$ and verify,

$$I[\mathcal{B}_1, \mathcal{B}_2]\,T[\mathcal{B}_1, \mathcal{B}_1]\,I[\mathcal{B}_2, \mathcal{B}_1] = T[\mathcal{B}_2, \mathcal{B}_2].$$

Check that,

$$T[\mathcal{B}_1, \mathcal{B}_1]\,I[\mathcal{B}_2, \mathcal{B}_1] = I[\mathcal{B}_2, \mathcal{B}_1]\,T[\mathcal{B}_2, \mathcal{B}_2] = \begin{bmatrix}2 & -2 & -2\\ -2 & 4 & 5\\ 2 & 1 & 0\end{bmatrix}.$$

Exercise 4.4.11 1. Let V be an n-dimensional vector space and let $T : V \longrightarrow V$ be a linear transformation. Suppose T has the property that $T^{n-1} \neq \mathbf{0}$ but $T^n = \mathbf{0}$.

(a) Then prove that there exists a vector $\mathbf{u} \in V$ such that the set

$$\{\mathbf{u}, T(\mathbf{u}), \ldots, T^{n-1}(\mathbf{u})\}$$

is a basis of V.

(b) Let $\mathcal{B} = (\mathbf{u}, T(\mathbf{u}), \ldots, T^{n-1}(\mathbf{u}))$. Then prove that

$$T[\mathcal{B}, \mathcal{B}] = \begin{bmatrix}0 & 0 & 0 & \cdots & 0\\ 1 & 0 & 0 & \cdots & 0\\ 0 & 1 & 0 & \cdots & 0\\ \vdots & & \ddots & \ddots & \vdots\\ 0 & 0 & \cdots & 1 & 0\end{bmatrix}.$$

(c) Let A be an $n \times n$ matrix with the property that $A^{n-1} \neq \mathbf{0}$ but $A^n = \mathbf{0}$. Then prove that A is similar to the matrix given above.

2. Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be a linear transformation given by

$$T((x, y, z)) = (x+y+2z, x-y-3z, 2x+3y+z).$$

Let $\mathcal{B}$ be the standard basis and $\mathcal{B}_1 = \big((1,1,1), (1,-1,1), (1,1,2)\big)$ be another ordered basis.

(a) Find the matrices $T[\mathcal{B}, \mathcal{B}]$ and $T[\mathcal{B}_1, \mathcal{B}_1]$.

(b) Find the matrix P such that $P^{-1}T[\mathcal{B}, \mathcal{B}]\,P = T[\mathcal{B}_1, \mathcal{B}_1]$.

3. Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be a linear transformation given by

$$T((x, y, z)) = (x, x+y, x+y+z).$$

Let $\mathcal{B}$ be the standard basis and $\mathcal{B}_1 = \big((1,0,0), (1,1,0), (1,1,1)\big)$ be another ordered basis.

(a) Find the matrices $T[\mathcal{B}, \mathcal{B}]$ and $T[\mathcal{B}_1, \mathcal{B}_1]$.

(b) Find the matrix P such that $P^{-1}T[\mathcal{B}, \mathcal{B}]\,P = T[\mathcal{B}_1, \mathcal{B}_1]$.

4. Let $\mathcal{B}_1 = \big((1,2,0), (1,3,2), (0,1,3)\big)$ and $\mathcal{B}_2 = \big((1,2,1), (0,1,2), (1,4,6)\big)$ be two ordered bases of $\mathbb{R}^3$.

(a) Find the change of basis matrix P from $\mathcal{B}_1$ to $\mathcal{B}_2$.

(b) Find the change of basis matrix Q from $\mathcal{B}_2$ to $\mathcal{B}_1$.

(c) Verify that $PQ = I = QP$.

(d) Find the change of basis matrix from the standard basis of $\mathbb{R}^3$ to $\mathcal{B}_1$. What do you notice?

Chapter 5

Inner Product Spaces

We had learned that given vectors $\vec{i}$ and $\vec{j}$ (which are at an angle of 90°) in a plane, any vector in the plane is a linear combination of the vectors $\vec{i}$ and $\vec{j}$. In this section, we investigate a method by which any basis of a finite dimensional vector can be transferred to another basis in such a way that the vectors in the new basis are at an angle of 90° to each other. To do this, we start by defining a notion of INNER PRODUCT (dot product) in a vector space. This helps us in finding out whether two vectors are at 90° or not.

5.1 Definition and Basic Properties

In $\mathbb{R}^2$, given two vectors $\mathbf{x} = (x_1, x_2),\ \mathbf{y} = (y_1, y_2)$, we know the inner product $\mathbf{x} \cdot \mathbf{y} = x_1y_1 + x_2y_2$. Note that for any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^2$ and $\alpha \in \mathbb{R}$, this inner product satisfies the conditions

$$\mathbf{x} \cdot (\mathbf{y} + \alpha\mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \alpha\mathbf{x} \cdot \mathbf{z},\ \mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x},\ \text{ and }\ \mathbf{x} \cdot \mathbf{x} \geq 0$$

and $\mathbf{x} \cdot \mathbf{x} = 0$ if and only if $\mathbf{x} = \mathbf{0}$. Thus, we are motivated to define an inner product on an arbitrary vector space.

Definition 5.1.1 (Inner Product) Let $V(\mathbb{F})$ be a vector space over $\mathbb{F}$. An inner product over $V(\mathbb{F})$, denoted by $\langle\ ,\ \rangle$, is a map,

$$\langle\ ,\ \rangle : V \times V \longrightarrow \mathbb{F}$$

such that for $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $a, b \in \mathbb{F}$

1. $\langle a\mathbf{u} + b\mathbf{v}, \mathbf{w}\rangle = a\langle \mathbf{u}, \mathbf{w}\rangle + b\langle \mathbf{v}, \mathbf{w}\rangle$,
2. $\langle \mathbf{u}, \mathbf{v}\rangle = \overline{\langle \mathbf{v}, \mathbf{u}\rangle}$, the complex conjugate of $\langle \mathbf{u}, \mathbf{v}\rangle$, and
3. $\langle \mathbf{u}, \mathbf{u}\rangle \geq 0$ for all $\mathbf{u} \in V$ and equality holds if and only if $\mathbf{u} = \mathbf{0}$.

Definition 5.1.2 (Inner Product Space) Let V be a vector space with an inner product $\langle\ ,\ \rangle$. Then $(V, \langle\ ,\ \rangle)$ is called an inner product space, in short denoted by IPS.

Example 5.1.3 The first two examples given below are called the STANDARD INNER PRODUCT or the DOT PRODUCT on $\mathbb{R}^n$ and $\mathbb{C}^n$, respectively..

1. Let $V = \mathbb{R}^n$ be the real vector space of dimension n. Given two vectors $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ of V, we define

   $$\langle u, v\rangle = u_1v_1 + u_2v_2 + \cdots + u_nv_n = \mathbf{u}\mathbf{v}^t.$$

   Verify $\langle\ ,\ \rangle$ is an inner product.

2. Let $V = \mathbb{C}^n$ be a complex vector space of dimension n. Then for $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ in V, check that

   $$\langle u, v\rangle = u_1\overline{v_1} + u_2\overline{v_2} + \cdots + u_n\overline{v_n} = \mathbf{u}\mathbf{v}^*$$

   is an inner product.

3. Let $V = \mathbb{R}^2$ and let $A = \begin{bmatrix} 4 & -1 \\ -1 & 2 \end{bmatrix}$. Define $\langle \mathbf{x}, \mathbf{y}\rangle = \mathbf{x}A\mathbf{y}^t$. Check that $\langle\ ,\ \rangle$ is an inner product. *Hint: Note that* $\mathbf{x}A\mathbf{y}^t = 4x_1y_1 - x_1y_2 - x_2y_1 + 2x_2y_2$.

4. let $\mathbf{x} = (x_1, x_2, x_3),\ \mathbf{y} = (y_1, y_2, y_3) \in \mathbb{R}^3$., Show that $\langle \mathbf{x}, \mathbf{y}\rangle = 10x_1y_1 + 3x_1y_2 + 3x_2y_1 + 2x_2y_2 + x_2y_3 + x_3y_2 + x_3y_3$ is an inner product in $\mathbb{R}^3(\mathbb{R})$.

5. Consider the real vector space $\mathbb{R}^2$. In this example, we define three products that satisfy two conditions out of the three conditions for an inner product. Hence the three products are not inner products.

   (a) Define $\langle \mathbf{x}, \mathbf{y}\rangle = \langle (x_1, x_2), (y_1, y_2)\rangle = x_1y_1$. Then it is easy to verify that the third condition is not valid whereas the first two conditions are valid.

   (b) Define $\langle \mathbf{x}, \mathbf{y}\rangle = \langle (x_1, x_2), (y_1, y_2)\rangle = x_1^2 + y_1^2 + x_2^2 + y_2^2$. Then it is easy to verify that the first condition is not valid whereas the second and third conditions are valid.

   (c) Define $\langle \mathbf{x}, \mathbf{y}\rangle = \langle (x_1, x_2), (y_1, y_2)\rangle = x_1y_1^3 + x_2y_2^3$. Then it is easy to verify that the second condition is not valid whereas the first and third conditions are valid.

Remark 5.1.4 *Note that in parts* 1 *and* 2 *of Example 5.1.3, the inner products are* $\mathbf{u}\mathbf{v}^t$ *and* $\mathbf{u}\mathbf{v}^*$, *respectively. This occurs because the vectors* $\mathbf{u}$ *and* $\mathbf{v}$ *are row vectors. In general,* $\mathbf{u}$ *and* $\mathbf{v}$ *are taken as column vectors and hence one uses the notation* $\mathbf{u}^t\mathbf{v}$ *or* $\mathbf{u}^*\mathbf{v}$.

Exercise 5.1.5 Verify that inner products defined in parts 3 and 4 of Example 5.1.3, are indeed inner products.

Definition 5.1.6 (Length/Norm of a Vector) For $\mathbf{u} \in V$, we define the length (norm) of $\mathbf{u}$, denoted $\|\mathbf{u}\|$, by $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u}\rangle}$, the positive square root.

A very useful and a fundamental inequality concerning the inner product is due to Cauchy and Schwartz. The next theorem gives the statement and a proof of this inequality.

Theorem 5.1.7 (Cauchy-Schwartz inequality) Let $V(\mathbb{F})$ be an inner product space. Then for any $\mathbf{u}, \mathbf{v} \in V$

$$|\langle \mathbf{u}, \mathbf{v}\rangle| \leq \|\mathbf{u}\|\ \|\mathbf{v}\|.$$

The equality holds if and only if the vectors $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent. Further, if $\mathbf{u} \neq \mathbf{0}$, then $\mathbf{v} = \langle \mathbf{v}, \frac{\mathbf{u}}{\|\mathbf{u}\|}\rangle \frac{\mathbf{u}}{\|\mathbf{u}\|}$.

PROOF. If $\mathbf{u} = \mathbf{0}$, then the inequality holds. Let $\mathbf{u} \neq \mathbf{0}$. Note that $\langle \lambda\mathbf{u} + \mathbf{v}, \lambda\mathbf{u} + \mathbf{v}\rangle \geq 0$ for all $\lambda \in \mathbb{F}$. In particular, for $\lambda = -\frac{\langle \mathbf{v}, \mathbf{u}\rangle}{\|\mathbf{u}\|^2}$, we get

$$\begin{aligned} 0 &\leq \langle \lambda\mathbf{u} + \mathbf{v}, \lambda\mathbf{u} + \mathbf{v}\rangle \\ &= \lambda\overline{\lambda}\|\mathbf{u}\|^2 + \lambda\langle \mathbf{u}, \mathbf{v}\rangle + \overline{\lambda}\langle \mathbf{v}, \mathbf{u}\rangle + \|\mathbf{v}\|^2 \\ &= \frac{\langle \mathbf{v}, \mathbf{u}\rangle}{\|\mathbf{u}\|^2}\frac{\overline{\langle \mathbf{v}, \mathbf{u}\rangle}}{\|\mathbf{u}\|^2}\|\mathbf{u}\|^2 - \frac{\langle \mathbf{v}, \mathbf{u}\rangle}{\|\mathbf{u}\|^2}\langle \mathbf{u}, \mathbf{v}\rangle - \frac{\overline{\langle \mathbf{v}, \mathbf{u}\rangle}}{\|\mathbf{u}\|^2}\langle \mathbf{v}, \mathbf{u}\rangle + \|\mathbf{v}\|^2 \\ &= \|\mathbf{v}\|^2 - \frac{|\langle \mathbf{v}, \mathbf{u}\rangle|^2}{\|\mathbf{u}\|^2}. \end{aligned}$$

Or, in other words

$$|\langle \mathbf{v}, \mathbf{u}\rangle|^2 \leq \|\mathbf{u}\|^2\|\mathbf{v}\|^2$$

and the proof of the inequality is over.

Observe that if $\mathbf{u} \neq \mathbf{0}$ then the equality holds if and only of $\lambda\mathbf{u} + \mathbf{v} = \mathbf{0}$ for $\lambda = -\frac{\langle \mathbf{v}, \mathbf{u}\rangle}{\|\mathbf{u}\|^2}$. That is, $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent. We leave it for the reader to prove

$$\mathbf{v} = \langle \mathbf{v}, \frac{\mathbf{u}}{\|\mathbf{u}\|}\rangle \frac{\mathbf{u}}{\|\mathbf{u}\|}.$$

□

Definition 5.1.8 (Angle between two vectors) Let V be a real vector space. Then for every $\mathbf{u}, \mathbf{v} \in V$, by the Cauchy-Schwartz inequality, we have

$$-1 \leq \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|\ \|\mathbf{v}\|} \leq 1.$$

We know that $\cos : [0, \pi] \longrightarrow [-1,\ 1]$ is an one-one and onto function. Therefore, for every real number $\frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|\ \|\mathbf{v}\|}$, there exists a unique $\theta,\ 0 \leq \theta \leq \pi$, such that

$$\cos\theta = \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|\ \|\mathbf{v}\|}.$$

1. The real number θ with $0 \leq \theta \leq \pi$ and satisfying $\cos\theta = \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|\ \|\mathbf{v}\|}$ is called the angle between the two vectors $\mathbf{u}$ and $\mathbf{v}$ in V.
2. The vectors $\mathbf{u}$ and $\mathbf{v}$ in V are said to be orthogonal if $\langle \mathbf{u}, \mathbf{v}\rangle = 0$.
3. A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is called mutually orthogonal if $\langle \mathbf{u}_i, \mathbf{u}_j\rangle = 0$ for all $1 \leq i \neq j \leq n$.

Exercise 5.1.9 1. Let $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ be the standard basis of $\mathbb{R}^n$. Then prove that with respect to the standard inner product on $\mathbb{R}^n$, the vectors $\mathbf{e}_i$ satisfy the following:

(a) $\|\mathbf{e}_i\| = 1$ for $1 \leq i \leq n$.

(b) $\langle \mathbf{e}_i, \mathbf{e}_j\rangle = 0$ for $1 \leq i \neq j \leq n$.

2. Recall the following inner product on $\mathbb{R}^2$: for $\mathbf{x} = (x_1, x_2)^t$ and $\mathbf{y} = (y_1, y_2)^t$,

   $$\langle \mathbf{x}, \mathbf{y}\rangle = 4x_1y_1 - x_1y_2 - x_2y_1 + 2x_2y_2.$$

   (a) Find the angle between the vectors $e_1 = (1, 0)^t$ and $e_2 = (0, 1)^t$.

   (b) Let $\mathbf{u} = (1, 0)^t$. Find $\mathbf{v} \in \mathbb{R}^2$ such that $\langle \mathbf{v}, \mathbf{u}\rangle = 0$.

   (c) Find two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$, such that $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$ and $\langle \mathbf{x}, \mathbf{y}\rangle = 0$.

3. Find an inner product in $\mathbb{R}^2$ such that the following conditions hold:

   $$\|(1, 2)\| = \|(2, -1)\| = 1,\ \text{ and }\ \langle (1, 2),\ (2, -1)\rangle = 0.$$

   [Hint: Consider a symmetric matrix $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$. *Define* $\langle \mathbf{x}, \mathbf{y}\rangle = \mathbf{y}^tA\mathbf{x}$ *and solve a system of* 3 *equations for the unknowns* a, b, c.*]*

4. Let V be a complex vector space with $\dim(V) = n$. Fix an ordered basis $\mathcal{B} = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n)$. Define a map

   $$\langle\ ,\ \rangle : V \times V \longrightarrow \mathbb{C}\ \text{ by }\ \langle \mathbf{u}, \mathbf{v}\rangle = \sum_{i=1}^{n} a_i\overline{b_i}$$

   whenever $[\mathbf{u}]_{\mathcal{B}} = (a_1, a_2, \ldots, a_n)^t$ and $[\mathbf{v}]_{\mathcal{B}} = (b_1, b_2, \ldots, b_n)^t$. Show that the above defined map is indeed an inner product.

5. Let $\mathbf{x} = (x_1, x_2, x_3),\ \mathbf{y} = (y_1, y_2, y_3) \in \mathbb{R}^3$. Show that

   $$\langle \mathbf{x}, \mathbf{y}\rangle = 10x_1y_1 + 3x_1y_2 + 3x_2y_1 + 2x_2y_2 + x_2y_3 + x_3y_2 + x_3y_3$$

   is an inner product in $\mathbb{R}^3(\mathbb{R})$. With respect to this inner product, find the angle between the vectors $(1, 1, 1)$ and $(2, -5, 2)$.

6. Consider the set $M_{n\times n}(\mathbb{R})$ of all real square matrices of order n. For $A, B \in M_{n\times n}(\mathbb{R})$ we define $\langle A, B\rangle = tr(AB^t)$. Then

   $$\langle A + B, C\rangle = tr\big((A + B)C^t\big) = tr(AC^t) + tr(BC^t) = \langle A, C\rangle + \langle B, C\rangle.$$

   $$\langle A, B\rangle = tr(AB^t) = tr(\ (AB^t)^t\ ) = tr(BA^t) = \langle B, A\rangle.$$

   Let $A = (a_{ij})$. Then

   $$\langle A, A\rangle = tr(AA^t) = \sum_{i=1}^{n}(AA^t)_{ii} = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}a_{ij} = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2$$

   and therefore, $\langle A, A\rangle > 0$ for all non-zero matrices A. So, it is clear that $\langle A, B\rangle$ is an inner product on $M_{n\times n}(\mathbb{R})$.

7. Let V be the real vector space of all continuous functions with domain $[-2\pi, 2\pi]$. That is, $V = C[-2\pi,\ 2\pi]$. Then show that V is an inner product space with inner product $\int_{-1}^{1} f(x)g(x)dx$.
   For different values of m and n, find the angle between the functions $\cos(mx)$ and $\sin(nx)$.

8. Let V be an inner product space. Prove that

   $$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\| \quad \text{for every} \quad \mathbf{u}, \mathbf{v} \in V.$$

   This inequality is called the TRIANGLE INEQUALITY.

9. Let $z_1, z_2, \ldots, z_n \in \mathbb{C}$. Use the Cauchy-Schwartz inequality to prove that

   $$|z_1 + z_2 + \cdots + z_n| \leq \sqrt{n(|z_1|^2 + |z_2|^2 + \cdots + |z_n|^2)}.$$

   When does the equality hold?

10. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Observe that $\langle \mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{y}, \mathbf{x}\rangle$. Hence or otherwise prove the following:

    (a) $\langle \mathbf{x}, \mathbf{y}\rangle = 0 \Longleftrightarrow \|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$, (This is called PYTHAGORAS THEOREM).

    (b) $\|\mathbf{x}\| = \|\mathbf{y}\| \Longleftrightarrow \langle \mathbf{x} + \mathbf{y}, \mathbf{x} - \mathbf{y}\rangle = 0$, ($\mathbf{x}$ and $\mathbf{y}$ form adjacent sides of a rhombus as the diagonals $\mathbf{x} + \mathbf{y}$ and $\mathbf{x} - \mathbf{y}$ are orthogonal).

    (c) $\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2\|\mathbf{x}\|^2 + 2\|\mathbf{y}\|^2$, (This is called the PARALLELOGRAM LAW).

    (d) $4\langle \mathbf{x}, \mathbf{y}\rangle = \|\mathbf{x} + \mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2$ (This is called the POLARISATION IDENTITY).

    Remark 5.1.10 *i. Suppose the norm of a vector is given. Then, the polarisation identity can be used to define an inner product.*

ii. Observe that if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$ then the parallelogram spanned by the vectors $\mathbf{x}$ and $\mathbf{y}$ is a rectangle. The above equality tells us that the lengths of the two diagonals are equal.

Are these results true if $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n(\mathbb{C})$?

11. Let $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n(\mathbb{C})$. Prove that
    (a) $4\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}+\mathbf{y}\|^2 - \|\mathbf{x}-\mathbf{y}\|^2 + i\|\mathbf{x}+i\mathbf{y}\|^2 - i\|\mathbf{x}-i\mathbf{y}\|^2$.
    (b) If $\mathbf{x} \neq \mathbf{0}$ then $\|\mathbf{x}+i\mathbf{x}\|^2 = \|\mathbf{x}\|^2 + \|i\mathbf{x}\|^2$, even though $\langle \mathbf{x}, i\mathbf{x} \rangle \neq 0$.
    (c) If $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$ and $\|\mathbf{x}+i\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|i\mathbf{y}\|^2$ then show that $\langle \mathbf{x}, \mathbf{y} \rangle = 0$.

12. Let V be an n-dimensional inner product space, with an inner product $\langle\ ,\ \rangle$. Let $\mathbf{u} \in V$ be a fixed vector with $\|\mathbf{u}\| = 1$. Then give reasons for the following statements.
    (a) Let $S^{\perp} = \{\mathbf{v} \in V\ :\ \langle \mathbf{v}, \mathbf{u} \rangle = 0\}$. Then S is a subspace of V of dimension $n-1$.
    (b) Let $0 \neq \alpha \in \mathbb{F}$ and let $S = \{\mathbf{v} \in V\ :\ \langle \mathbf{v}, \mathbf{u} \rangle = \alpha\}$. Then S is not a subspace of V.
    (c) For any $\mathbf{v} \in S$, there exists a vector $\mathbf{v}_0 \in S^{\perp}$, such that $\mathbf{v} = \mathbf{v}_0 + \alpha\mathbf{u}$.

Theorem 5.1.11 Let V be an inner product space. Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be a set of non-zero, mutually orthogonal vectors of V.

1. Then the set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is linearly independent.
2. $\|\sum_{i=1}^{n} \alpha_i \mathbf{u}_i\|^2 = \sum_{i=1}^{n} |\alpha_i|^2 \|\mathbf{u}_i\|^2$;
3. Let $\dim(V) = n$ and also let $\|\mathbf{u}_i\| = 1$ for $i = 1, 2, \ldots, n$. Then for any $\mathbf{v} \in V$,
$$\mathbf{v} = \sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{u}_i \rangle \mathbf{u}_i.$$
In particular, $\langle \mathbf{v}, \mathbf{u}_i \rangle = 0$ for all $i = 1, 2, \ldots, n$ if and only if $\mathbf{v} = \mathbf{0}$.

PROOF. Consider the set of non-zero, mutually orthogonal vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$. Suppose there exist scalars $c_1, c_2, \ldots, c_n$ not all zero, such that
$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n = \mathbf{0}.$$
Then for $1 \leq i \leq n$, we have
$$0 = \langle \mathbf{0}, \mathbf{u}_i \rangle = \langle c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n, \mathbf{u}_i \rangle = \sum_{j=1}^{n} c_j \langle \mathbf{u}_j, \mathbf{u}_i \rangle = c_i$$
as $\langle \mathbf{u}_j, \mathbf{u}_i \rangle = 0$ for all $j \neq i$ and $\langle \mathbf{u}_i, \mathbf{u}_i \rangle = 1$. This gives a contradiction to our assumption that some of the c_i's are non-zero. This establishes the linear independence of a set of non-zero, mutually orthogonal vectors.

For the second part, using $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ \|\mathbf{u}_i\|^2 & \text{if } i = j \end{cases}$ for $1 \leq i, j \leq n$, we have
$$\begin{aligned} \|\sum_{i=1}^{n} \alpha_i \mathbf{u}_i\|^2 &= \langle \sum_{i=1}^{n} \alpha_i \mathbf{u}_i, \sum_{i=1}^{n} \alpha_i \mathbf{u}_i \rangle = \sum_{i=1}^{n} \alpha_i \langle \mathbf{u}_i, \sum_{j=1}^{n} \alpha_j \mathbf{u}_j \rangle \\ &= \sum_{i=1}^{n} \alpha_i \sum_{j=1}^{n} \overline{\alpha_j} \langle \mathbf{u}_i, \mathbf{u}_j \rangle = \sum_{i=1}^{n} \alpha_i \overline{\alpha_i} \langle \mathbf{u}_i, \mathbf{u}_i \rangle \\ &= \sum_{i=1}^{n} |\alpha_i|^2 \|\mathbf{u}_i\|^2. \end{aligned}$$

For the third part, observe from the first part, the linear independence of the non-zero mutually orthogonal vectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$. Since $\dim(V) = n$, they form a basis of V. Thus, for every vector $\mathbf{v} \in V$, there exist scalars α_i, $1 \leq i \leq n$, such that $\mathbf{v} = \sum_{i=1}^{n} \alpha_i \mathbf{u}_n$. Hence,
$$\langle \mathbf{v}, \mathbf{u}_j \rangle = \langle \sum_{i=1}^{n} \alpha_i \mathbf{u}_i, \mathbf{u}_j \rangle = \sum_{i=1}^{n} \alpha_i \langle \mathbf{u}_i, \mathbf{u}_j \rangle = \alpha_j.$$
Therefore, we have obtained the required result. □

Definition 5.1.12 (Orthonormal Set) Let V be an inner product space. A set of non-zero, mutually orthogonal vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ in V is called an orthonormal set if $\|\mathbf{v}_i\| = 1$ for $i = 1, 2, \ldots, n$.

If the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is also a basis of V, then the set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is called an orthonormal basis of V.

Example 5.1.13
1. Consider the vector space $\mathbb{R}^2$ with the standard inner product. Then the standard ordered basis $\mathcal{B} = \big((1,0),(0,1)\big)$ is an orthonormal set. Also, the basis $\mathcal{B}_1 = \big(\frac{1}{\sqrt{2}}(1,1), \frac{1}{\sqrt{2}}(1,-1)\big)$ is an orthonormal set.
2. Let $\mathbb{R}^n$ be endowed with the standard inner product. Then by Exercise 5.1.9.1, the standard ordered basis $(\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n)$ is an orthonormal set.

In view of Theorem 5.1.11, we inquire into the question of extracting an orthonormal basis from a given basis. In the next section, we describe a process (called the Gram-Schmidt Orthogonalisation process) that generates an orthonormal set from a given set containing finitely many vectors.

Remark 5.1.14 *The last part of the above theorem can be rephrased as "suppose $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an orthonormal basis of an inner product space V. Then for each $\mathbf{u} \in V$ the numbers $\langle \mathbf{u}, \mathbf{v}_i \rangle$ for $1 \leq i \leq n$ are the coordinates of $\mathbf{u}$ with respect to the above basis".*

That is, let $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ be an ordered basis. Then for any $\mathbf{u} \in V$,
$$[\mathbf{u}]_{\mathcal{B}} = (\langle \mathbf{u}, \mathbf{v}_1 \rangle, \langle \mathbf{u}, \mathbf{v}_2 \rangle, \ldots, \langle \mathbf{u}, \mathbf{v}_n \rangle)^t.$$

5.2 Gram-Schmidt Orthogonalisation Process

Let V be a finite dimensional inner product space. Suppose $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ is a linearly independent subset of V. Then the Gram-Schmidt orthogonalisation process uses the vectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ to construct new vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ such that $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for $i \neq j$, $\|\mathbf{v}_i\| = 1$ and Span $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_i\}$ = Span $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_i\}$ for $i = 1, 2, \ldots, n$. This process proceeds with the following idea.

Suppose we are given two vectors $\mathbf{u}$ and $\mathbf{v}$ in a plane. If we want to get vectors $\mathbf{z}$ and $\mathbf{y}$ such that $\mathbf{z}$ is a unit vector in the direction of $\mathbf{u}$ and $\mathbf{y}$ is a unit vector perpendicular to $\mathbf{z}$, then they can be obtained in the following way:

Take the first vector $\mathbf{z} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$. Let θ be the angle between the vectors $\mathbf{u}$ and $\mathbf{v}$. Then $\cos(\theta) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|u\|\ \|v\|}$.

Defined $\alpha = \|\mathbf{v}\|\ \cos(\theta) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|} = \langle \mathbf{z}, \mathbf{v} \rangle$. Then $\mathbf{w} = \mathbf{v} - \alpha\ \mathbf{z}$ is a vector perpendicular to the unit vector $\mathbf{z}$, as we have removed the component of $\mathbf{z}$ from $\mathbf{v}$. So, the vectors that we are interested in are $\mathbf{z}$ and $\mathbf{y} = \frac{\mathbf{w}}{\|\mathbf{w}\|}$.

This idea is used to give the Gram-Schmidt Orthogonalization process which we now describe.

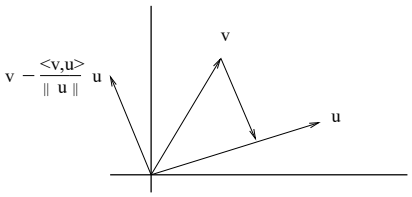


Figure 5.1: Gram-Schmidt Process

Theorem 5.2.1 (Gram-Schmidt Orthogonalization Process) Let V be an inner product space. Suppose $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is a set of linearly independent vectors of V. Then there exists a set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ of vectors of V satisfying the following:

1. $\|\mathbf{v}_i\| = 1$ for $1 \leq i \leq n$,
2. $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for $1 \leq i, j \leq n$, $i \neq j$ and
3. $L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_i) = L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_i)$ for $1 \leq i \leq n$.

PROOF. We successively define the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ as follows.

$\mathbf{v}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$.

Calculate $\mathbf{w}_2 = \mathbf{u}_2 - \langle \mathbf{u}_2, \mathbf{v}_1 \rangle \mathbf{v}_1$, and let $\mathbf{v}_2 = \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|}$.

Obtain $\mathbf{w}_3 = \mathbf{u}_3 - \langle \mathbf{u}_3, \mathbf{v}_1 \rangle \mathbf{v}_1 - \langle \mathbf{u}_3, \mathbf{v}_2 \rangle \mathbf{v}_2$, and let $\mathbf{v}_3 = \frac{\mathbf{w}_3}{\|\mathbf{w}_3\|}$.

In general, if $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \ldots, \mathbf{v}_{i-1}$ are already obtained, we compute
$$\mathbf{w}_i = \mathbf{u}_i - \langle \mathbf{u}_i, \mathbf{v}_1 \rangle \mathbf{v}_1 - \langle \mathbf{u}_i, \mathbf{v}_2 \rangle \mathbf{v}_2 - \cdots - \langle \mathbf{u}_i, \mathbf{v}_{i-1} \rangle \mathbf{v}_{i-1}, \tag{5.2.1}$$
and define
$$\mathbf{v}_i = \frac{\mathbf{w}_i}{\|\mathbf{w}_i\|}.$$

We prove the theorem by induction on n, the number of linearly independent vectors.

For $n = 1$, we have $\mathbf{v}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$. Since $\mathbf{u}_1 \neq \mathbf{0}$, $\mathbf{v}_1 \neq \mathbf{0}$ and
$$\|\mathbf{v}_1\|^2 = \langle \mathbf{v}_1, \mathbf{v}_1 \rangle = \langle \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}, \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} \rangle = \frac{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle}{\|\mathbf{u}_1\|^2} = 1.$$
Hence, the result holds for $n = 1$.

Let the result hold for all $k \leq n-1$. That is, suppose we are given any set of k, $1 \leq k \leq n-1$ linearly independent vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ of V. Then by the inductive assumption, there exists a set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ of vectors satisfying the following:

1. $\|\mathbf{v}_i\| = 1$ for $1 \leq i \leq k$,
2. $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for $1 \leq i \neq j \leq k$, and

3. $L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_i) = L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_i)$ for $1 \leq i \leq k$.

Now, let us assume that we are given a set of n linearly independent vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ of V. Then by the inductive assumption, we already have vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n-1}$ satisfying

1. $\|\mathbf{v}_i\| = 1$ for $1 \leq i \leq n-1$,
2. $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for $1 \leq i \neq j \leq n-1$, and
3. $L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_i) = L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_i)$ for $1 \leq i \leq n-1$.

Using (5.2.1), we define
$$\mathbf{w}_n = \mathbf{u}_n - \langle \mathbf{u}_n, \mathbf{v}_1 \rangle \mathbf{v}_1 - \langle \mathbf{u}_n, \mathbf{v}_2 \rangle \mathbf{v}_2 - \cdots - \langle \mathbf{u}_n, \mathbf{v}_{n-1} \rangle \mathbf{v}_{n-1}. \tag{5.2.2}$$
We first show that $\mathbf{w}_n \notin L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n-1})$. This will also imply that $\mathbf{w}_n \neq \mathbf{0}$ and hence $\mathbf{v}_n = \frac{\mathbf{w}_n}{\|\mathbf{w}_n\|}$ is well defined.

On the contrary, assume that $\mathbf{w}_n \in L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n-1})$. Then there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_{n-1}$ such that
$$\mathbf{w}_n = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_{n-1}\mathbf{v}_{n-1}.$$
So, by (5.2.2)
$$\mathbf{u}_n = \big(\alpha_1 + \langle \mathbf{u}_n, \mathbf{v}_1 \rangle\big)\mathbf{v}_1 + \big(\alpha_2 + \langle \mathbf{u}_n, \mathbf{v}_2 \rangle\big)\mathbf{v}_2 + \cdots + \big((\alpha_{n-1} + \langle \mathbf{u}_n, \mathbf{v}_{n-1} \rangle\big)\mathbf{v}_{n-1}.$$
Thus, by the third induction assumption,
$$\mathbf{u}_n \in L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n-1}) = L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{n-1}).$$
This gives a contradiction to the given assumption that the set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is linear independent.

So, $\mathbf{w}_n \neq \mathbf{0}$. Define $\mathbf{v}_n = \frac{\mathbf{w}_n}{\|\mathbf{w}_n\|}$. Then $\|\mathbf{v}_n\| = 1$. Also, it can be easily verified that $\langle \mathbf{v}_n, \mathbf{v}_i \rangle = 0$ for $1 \leq i \leq n-1$. Hence, by the principle of mathematical induction, the proof of the theorem is complete. □

We illustrate the Gram-Schmidt process by the following example.

Example 5.2.2 Let $\{(1,-1,1,1),(1,0,1,0),(0,1,0,1)\}$ be a linearly independent set in $\mathbb{R}^4(\mathbb{R})$. Find an orthonormal set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ such that $L(\,(1,-1,1,1),(1,0,1,0),(0,1,0,1)\,) = L(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

Solution: Let $\mathbf{u}_1 = (1,0,1,0)$. Define $\mathbf{v}_1 = \frac{(1,0,1,0)}{\sqrt{2}}$. Let $\mathbf{u}_2 = (0,1,0,1)$. Then
$$\mathbf{w}_2 = (0,1,0,1) - \langle (0,1,0,1), \frac{(1,0,1,0)}{\sqrt{2}} \rangle \mathbf{v}_1 = (0,1,0,1).$$
Hence, $\mathbf{v}_2 = \frac{(0,1,0,1)}{\sqrt{2}}$. Let $\mathbf{u}_3 = (1,-1,1,1)$. Then
$$\begin{aligned} \mathbf{w}_3 &= (1,-1,1,1) - \langle (1,-1,1,1), \frac{(1,0,1,0)}{\sqrt{2}} \rangle \mathbf{v}_1 - \langle (1,-1,1,1), \frac{(0,1,0,1)}{\sqrt{2}} \rangle \mathbf{v}_2 \\ &= (0,-1,0,1) \end{aligned}$$
and $\mathbf{v}_3 = \frac{(0,-1,0,1)}{\sqrt{2}}$.

Remark 5.2.3 *1. Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ be any basis of a k-dimensional subspace W of $\mathbb{R}^n$. Then by Gram-Schmidt orthogonalisation process, we get an orthonormal set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\} \subset \mathbb{R}^n$ with $W = L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$, and for $1 \le i \le k$,*

$$L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_i) = L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_i).$$

2. Suppose we are given a set of n vectors, $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ of V that are linearly dependent. Then by Corollary 3.2.5, there exists a smallest k, $2 \le k \le n$ such that

$$L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k) = L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{k-1}).$$

We claim that in this case, $\mathbf{w}_k = \mathbf{0}$.

Since, we have chosen the smallest k satisfying

$$L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_i) = L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{i-1}),$$

for $2 \le i \le n$, the set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{k-1}\}$ is linearly independent (use Corollary 3.2.5). So, by Theorem 5.2.1, there exists an orthonormal set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{k-1}\}$ such that

$$L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_{k-1}) = L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{k-1}).$$

As $\mathbf{u}_k \in L(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{k-1})$, by Remark 5.1.14

$$\mathbf{u}_k = \langle \mathbf{u}_k, \mathbf{v}_1\rangle \mathbf{v}_1 + \langle \mathbf{u}_k, \mathbf{v}_2\rangle \mathbf{v}_2 + \cdots + \langle \mathbf{u}_k, \mathbf{v}_{k-1}\rangle \mathbf{v}_{n-1}.$$

So, by definition of $\mathbf{w}_k$, $\mathbf{w}_k = \mathbf{0}$.

Therefore, in this case, we can continue with the Gram-Schmidt process by replacing $\mathbf{u}_k$ by $\mathbf{u}_{k+1}$.

3. Let S be a countably infinite set of linearly independent vectors. Then one can apply the Gram-Schmidt process to get a countably infinite orthonormal set.

4. Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ be an orthonormal subset of $\mathbb{R}^n$. Let $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n)$ be the standard ordered basis of $\mathbb{R}^n$. Then there exist real numbers α_{ij}, $1 \le i \le k$, $1 \le j \le n$ such that

$$[\mathbf{v}_i]_{\mathcal{B}} = (\alpha_{1i}, \alpha_{2i}, \ldots, \alpha_{ni})^t.$$

Let $A = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k]$. Then in the ordered basis $\mathcal{B}$, we have

$$A = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1k} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nk} \end{bmatrix}$$

is an $n \times k$ matrix.

Also, observe that the conditions $\|\mathbf{v}_i\| = 1$ and $\langle \mathbf{v}_i, \mathbf{v}_j\rangle = 0$ for $1 \le i \ne j \le n$, implies that

$$\left.\begin{aligned} 1 = \|\mathbf{v}_i\| = \|\mathbf{v}_i\|^2 = \langle \mathbf{v}_i, \mathbf{v}_i\rangle = \sum_{j=1}^{n} \alpha_{ji}^2, \\ \text{and} \quad 0 = \langle \mathbf{v}_i, \mathbf{v}_j\rangle = \sum_{s=1}^{n} \alpha_{si}\alpha_{sj}. \end{aligned}\right\} \tag{5.2.3}$$

Note that,

$$A^tA = \begin{bmatrix} \mathbf{v}_1^t \\ \mathbf{v}_2^t \\ \vdots \\ \mathbf{v}_k^t \end{bmatrix} [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k] = \begin{bmatrix} \|\mathbf{v}_1\|^2 & \langle \mathbf{v}_1, \mathbf{v}_2\rangle & \cdots & \langle \mathbf{v}_1, \mathbf{v}_k\rangle \\ \langle \mathbf{v}_2, \mathbf{v}_1\rangle & \|\mathbf{v}_2\|^2 & \cdots & \langle \mathbf{v}_2, \mathbf{v}_k\rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{v}_k, \mathbf{v}_1\rangle & \langle \mathbf{v}_k, \mathbf{v}_2\rangle & \cdots & \|\mathbf{v}_k\|^2 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I_k.$$

Or using (5.2.3), in the language of matrices, we get

$$A^tA = \begin{bmatrix} \alpha_{11} & \alpha_{21} & \cdots & \alpha_{n1} \\ \alpha_{12} & \alpha_{22} & \cdots & \alpha_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{1k} & \alpha_{2k} & \cdots & \alpha_{nk} \end{bmatrix} \begin{bmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1k} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \cdots & \alpha_{nk} \end{bmatrix} = I_k.$$

Perhaps the readers must have noticed that the inverse of A is its transpose. Such matrices are called orthogonal matrices and they have a special role to play.

Definition 5.2.4 (Orthogonal Matrix) A $n \times n$ real matrix A is said to be an orthogonal matrix if $A\,A^t = A^tA = I_n$.

It is worthwhile to solve the following exercises.

Exercise 5.2.5 1. Let A and B be two $n \times n$ orthogonal matrices. Then prove that AB and BA are both orthogonal matrices.

2. Let A be an $n \times n$ orthogonal matrix. Then prove that
   (a) the rows of A form an orthonormal basis of $\mathbb{R}^n$.
   (b) the columns of A form an orthonormal basis of $\mathbb{R}^n$.
   (c) for any two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{n\times 1}$, $\langle A\mathbf{x}, A\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$.
   (d) for any vector $\mathbf{x} \in \mathbb{R}^{n\times 1}$, $\|A\mathbf{x}\| = \|\mathbf{x}\|$.

3. Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be an orthonormal basis of $\mathbb{R}^n$. Let $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n)$ be the standard basis of $\mathbb{R}^n$. Construct an $n \times n$ matrix A by

$$A = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

where

$$\mathbf{u}_i = \sum_{j=1}^{n} a_{ji}\mathbf{e}_j, \quad \text{for } 1 \le i \le n.$$

Prove that $A^tA = I_n$. Hence deduce that A is an orthogonal matrix.

4. Let A be an $n \times n$ upper triangular matrix. If A is also an orthogonal matrix, then prove that $A = I_n$.

Theorem 5.2.6 (QR Decomposition) Let A be a square matrix of order n. Then there exist matrices Q and R such that Q is orthogonal and R is upper triangular with $A = QR$.

In case, A is non-singular, the diagonal entries of R can be chosen to be positive. Also, in this case, the decomposition is unique.

PROOF. We prove the theorem when A is non-singular. The proof for the singular case is left as an exercise.

Let the columns of A be $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$. The Gram-Schmidt orthogonalisation process applied to the vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ gives the vectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ satisfying

$$\left.\begin{aligned} L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_i) = L(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_i), \\ \|\mathbf{u}_i\| = 1,\ \langle \mathbf{u}_i, \mathbf{u}_j\rangle = 0, \end{aligned}\right\} \text{ for } 1 \le i \ne j \le n. \tag{5.2.4}$$

Now, consider the ordered basis $\mathcal{B} = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n)$. From (5.2.4), for $1 \le i \le n$, we have $L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_i) = L(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_i)$. So, we can find scalars $\alpha_{ji}, 1 \le j \le i$ such that

$$\mathbf{x}_i = \alpha_{1i}\mathbf{u}_1 + \alpha_{2i}\mathbf{u}_2 + \cdots + \alpha_{ii}\mathbf{u}_i = \left[(\alpha_{1i}, \ldots, \alpha_{ii}, 0 \ldots, 0)^t\right]_{\mathcal{B}}. \tag{5.2.5}$$

Let $Q = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n]$. Then by Exercise 5.2.5.3, Q is an orthogonal matrix. We now define an $n \times n$ upper triangular matrix R by

$$R = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ 0 & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_{nn} \end{bmatrix}.$$

By using (5.2.5), we get

$$\begin{aligned} QR &= [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n] \begin{bmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ 0 & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_{nn} \end{bmatrix} \\ &= \left[\alpha_{11}\mathbf{u}_1,\ \alpha_{12}\mathbf{u}_1 + \alpha_{22}\mathbf{u}_2, \ldots, \sum_{i=1}^{n} \alpha_{in}\mathbf{u}_i\right] \\ &= [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] = A. \end{aligned}$$

Thus, we see that $A = QR$, where Q is an orthogonal matrix (see Remark 5.2.3.4) and R is an upper triangular matrix.

The proof doesn't guarantee that for $1 \le i \le n$, α_{ii} is positive. But this can be achieved by replacing the vector $\mathbf{u}_i$ by $-\mathbf{u}_i$ whenever α_{ii} is negative.

Uniqueness: suppose $Q_1R_1 = Q_2R_2$ then $Q_2^{-1}Q_1 = R_2R_1^{-1}$. Observe the following properties of upper triangular matrices.

1. The inverse of an upper triangular matrix is also an upper triangular matrix, and
2. product of upper triangular matrices is also upper triangular.

Thus the matrix $R_2R_1^{-1}$ is an upper triangular matrix. Also, by Exercise 5.2.5.1, the matrix $Q_2^{-1}Q_1$ is an orthogonal matrix. Hence, by Exercise 5.2.5.4, $R_2R_1^{-1} = I_n$. So, $R_2 = R_1$ and therefore $Q_2 = Q_1$. □

Suppose we have matrix $A = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k]$ of dimension $n \times k$ with rank $(A) = r$. Then by Remark 5.2.3.2, the application of the Gram-Schmidt orthogonalisation process yields a set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_r\}$ of

orthonormal vectors of $\mathbb{R}^n$. In this case, for each i, $1 \le i \le r$, we have

$$L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_i) = L(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_j), \quad \text{for some } j,\ i \le j \le k.$$

Hence, proceeding on the lines of the above theorem, we have the following result.

Theorem 5.2.7 (Generalised QR Decomposition) Let A be an $n \times k$ matrix of rank r. Then $A = QR$, where

1. Q is an $n \times r$ matrix with $Q^tQ = I_r$. That is, the columns of Q form an orthonormal set,
2. If $Q = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_r]$, then $L(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_r) = L(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k)$, and
3. R is an $r \times k$ matrix with rank $(R) = r$.

Example 5.2.8 1. Let $A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$. Find an orthogonal matrix Q and an upper triangular matrix R such that $A = QR$.

Solution: From Example 5.2.2, we know that

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 0, 1, 0),\ \mathbf{v}_2 = \frac{1}{\sqrt{2}}(0, 1, 0, 1),\ \mathbf{v}_3 = \frac{1}{\sqrt{2}}(0, -1, 0, 1). \tag{5.2.6}$$

We now compute $\mathbf{w}_4$. If we denote $\mathbf{u}_4 = (2, 1, 1, 1)^t$ then by the Gram-Schmidt process,

$$\begin{aligned} \mathbf{w}_4 &= \mathbf{u}_4 - \langle \mathbf{u}_4, \mathbf{v}_1\rangle \mathbf{v}_1 - \langle \mathbf{u}_4, \mathbf{v}_2\rangle \mathbf{v}_2 - \langle \mathbf{u}_4, \mathbf{v}_3\rangle \mathbf{v}_3 \\ &= \frac{1}{2}(1, 0, -1, 0)^t. \end{aligned} \tag{5.2.7}$$

Thus, using (5.2.6) and (5.2.7), we get

$$Q = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4] = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & 0 & \frac{-1}{2} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

and

$$R = \begin{bmatrix} \sqrt{2} & 0 & \sqrt{2} & \frac{3}{\sqrt{2}} \\ 0 & \sqrt{2} & 0 & \sqrt{2} \\ 0 & 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 & \frac{-1}{\sqrt{2}} \end{bmatrix}.$$

The readers are advised to check that $A = QR$ is indeed correct.

2. Let $A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & -2 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 2 & 1 \end{bmatrix}$. Find a 4×3 matrix Q satisfying $Q^tQ = I_3$ and an upper triangular matrix R such that $A = QR$.

Solution: Let us apply the Gram Schmidt orthogonalisation to the columns of A. Or equivalently to the rows of A^t. So, we need to apply the process to the subset $\{(1, -1, 1, 1), (1, 0, 1, 0), (1, -2, 1, 2), (0, 1, 0, 1)\}$ of $\mathbb{R}^4$.

Let $\mathbf{u}_1 = (1,-1,1,1)$. Define $\mathbf{v}_1 = \frac{\mathbf{u}_1}{2}$. Let $\mathbf{u}_2 = (1,0,1,0)$. Then

$$\mathbf{w}_2 = (1,0,1,0) - \langle \mathbf{u}_2, \mathbf{v}_1\rangle \mathbf{v}_1 = (1,0,1,0) - \mathbf{v}_1 = \frac{1}{2}(1,1,1,-1).$$

Hence, $\mathbf{v}_2 = \dfrac{(1,1,1,-1)}{2}$. Let $\mathbf{u}_3 = (1,-2,1,2)$. Then

$$\mathbf{w}_3 = \mathbf{u}_3 - \langle \mathbf{u}_3, \mathbf{v}_1\rangle \mathbf{v}_1 - \langle \mathbf{u}_3, \mathbf{v}_2\rangle \mathbf{v}_2 = \mathbf{u}_3 - 3\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{0}.$$

So, we again take $\mathbf{u}_3 = (0,1,0,1)$. Then

$$\mathbf{w}_3 = \mathbf{u}_3 - \langle \mathbf{u}_3, \mathbf{v}_1\rangle \mathbf{v}_1 - \langle \mathbf{u}_3, \mathbf{v}_2\rangle \mathbf{v}_2 = \mathbf{u}_3 - 0\mathbf{v}_1 - 0\mathbf{v}_2 = \mathbf{u}_3.$$

So, $\mathbf{v}_3 = \dfrac{(0,1,0,1)}{\sqrt{2}}$. Hence,

$$Q = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{-1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{-1}{2} & \frac{1}{\sqrt{2}} \end{bmatrix}, \text{ and } R = \begin{bmatrix} 2 & 1 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & \sqrt{2} \end{bmatrix}.$$

The readers are advised to check the following:

(a) rank $(A) = 3$,

(b) $A = QR$ with $Q^tQ = I_3$, and

(c) R a 3×4 upper triangular matrix with rank $(R) = 3$.

Exercise 5.2.9 1. Determine an orthonormal basis of $\mathbb{R}^4$ containing the vectors $(1,-2,1,3)$ and $(2,1,-3,1)$.

2. Prove that the polynomials $1, x, \frac{3}{2}x^2 - \frac{1}{2}, \frac{5}{2}x^3 - \frac{3}{2}x$ form an orthogonal set of functions in the inner product space $C[-1,1]$ with the inner product $\langle f, g\rangle = \int_{-1}^{1} f(t)\overline{g(t)}dt$. Find the corresponding functions, $f(x)$ with $\|f(x)\| = 1$.

3. Consider the vector space $C[-\pi, \pi]$ with the standard inner product defined in the above exercise. Find an orthonormal basis for the subspace spanned by x, $\sin x$ and $\sin(x+1)$.

4. Let M be a subspace of $\mathbb{R}^n$ and $\dim M = m$. A vector $x \in \mathbb{R}^n$ is said to be orthogonal to M if $\langle x, y\rangle = 0$ for every $y \in M$.

   (a) How many linearly independent vectors can be orthogonal to M?

   (b) If $M = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$, determine a maximal set of linearly independent vectors orthogonal to M in $\mathbb{R}^3$.

5. Determine an orthogonal basis of vector subspace spanned by $\{(1,1,0,1), (-1,1,1,-1), (0,2,1,0), (1,0,0,0)\}$ in $\mathbb{R}^4$.

6. Let $S = \{(1,1,1,1), (1,2,0,1), (2,2,4,0)\}$. Find an orthonormal basis of $L(S)$ in $\mathbb{R}^4$.

7. Let $\mathbb{R}^n$ be endowed with the standard inner product. Suppose we have a vector $\mathbf{x}^t = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$, with $\|\mathbf{x}\| = 1$. Then prove the following:

   (a) the set $\{\mathbf{x}\}$ can always be extended to form an orthonormal basis of $\mathbb{R}^n$.

   (b) Let this basis be $\{\mathbf{x}, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$. Suppose $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n)$ is the standard basis of $\mathbb{R}^n$. Let $A = \Big[[\mathbf{x}]_{\mathcal{B}},\ [\mathbf{x}_2]_{\mathcal{B}},\ \ldots,\ [\mathbf{x}_n]_{\mathcal{B}}\Big]$. Then prove that A is an orthogonal matrix.

8. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n, n \geq 1$ with $\|\mathbf{u}\| = \|\mathbf{w}\| = 1$. Prove that there exists an orthogonal matrix A such that $A\mathbf{v} = \mathbf{w}$. Prove also that A can be chosen such that $\det(A) = 1$.

5.3 Orthogonal Projections and Applications

Recall that given a k-dimensional vector subspace of a vector space V of dimension n, one can always find an $(n-k)$-dimensional vector subspace W_0 of V (see Exercise 3.3.19.9) satisfying

$$W + W_0 = V \quad \text{and} \quad W \cap W_0 = \{\mathbf{0}\}.$$

The subspace W_0 is called the complementary subspace of W in V. We now define an important class of linear transformations on an inner product space, called orthogonal projections.

Definition 5.3.1 (Projection Operator) Let V be an n-dimensional vector space and let W be a k-dimensional subspace of V. Let W_0 be a complement of W in V. Then we define a map $P_W : V \longrightarrow V$ by

$$P_W(\mathbf{v}) = \mathbf{w}, \ \text{ whenever } \ \mathbf{v} = \mathbf{w} + \mathbf{w}_0, \ \mathbf{w} \in W, \ \mathbf{w}_0 \in W_0.$$

The map P_W is called the projection of V onto W along W_0.

Remark 5.3.2 *The map P is well defined due to the following reasons:*

1. *$W + W_0 = V$ implies that for every $\mathbf{v} \in V$, we can find $w \in W$ and $w_0 \in W_0$ such that $\mathbf{v} = \mathbf{w} + \mathbf{w}_0$.*

2. *$W \cap W_0 = \{\mathbf{0}\}$ implies that the expression $\mathbf{v} = \mathbf{w} + \mathbf{w}_0$ is unique for every $\mathbf{v} \in V$.*

The next proposition states that the map defined above is a linear transformation from V to V. We omit the proof, as it follows directly from the above remarks.

Proposition 5.3.3 The map $P_W : V \longrightarrow V$ defined above is a linear transformation.

Example 5.3.4 Let $V = \mathbb{R}^3$ and $W = \{(x,y,z) \in \mathbb{R}^3 : x + y - z = 0\}$.

1. Let $W_0 = L(\ (1,2,2)\ )$. Then $W \cap W_0 = \{\mathbf{0}\}$ and $W + W_0 = \mathbb{R}^3$. Also, for any vector $(x,y,z) \in \mathbb{R}^3$, note that $(x,y,z) = \mathbf{w} + \mathbf{w}_0$, where

   $$\mathbf{w} = (z-y, 2z-2x-y, 3z-2x-2y), \ \text{ and } \ \mathbf{w}_0 = (x+y-z)(1,2,2).$$

   So, by definition,

   $$P_W((x,y,z)) = (z-y, 2z-2x-y, 3z-2x-2y) = \begin{bmatrix} 0 & -1 & 1 \\ -2 & -1 & 2 \\ -2 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

2. Let $W_0 = L(\ (1,1,1)\ )$. Then $W \cap W_0 = \{\mathbf{0}\}$ and $W + W_0 = \mathbb{R}^3$. Also, for any vector $(x,y,z) \in \mathbb{R}^3$, note that $(x,y,z) = \mathbf{w} + \mathbf{w}_0$, where

   $$\mathbf{w} = (z-y, z-x, 2z-x-y), \ \text{ and } \ \mathbf{w}_0 = (x+y-z)(1,1,1).$$

   So, by definition,

   $$P_W(\ (x,y,z)\ ) = (z-y, z-x, 2z-x-y) = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Remark 5.3.5 *1. The projection map P_W depends on the complementary subspace W_0.*

2. Observe that for a fixed subspace W, there are infinitely many choices for the complementary subspace W_0.

3. It will be shown later that if V is an inner product space with inner product, $\langle\ ,\ \rangle$, then the subspace W_0 is unique if we put an additional condition that $W_0 = \{\mathbf{v} \in V\ : \langle \mathbf{v}, \mathbf{w}\rangle = 0$ for all $\mathbf{w} \in W\}$.

We now prove some basic properties about projection maps.

Theorem 5.3.6 Let W and W_0 be complementary subspaces of a vector space V. Let $P_W : V \longrightarrow V$ be a projection operator of V onto W along W_0. Then

1. the null space of P_W, $\mathcal{N}(P_W) = \{\mathbf{v} \in V : P_W(\mathbf{v}) = \mathbf{0}\} = W_0$.

2. the range space of P_W, $\mathcal{R}(P_W) = \{P_W(\mathbf{v}) : \mathbf{v} \in V\} = W$.

3. $P_W^2 = P_W$. The condition $P_W^2 = P_W$ is equivalent to $P_W(I - P_W) = \mathbf{0} = (I - P_W)P_W$.

PROOF. We only prove the first part of the theorem.
Let $\mathbf{w}_0 \in W_0$. Then $\mathbf{w}_0 = \mathbf{0} + \mathbf{w}_0$ for $\mathbf{0} \in W$. So, by definition, $P(\mathbf{w}_0) = \mathbf{0}$. Hence, $W_0 \subset \mathcal{N}(P_W)$.

Also, for any $\mathbf{v} \in V$, let $P_W(\mathbf{v}) = \mathbf{0}$ with $\mathbf{v} = \mathbf{w} + \mathbf{w}_0$ for some $\mathbf{w}_0 \in W_0$ and $\mathbf{w} \in W$. Then by definition $\mathbf{0} = P_W(\mathbf{v}) = \mathbf{w}$. That is, $\mathbf{w} = \mathbf{0}$ and $\mathbf{v} = \mathbf{w}_0$. Thus, $\mathbf{v} \in W_0$. Hence $\mathcal{N}(P_W) = W_0$. □

Exercise 5.3.7 1. Let A be an $n \times n$ real matrix with $A^2 = A$. Consider the linear transformation $T_A : \mathbb{R}^n \longrightarrow \mathbb{R}^n$, defined by $T_A(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$. Prove that

   (a) $T_A \circ T_A = T_A$ (use the condition $A^2 = A$).

   (b) $\mathcal{N}(T_A) \cap \mathcal{R}(T_A) = \{\mathbf{0}\}$.
   Hint: Let $\mathbf{x} \in \mathcal{N}(T_A) \cap \mathcal{R}(T_A)$. This implies $T_A(\mathbf{x}) = \mathbf{0}$ and $\mathbf{x} = T_A(\mathbf{y})$ for some $\mathbf{y} \in \mathbb{R}^n$. So,

   $$\mathbf{x} = T_A(\mathbf{y}) = (T_A \circ T_A)(\mathbf{y}) = T_A\big(T_A(\mathbf{y})\big) = T_A(\mathbf{x}) = \mathbf{0}.$$

   (c) $\mathbb{R}^n = \mathcal{N}(T_A) + \mathcal{R}(T_A)$.
   Hint: Let $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ be a basis of $\mathcal{N}(T_A)$. Extend it to get a basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_k, \mathbf{v}_{k+1}, \ldots, \mathbf{v}_n\}$ of $\mathbb{R}^n$. Then by Rank-nullity Theorem 4.3.6, $\{T_A(\mathbf{v}_{k+1}), \ldots, T_A(\mathbf{v}_n)\}$ is a basis of $\mathcal{R}(T_A)$.

   (d) Define $W = \mathcal{R}(T_A)$ and $W_0 = \mathcal{N}(T_A)$. Then T_A is a projection operator of $\mathbb{R}^n$ onto W along W_0.
   Recall that the first three parts of this exercise was also given in Exercise 4.3.10.7.

2. Find all 2×2 real matrices A such that $A^2 = A$. Hence or otherwise, determine all projection operators of $\mathbb{R}^2$.

The next result uses the Gram-Schmidt orthogonalisation process to get the complementary subspace in such a way that the vectors in different subspaces are orthogonal.

Definition 5.3.8 (Orthogonal Subspace of a Set) Let V be an inner product space. Let S be a non-empty subset of V. We define

$$S^{\perp} = \{\mathbf{v} \in V\ : \langle \mathbf{v}, \mathbf{s}\rangle = 0 \text{ for all } \mathbf{s} \in S\}.$$

Example 5.3.9 Let $V = \mathbb{R}$.

1. $S = \{0\}$. Then $S^{\perp} = \mathbb{R}$.

2. $S = \mathbb{R}$, Then $S^{\perp} = \{0\}$.

3. Let S be any subset of $\mathbb{R}$ containing a non-zero real number. Then $S^{\perp} = \{0\}$.

Theorem 5.3.10 Let S be a subset of a finite dimensional inner product space V, with inner product $\langle\ ,\ \rangle$. Then

1. $S^{\perp}$ is a subspace of V.

2. Let S be equal to a subspace W. Then the subspaces W and $W^{\perp}$ are complementary. Moreover, if $\mathbf{w} \in W$ and $\mathbf{u} \in W^{\perp}$, then $\langle \mathbf{u}, \mathbf{w}\rangle = 0$ and $V = W + W^{\perp}$.

PROOF. We leave the prove of the first part for the reader. The prove of the second part is as follows: Let $\dim(V) = n$ and $\dim(W) = k$. Let $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k\}$ be a basis of W. By Gram-Schmidt orthogonalisation process, we get an orthonormal basis, say, $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ of W. Then, for any $\mathbf{v} \in V$,

$$\mathbf{v} - \sum_{i=1}^{k} \langle \mathbf{v}, \mathbf{v}_i\rangle \mathbf{v}_i \in W^{\perp}.$$

So, $V \subset W + W^{\perp}$. Also, for any $\mathbf{v} \in W \cap W^{\perp}$, by definition of $W^{\perp}$, $\ 0 = \langle \mathbf{v}, \mathbf{v}\rangle = \|\mathbf{v}\|^2$. So, $\mathbf{v} = \mathbf{0}$. That is, $W \cap W^{\perp} = \{\mathbf{0}\}$. □

Definition 5.3.11 (Orthogonal Complement) Let W be a subspace of a vector space V. The subspace $W^{\perp}$ is called the orthogonal complement of W in V.

Exercise 5.3.12 1. Let $W = \{(x,y,z) \in \mathbb{R}^3 : x + y + z = 0\}$. Find $W^{\perp}$ with respect to the standard inner product.

2. Let W be a subspace of a finite dimensional inner product space V. Prove that $(W^{\perp})^{\perp} = W$.

3. Let V be the vector space of all $n \times n$ real matrices. Then Exercise5.1.9.6 shows that V is a real inner product space with the inner product given by $\langle A, B\rangle = \text{tr}(AB^t)$. If W is the subspace given by $W = \{A \in V\ :\ A^t = A\}$, determine $W^{\perp}$.

Definition 5.3.13 (Orthogonal Projection) Let W be a subspace of a finite dimensional inner product space V, with inner product $\langle\ ,\ \rangle$. Let $W^{\perp}$ be the orthogonal complement of W in V. Define $P_W : V \longrightarrow V$ by

$$P_W(\mathbf{v}) = \mathbf{w} \ \text{ where } \ \mathbf{v} = \mathbf{w} + \mathbf{u}, \ \text{ with } \ \mathbf{w} \in W, \ \text{ and } \ \mathbf{u} \in W^{\perp}.$$

Then P_W is called the orthogonal projection of V onto W along $W^{\perp}$.

Definition 5.3.14 (Self-Adjoint Transformation/Operator) Let V be an inner product space with inner product $\langle\ ,\ \rangle$. A linear transformation $T : V \longrightarrow V$ is called a self-adjoint operator if $\langle T(\mathbf{v}), \mathbf{u}\rangle = \langle \mathbf{v}, T(\mathbf{u})\rangle$ for every $\mathbf{u}, \mathbf{v} \in V$.

Example 5.3.15 1. Let A be an $n \times n$ real symmetric matrix. That is, $A^t = A$. Then show that the linear transformation $T_A : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$ for every $\mathbf{x}^t \in \mathbb{R}^n$ is a self-adjoint operator.
Solution: By definition, for every $\mathbf{x}^t, \mathbf{y}^t \in \mathbb{R}^n$,

$$\langle T_A(\mathbf{x}), \mathbf{y}\rangle = (\mathbf{y})^t A\mathbf{x} = (\mathbf{y})^t A^t \mathbf{x} = (A\mathbf{y})^t \mathbf{x} = \langle \mathbf{x}, T_A(\mathbf{y})\rangle.$$

Hence, the result follows.

2. Let A be an $n \times n$ Hermitian matrix, that is, $A^* = A$. Then the linear transformation $T_A : \mathbb{C}^n \longrightarrow \mathbb{C}^n$ defined by $T_A(\mathbf{z}) = A\mathbf{z}$ for every $\mathbf{z}^t \in \mathbb{C}^n$ is a self-adjoint operator.

Remark 5.3.16 *1. By Proposition 5.3.3, the map P_W defined above is a linear transformation.*

2. $P_W^2 = P_W$, $(I - P_W)P_W = \mathbf{0} = P_W(I - P_W)$.

3. *Let $\mathbf{u}, \mathbf{v} \in V$ with $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$ and $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$ for some $\mathbf{u}_1, \mathbf{v}_1 \in W$ and $\mathbf{u}_2, \mathbf{v}_2 \in W^\perp$. Then we know that $\langle \mathbf{u}_i, \mathbf{v}_j \rangle = 0$ whenever $1 \leq i \neq j \leq 2$. Therefore, for every $\mathbf{u}, \mathbf{v} \in V$,*

$$\begin{aligned} \langle P_W(\mathbf{u}), \mathbf{v} \rangle &= \langle \mathbf{u}_1, \mathbf{v} \rangle = \langle \mathbf{u}_1, \mathbf{v}_1 + \mathbf{v}_2 \rangle = \langle \mathbf{u}_1, \mathbf{v}_1 \rangle = \langle \mathbf{u}_1 + \mathbf{u}_2, \mathbf{v}_1 \rangle \\ &= \langle \mathbf{u}, P_W(\mathbf{v}) \rangle. \end{aligned}$$

Thus, the orthogonal projection operator is a self-adjoint operator.

4. *Let $\mathbf{v} \in V$ and $\mathbf{w} \in W$. Then $P_W(\mathbf{w}) = \mathbf{w}$ for all $\mathbf{w} \in W$. Therefore, using Remarks 5.3.16.2 and 5.3.16.3, we get*

$$\begin{aligned} \langle \mathbf{v} - P_W(\mathbf{v}), \mathbf{w} \rangle &= \langle \big(I - P_W\big)(\mathbf{v}), P_W(\mathbf{w}) \rangle = \langle P_W\big(I - P_W\big)(\mathbf{v}), \mathbf{w} \rangle \\ &= \langle \mathbf{0}(\mathbf{v}), \mathbf{w} \rangle = \langle \mathbf{0}, \mathbf{w} \rangle = 0 \end{aligned}$$

for every $\mathbf{w} \in W$.

5. *In particular, $\langle \mathbf{v} - P_W(\mathbf{v}), P_W(\mathbf{v}) - \mathbf{w} \rangle = 0$ as $P_W(\mathbf{v}) \in W$. Thus, $\langle \mathbf{v} - P_W(\mathbf{v}), P_W(\mathbf{v}) - \mathbf{w}' \rangle = 0$, for every $\mathbf{w}' \in W$. Hence, for any $\mathbf{v} \in V$ and $\mathbf{w} \in W$, we have*

$$\begin{aligned} \|\mathbf{v} - \mathbf{w}\|^2 &= \|\mathbf{v} - P_W(\mathbf{v}) + P_W(\mathbf{v}) - \mathbf{w}\|^2 \\ &= \|\mathbf{v} - P_W(\mathbf{v})\|^2 + \|P_W(\mathbf{v}) - \mathbf{w}\|^2 \\ &\qquad +2\langle \mathbf{v} - P_W(\mathbf{v}), P_W(\mathbf{v}) - \mathbf{w} \rangle \\ &= \|\mathbf{v} - P_W(\mathbf{v})\|^2 + \|P_W(\mathbf{v}) - \mathbf{w}\|^2. \end{aligned}$$

Therefore,

$$\|\mathbf{v} - \mathbf{w}\| \geq \|\mathbf{v} - P_W(\mathbf{v})\|$$

and the equality holds if and only if $\mathbf{w} = P_W(\mathbf{v})$. Since $P_W(\mathbf{v}) \in W$, we see that

$$d(\mathbf{v}, W) = \inf \{\|\mathbf{v} - \mathbf{w}\| \ : \mathbf{w} \in W\} = \|\mathbf{v} - P_W(\mathbf{v})\|.$$

That is, $P_W(\mathbf{v})$ is the vector nearest to $\mathbf{v} \in W$. This can also be stated as: the vector $P_W(\mathbf{v})$ solves the following minimisation problem:

$$\inf_{\mathbf{w} \in W} \|\mathbf{v} - \mathbf{w}\| = \|\mathbf{v} - P_W(\mathbf{v})\|.$$

5.3.1 Matrix of the Orthogonal Projection

The minimization problem stated above arises in lot of applications. So, it will be very helpful if the matrix of the orthogonal projection can be obtained under a given basis.

To this end, let W be a k-dimensional subspace of $\mathbb{R}^n$ with $W^\perp$ as its orthogonal complement. Let $P_W : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be the orthogonal projection of $\mathbb{R}^n$ onto W. Suppose, we are given an orthonormal basis $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$ of W. Under the assumption that $\mathcal{B}$ is known, we explicitly give the matrix of P_W with respect to an extended ordered basis of $\mathbb{R}^n$.

Let us extend the given ordered orthonormal basis $\mathcal{B}$ of W to get an orthonormal ordered basis $\mathcal{B}_1 = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k, \mathbf{v}_{k+1} \ldots, \mathbf{v}_n)$ of $\mathbb{R}^n$. Then by Theorem 5.1.11, for any $\mathbf{v} \in \mathbb{R}^n$, $\mathbf{v} = \sum_{i=1}^{n} \langle \mathbf{v}, \mathbf{v}_i \rangle \mathbf{v}_i$. Thus, by definition, $P_W(\mathbf{v}) = \sum_{i=1}^{k} \langle \mathbf{v}, \mathbf{v}_i \rangle \mathbf{v}_i$. Let $A = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k]$. Consider the standard orthogonal

ordered basis $\mathcal{B}_2 = (\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n)$ of $\mathbb{R}^n$. Therefore, if $\mathbf{v}_i = \sum_{j=1}^{n} a_{ji} \mathbf{e}_j$, for $1 \leq i \leq k$, then

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nk} \end{bmatrix}, \ [\mathbf{v}]_{\mathcal{B}_2} = \begin{bmatrix} \sum_{i=1}^{n} a_{1i} \langle \mathbf{v}, \mathbf{v}_i \rangle \\ \sum_{i=1}^{n} a_{2i} \langle \mathbf{v}, \mathbf{v}_i \rangle \\ \vdots \\ \sum_{i=1}^{n} a_{ni} \langle \mathbf{v}, \mathbf{v}_i \rangle \end{bmatrix}$$

and

$$[P_W(\mathbf{v})]_{\mathcal{B}_2} = \begin{bmatrix} \sum_{i=1}^{k} a_{1i} \langle \mathbf{v}, \mathbf{v}_i \rangle \\ \sum_{i=1}^{k} a_{2i} \langle \mathbf{v}, \mathbf{v}_i \rangle \\ \vdots \\ \sum_{i=1}^{k} a_{ni} \langle \mathbf{v}, \mathbf{v}_i \rangle \end{bmatrix}.$$

Then as observed in Remark 5.2.3.4, $A^t A = I_k$. That is, for $1 \leq i, j \leq k$,

$$\sum_{s=1}^{n} a_{si} a_{sj} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases} \tag{5.3.1}$$

Thus, using the associativity of matrix product and (5.3.1), we get

$$\begin{aligned} (AA^t)(\mathbf{v}) &= A \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1k} & a_{2k} & \cdots & a_{nk} \end{bmatrix} \begin{bmatrix} \sum_{i=1}^{n} a_{1i} \langle \mathbf{v}, \mathbf{v}_i \rangle \\ \sum_{i=1}^{n} a_{2i} \langle \mathbf{v}, \mathbf{v}_i \rangle \\ \vdots \\ \sum_{i=1}^{n} a_{ni} \langle \mathbf{v}, \mathbf{v}_i \rangle \end{bmatrix} \\ &= A \begin{bmatrix} \sum_{s=1}^{n} a_{s1} \left(\sum_{i=1}^{n} a_{si} \langle \mathbf{v}, \mathbf{v}_i \rangle \right) \\ \sum_{s=1}^{n} a_{s2} \left(\sum_{i=1}^{n} a_{si} \langle \mathbf{v}, \mathbf{v}_i \rangle \right) \\ \vdots \\ \sum_{s=1}^{n} a_{sk} \left(\sum_{i=1}^{n} a_{si} \langle \mathbf{v}, \mathbf{v}_i \rangle \right) \end{bmatrix} = A \begin{bmatrix} \sum_{i=1}^{n} \left(\sum_{s=1}^{n} a_{s1} a_{si} \right) \langle \mathbf{v}, \mathbf{v}_i \rangle \\ \sum_{i=1}^{n} \left(\sum_{s=1}^{n} a_{s2} a_{si} \right) \langle \mathbf{v}, \mathbf{v}_i \rangle \\ \vdots \\ \sum_{i=1}^{n} \left(\sum_{s=1}^{n} a_{sk} a_{si} \right) \langle \mathbf{v}, \mathbf{v}_i \rangle \end{bmatrix} \\ &= A \begin{bmatrix} \langle \mathbf{v}, \mathbf{v}_1 \rangle \\ \langle \mathbf{v}, \mathbf{v}_2 \rangle \\ \vdots \\ \langle \mathbf{v}, \mathbf{v}_k \rangle \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{k} a_{1i} \langle \mathbf{v}, \mathbf{v}_i \rangle \\ \sum_{i=1}^{k} a_{2i} \langle \mathbf{v}, \mathbf{v}_i \rangle \\ \vdots \\ \sum_{i=1}^{k} a_{ni} \langle \mathbf{v}, \mathbf{v}_i \rangle \end{bmatrix} \\ &= [P_W(\mathbf{v})]_{\mathcal{B}_2}. \end{aligned}$$

Thus $P_W[\mathcal{B}_2, \mathcal{B}_2] = AA^t$. Thus, we have proved the following theorem.

Theorem 5.3.17 Let W be a k-dimensional subspace of $\mathbb{R}^n$ and let $P_W : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be the orthogonal projection of $\mathbb{R}^n$ onto W along $W^\perp$. Suppose, $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$ is an orthonormal ordered basis of W. Define an $n \times k$ matrix $A = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k]$. Then the matrix of the linear transformation P_W in the standard orthogonal ordered basis $(\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n)$ is AA^t.

Example 5.3.18 Let $W = \{(x, y, z, w) \in \mathbb{R}^4 : x = y, z = w\}$ be a subspace of W. Then an orthonormal ordered basis of W is

$$\big(\frac{1}{\sqrt{2}}(1, 1, 0, 0), \frac{1}{\sqrt{2}}(0, 0, 1, 1)\big),$$

and that of $W^\perp$ is

$$\big(\frac{1}{\sqrt{2}}(1, -1, 0, 0), \frac{1}{\sqrt{2}}(0, 0, 1, -1)\big).$$

Therefore, if $P_W : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$ is an orthogonal projection of $\mathbb{R}^4$ onto W along $W^\perp$, then the corresponding matrix A is given by

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Hence, the matrix of the orthogonal projection P_W in the ordered basis

$$\mathcal{B} = \big(\frac{1}{\sqrt{2}}(1, 1, 0, 0), \frac{1}{\sqrt{2}}(0, 0, 1, 1), \frac{1}{\sqrt{2}}(1, -1, 0, 0), \frac{1}{\sqrt{2}}(0, 0, 1, -1)\big)$$

is

$$P_W[\mathcal{B}, \mathcal{B}] = AA^t = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

It is easy to see that

1. the matrix $P_W[\mathcal{B}, \mathcal{B}]$ is symmetric,
2. $P_W[\mathcal{B}, \mathcal{B}]^2 = P_W[\mathcal{B}, \mathcal{B}]$, and
3. $\big(I_4 - P_W[\mathcal{B}, \mathcal{B}]\big)P_W[\mathcal{B}, \mathcal{B}] = \mathbf{0} = P_W[\mathcal{B}, \mathcal{B}]\big(I_4 - P_W[\mathcal{B}, \mathcal{B}]\big)$.

Also, for any $(x, y, z, w) \in \mathbb{R}^4$, we have

$$[(x, y, z, w)]_{\mathcal{B}} = \left(\frac{x+y}{\sqrt{2}}, \frac{z+w}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}, \frac{z-w}{\sqrt{2}} \right)^t.$$

Thus, $P_W\big((x, y, z, w)\big) = \dfrac{x+y}{2}(1, 1, 0, 0) + \dfrac{z+w}{2}(0, 0, 1, 1)$ is the closest vector to the subspace W for any vector $(x, y, z, w) \in \mathbb{R}^4$.

Exercise 5.3.19

1. Show that for any non-zero vector $\mathbf{v}^t \in \mathbb{R}^n$, the rank of the matrix $\mathbf{v}\mathbf{v}^t$ is 1.
2. Let W be a subspace of a vector space V and let $P : V \longrightarrow V$ be the orthogonal projection of V onto W along $W^\perp$. Let $\mathcal{B}$ be an orthonormal ordered basis of V. Then prove that corresponding matrix satisfies $P[\mathcal{B}, \mathcal{B}]^t = P[\mathcal{B}, \mathcal{B}]$.
3. Let A be an $n \times n$ matrix with $A^2 = A$ and $A^t = A$. Consider the associated linear transformation $T_A : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ defined by $T_A(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v}^t \in \mathbb{R}^n$. Then prove that there exists a subspace W of $\mathbb{R}^n$ such that T_A is the orthogonal projection of $\mathbb{R}^n$ onto W along $W^\perp$.
4. Let W_1 and W_2 be two distinct subspaces of a finite dimensional vector space V. Let P_{W_1} and P_{W_2} be the corresponding orthogonal projection operators of V along $W_1^\perp$ and $W_2^\perp$, respectively. Then by constructing an example in $\mathbb{R}^2$, show that the map $P_{W_1} \circ P_{W_2}$ is a projection but not an orthogonal projection.

5. Let W be an $(n-1)$-dimensional vector subspace of $\mathbb{R}^n$ and let $W^\perp$ be its orthogonal complement. Let $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n-1}, \mathbf{v}_n)$ be an orthogonal ordered basis of $\mathbb{R}^n$ with $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n-1})$ an ordered basis of W. Define a map

$$T : \mathbb{R}^n \longrightarrow \mathbb{R}^n \quad \text{by } T(\mathbf{v}) = \mathbf{w}_0 - \mathbf{w}$$

whenever $\mathbf{v} = \mathbf{w} + \mathbf{w}_0$ for some $\mathbf{w} \in W$ and $\mathbf{w}_0 \in W^\perp$. Then

(a) prove that T is a linear transformation,

(b) find the matrix, $T[\mathcal{B}, \mathcal{B}]$, and

(c) prove that $T[\mathcal{B}, \mathcal{B}]$ is an orthogonal matrix.

T is called the reflection along $W^\perp$.